Hydrogen and Power-to-X

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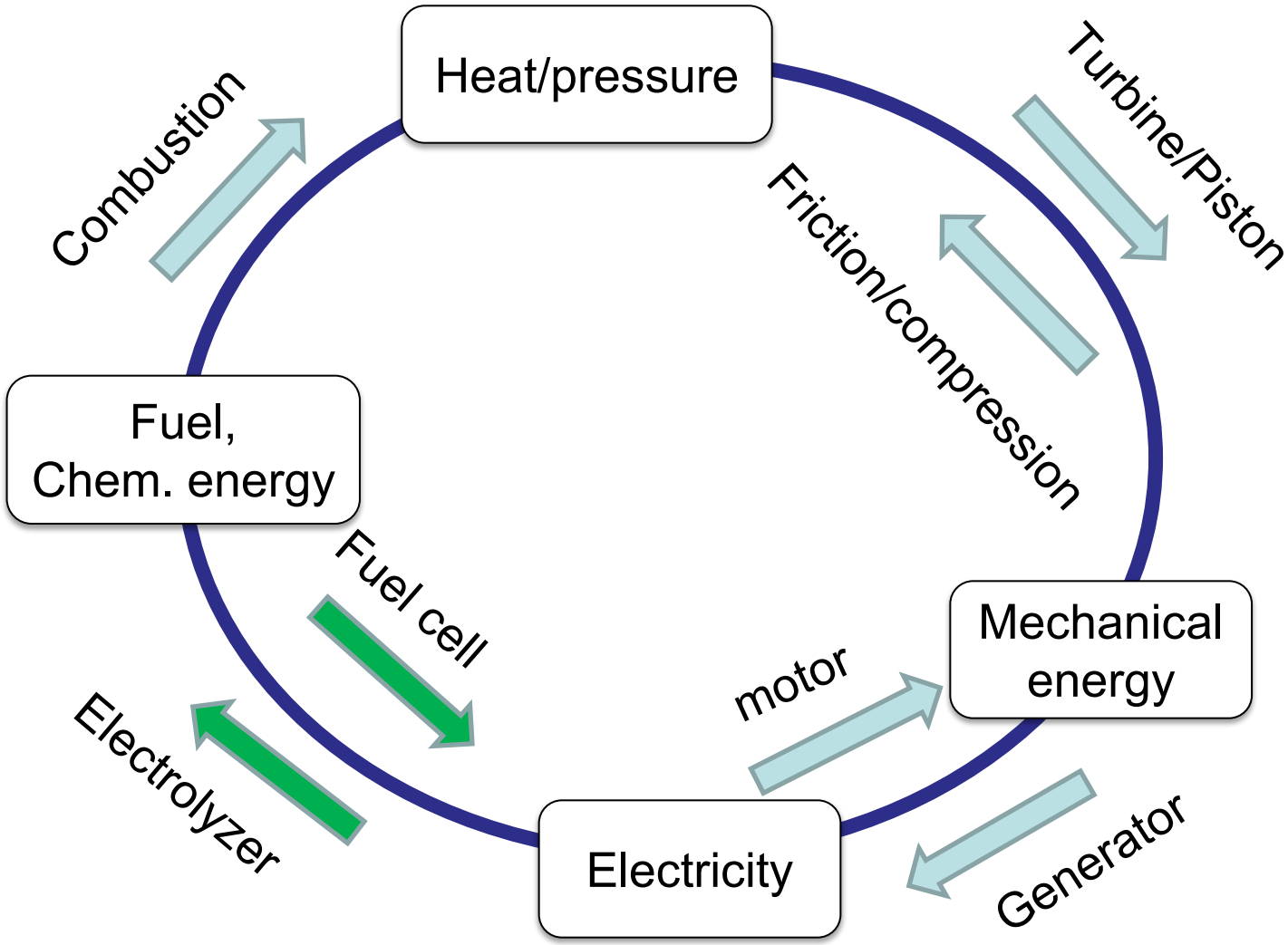
Outline

- The hydrogen energy concept
- Thermodynamics and efficiency
- Real cell voltages
- The function of a fuel cell
- The function of an electrolyzer
- Types of fuel cells
- Types of electrolyzers
- Implementation of fuel cells
- Implementation of electrolyzers
- Synthetic fuels
- Carbon capture

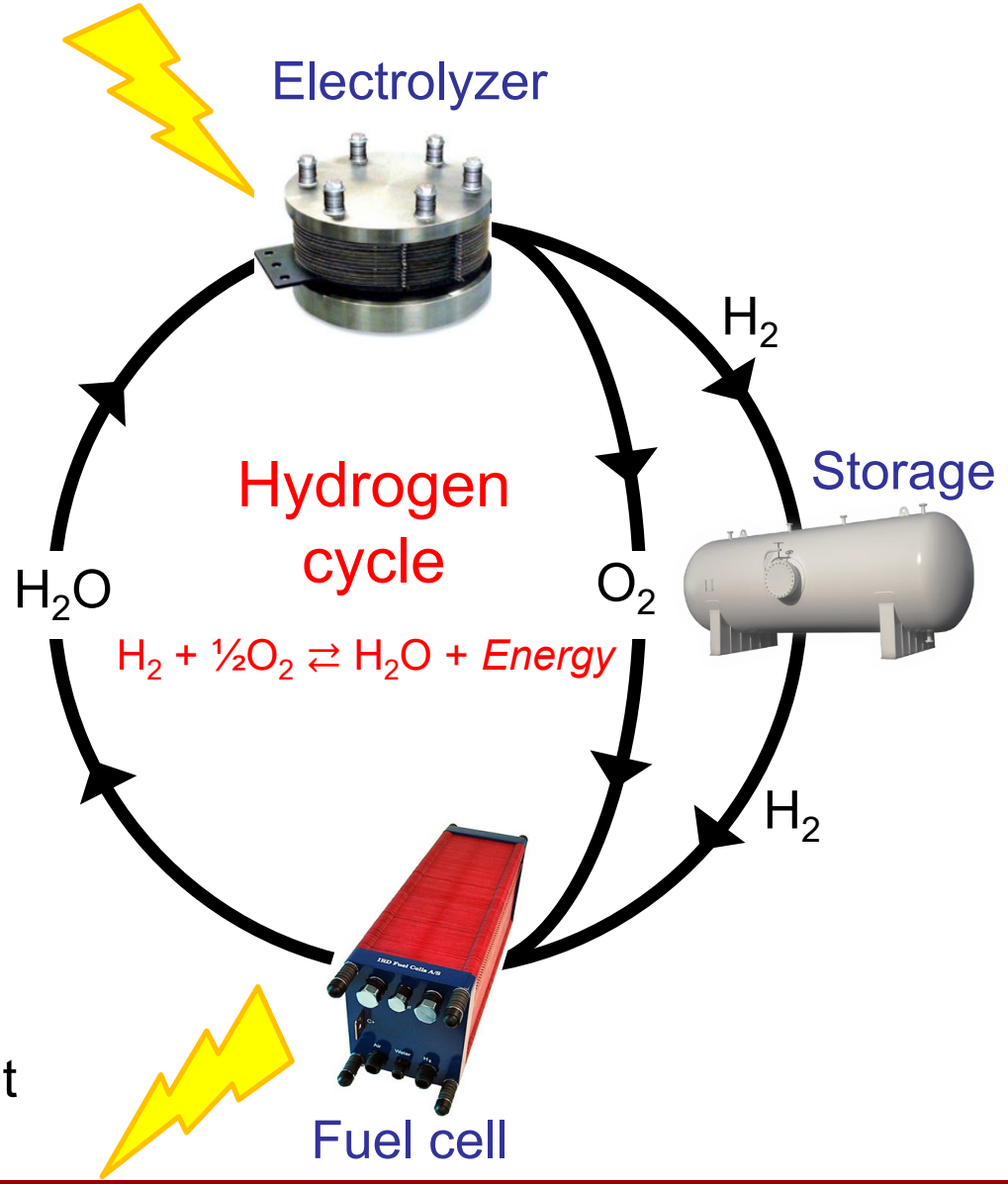
Hydrogen energy

Conversion - fuel to work



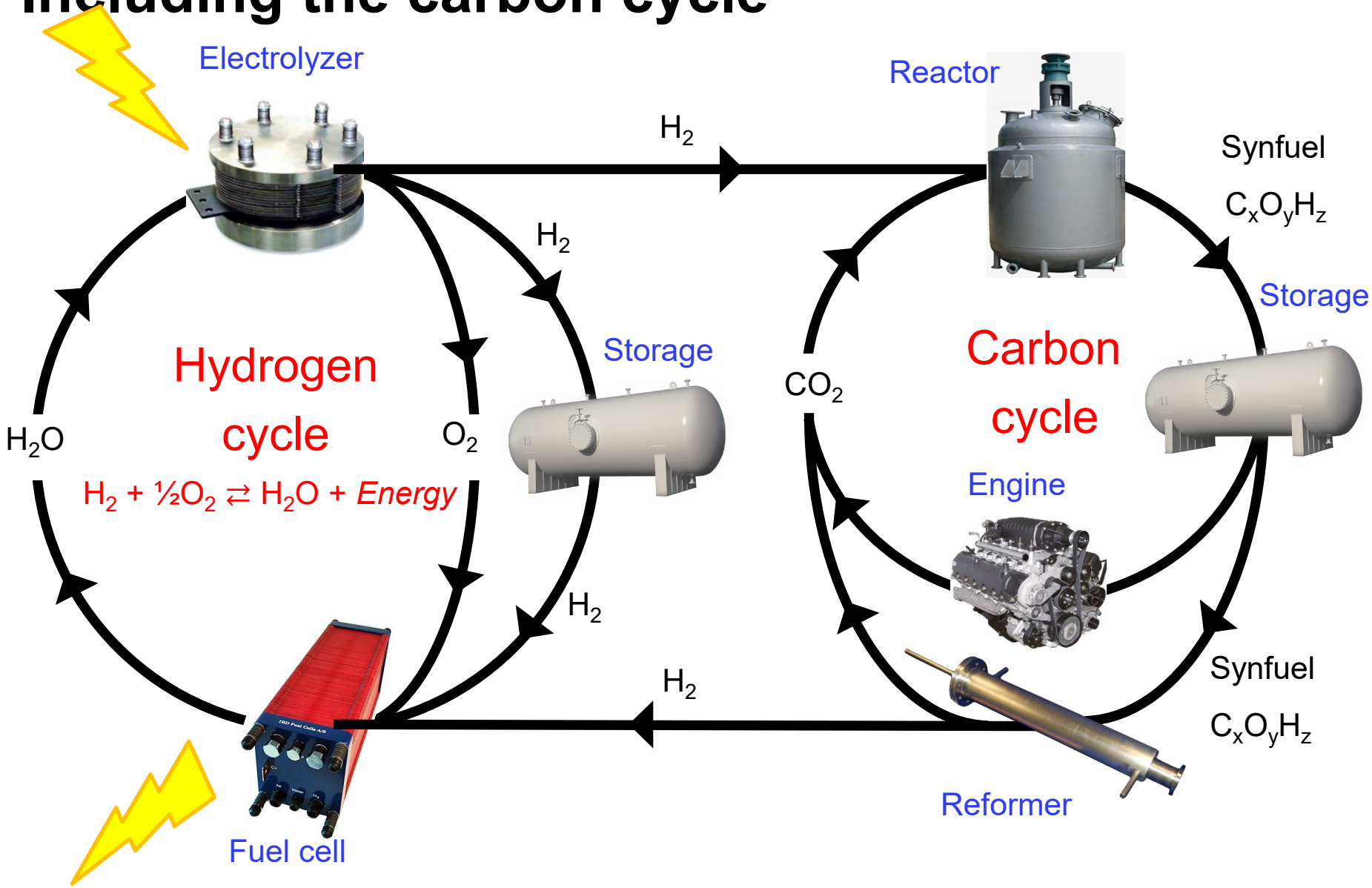
The hydrogen cycle

Electrical energy in

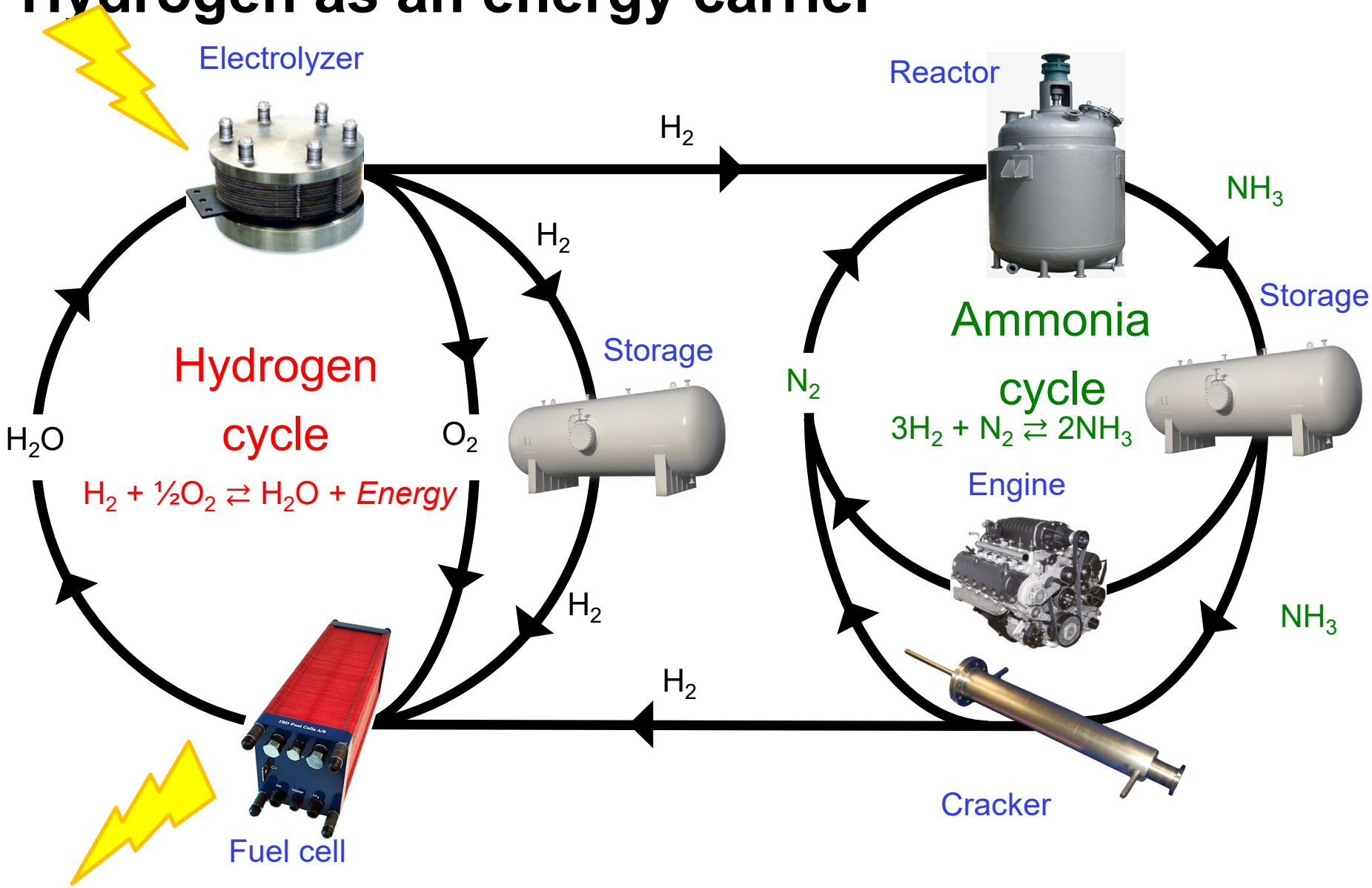


Electrical energy out

Including the carbon cycle



Hydrogen as an energy carrier



The colours of hydrogen

	Color	Definition
Overall common understanding	green	hydrogen directly or indirectly produced using solar, wind and water power without generating direct CO2 emissions or other waste
	grey	hydrogen from steam reforming of natural gas without the prevention or containment of CO2 emissions
	blue	hydrogen from steam reforming of natural gas with additional CCS to prevent CO2 emissions (regardless of the capture rate)
	turquoise	hydrogen from pyrolysis of natural gas with generation of solid carbon as waste product
Coal-based hydrogen	brown	hydrogen from gasification of lignite
	black	hydrogen from gasification of black coal
Nuclear power based hydrogen	pink	hydrogen from electrolysis using electrical energy from nuclear power plants
	purple	hydrogen from combined chemo-thermal electrolysis based on nuclear power
	red	hydrogen from high temperature catalytic splitting of water based on nuclear power
Other introduced colors	yellow	hydrogen from electrolysis using a mixed carbon electrical grid
	white	naturally occurring hydrogen
	orange	hydrogen from conversion of biomass

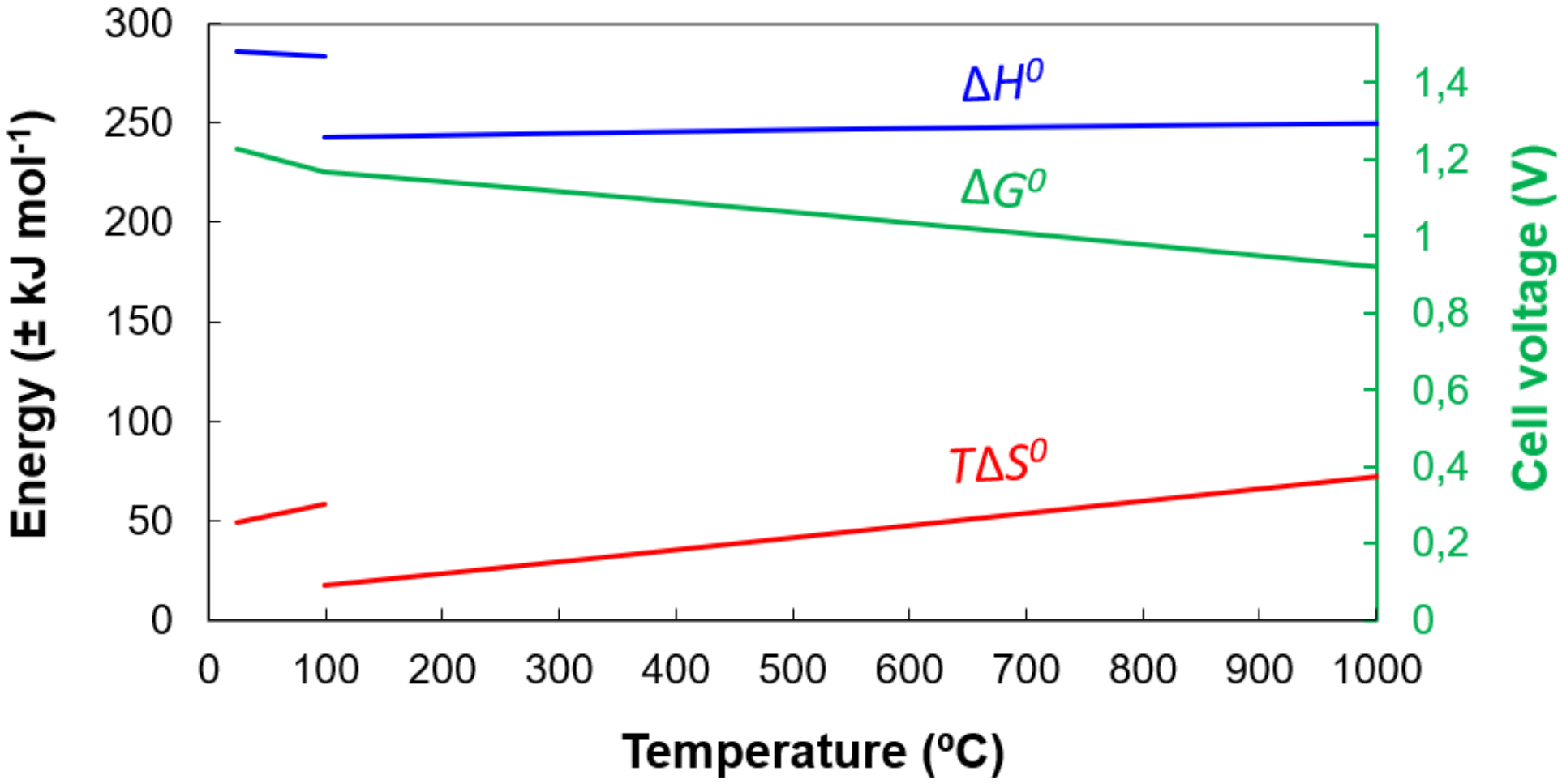
from Weisenburger 2025, DOI: 10.1016/j.erss.2025.104346, CC BY 4.0

The reactions in conversion cells

Proton conducting electrolyte (acidic). Ion conducted: H^+	
Fuel cell anode →/ Electrolyzer cathode ←:	$2H_2 \rightleftarrows 4H^+ + 4e^-$
Fuel cell cathode →/ Electrolyzer anode ←:	$4H^+ + O_2 + 4e^- \rightleftarrows 2H_2O$
Total:	$2H_2 + O_2 \rightleftarrows 2H_2O$
Oxide ion conducting electrolyte. Ion conducted: O^{2-}	
Fuel cell anode →/ Electrolyzer cathode ←:	$2H_2 + 2O^{2-} \rightleftarrows H_2O + 4e^-$
Fuel cell cathode →/ Electrolyzer anode ←:	$O_2 + 4e^- \rightleftarrows 2O^{2-}$
Total:	$2H_2 + O_2 \rightleftarrows 2H_2O$
Hydroxide ion conducting electrolyte (alkaline). Ion conducted: OH^-	
Fuel cell anode →/ Electrolyzer cathode ←:	$2H_2 + 4OH^- \rightleftarrows 4H_2O + 4e^-$
Fuel cell cathode →/ Electrolyzer anode ←:	$O_2 + 2H_2O + 4e^- \rightleftarrows 4OH^-$
Total:	$2H_2 + O_2 \rightleftarrows 2H_2O$

Thermodynamics and efficiency

Thermodynamic functions of $2H_2 + O_2 \rightleftarrows 2H_2O$



The available energy and maximum efficiency, η

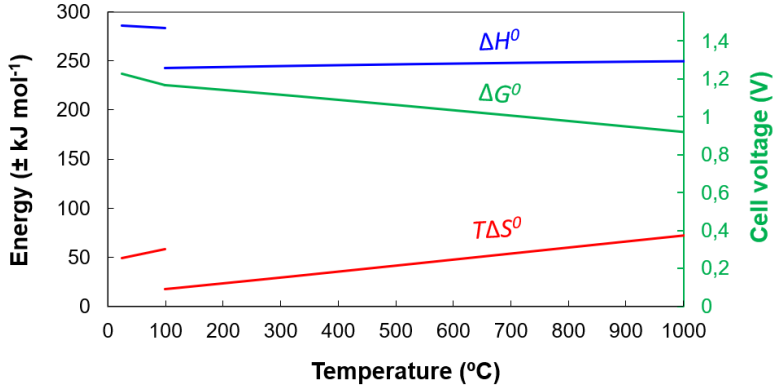
Carnot efficiency:
(any heat engine)

$$\eta_{\text{carnot}} = \frac{T_{\text{high}} - T_{\text{low}}}{T_{\text{high}}}$$

$\Delta G_{\text{r}}^{0}(T)$ is maximum work related to a fuel

ΔH_{r}^{0} is total energy related to a fuel

$$\eta = \frac{\text{What you get}}{\text{What you pay}}$$



Maximum fuel cell efficiency:

$$\eta_{\text{fuel cell}} = \frac{\Delta G(T)}{\Delta H°}$$

The maximum electrical efficiency

$\Delta G_r^0(T)$ is maximum work related to a fuel

ΔH_r^0 is total energy related to a fuel

$$\eta = \frac{\text{What you get}}{\text{What you pay}}$$

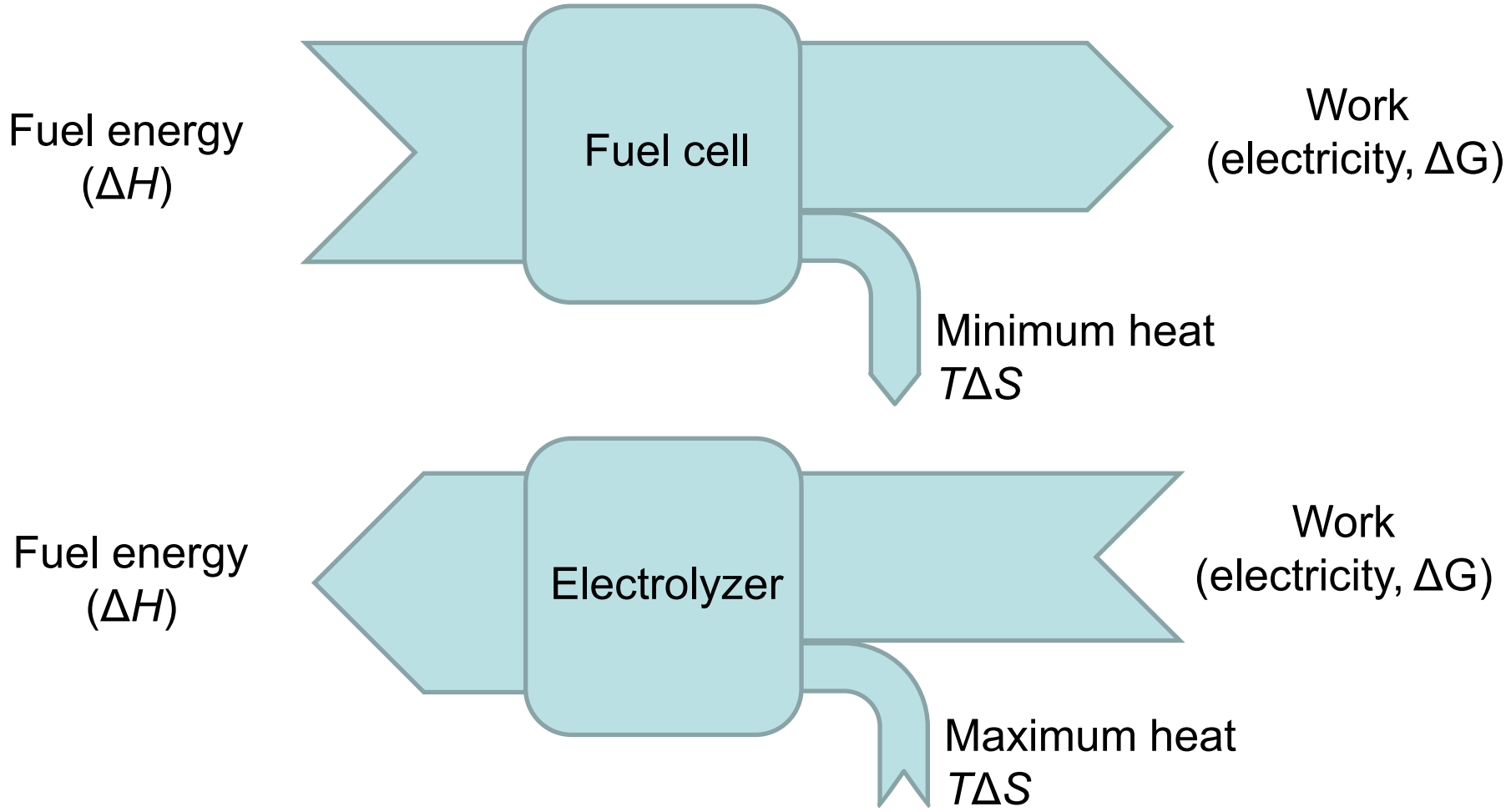
Maximum fuel cell efficiency:

$$\eta_{FC} = \frac{\Delta G_r^0(T)}{\Delta H_r^0}$$

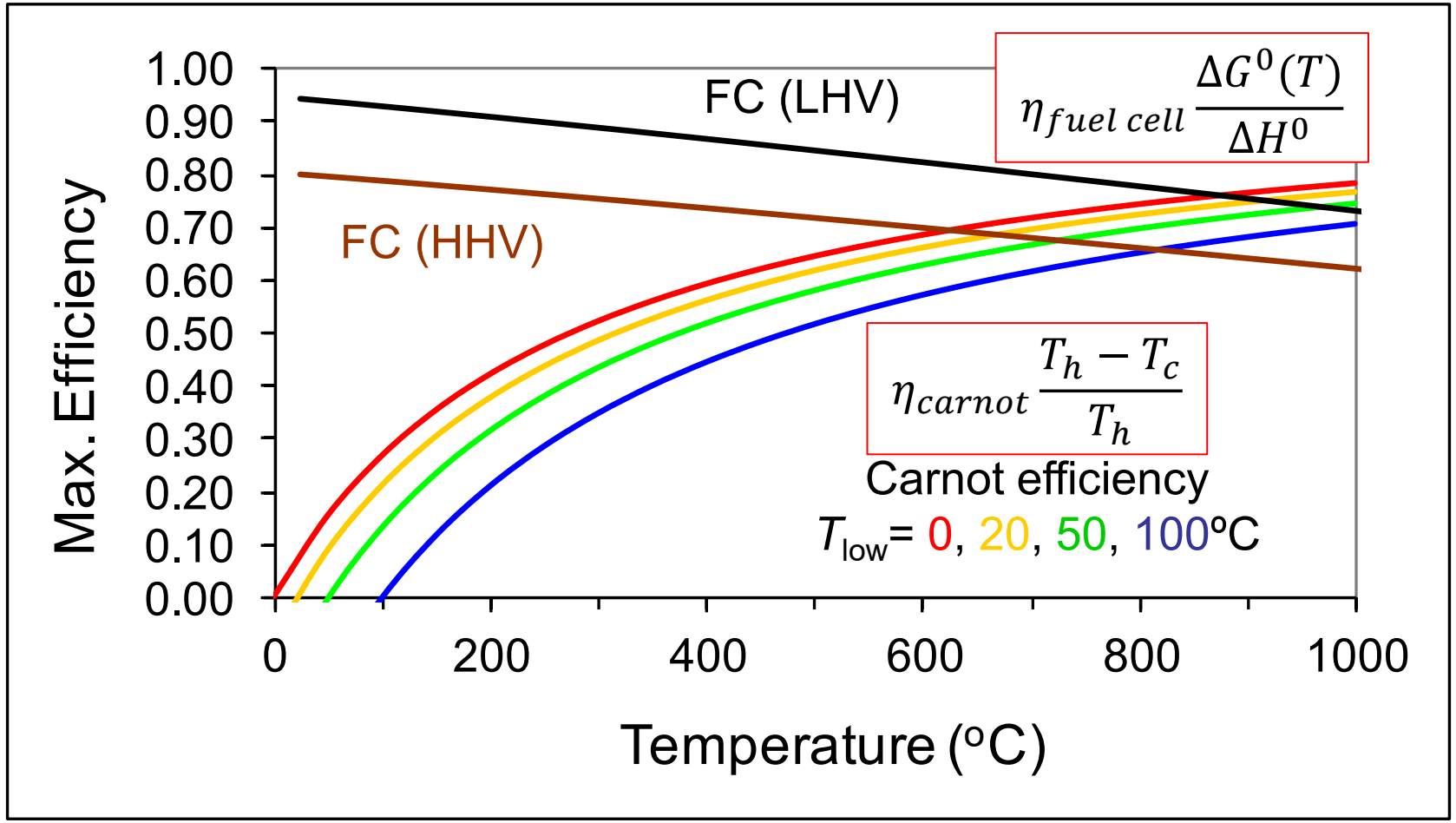
Maximum electrolyzer efficiency:

$$\eta_{EC} = \frac{\Delta H_r^0}{\Delta G_r^0(T)}$$

Fuel and work



Maximum fuel cell electrical efficiency



Key numbers for fuel cells and electrolyzers

	Fuel cell $H_2 + ½O_2 \rightarrow H_2O$		Electrolyzer $H_2O \rightarrow H_2 + ½O_2$	
	Water as liquid (HHV)	Water as vapour (LHV)	Water as liquid (HHV)	Water as vapour (LHV)
Standard enthalpy of reaction, ΔH_r^0.	-285.8 kJ mol^{-1}	-241.8 kJ mol^{-1}	285.8 kJ mol^{-1}	241.8 kJ mol^{-1}
Standard free energy of reaction, ΔG_r^0	-237.1 kJ mol^{-1}	-228.6 kJ mol^{-1}	237.1 kJ mol^{-1}	228.6 kJ mol^{-1}
Maximum voltage efficiency, η_{max}	$(\Delta G_r^0/\Delta H_r^0)$ 83%	$(\Delta G_r^0/\Delta H_r^0)$ 98%	$(\Delta H_r^0/\Delta G_r^0)$ 121%	$(\Delta H_r^0/\Delta G_r^0)$ 106%
Reversible voltage, E_{rev}^0	1.23 V	1.18 V	1.23 V	1.18 V
Thermoneutral voltage, E_{TN}^0	1.48 V	1.25 V	1.48 V	1.25 V
Minimum total energy for producing H_2 (Nm^3 at 1 atm, 0 °C)			39.38 kWh kg^{-1} 3.54 kWh Nm^{-3}	33.32 kWh kg^{-1} 3.00 kWh Nm^{-3}

Real cell voltages

Thermodynamics versus kinetics

Kinetics and thermodynamics are two different angles on processes

Thermodynamics governs the idealized energy

- Minimum electrical energy to run an electrolyzer
- Maximum electrical energy from a fuel cell

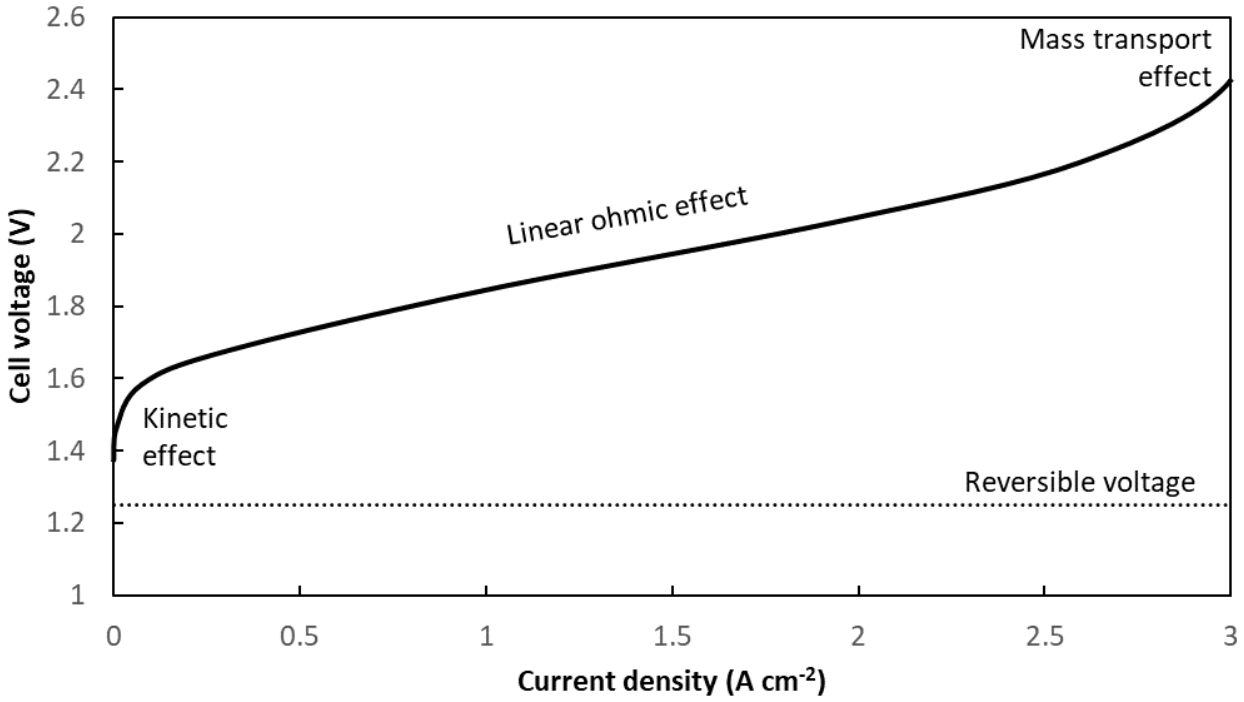
Kinetics governs the real power

- Real power to run an electrolyzer
- Real power from a fuel cell

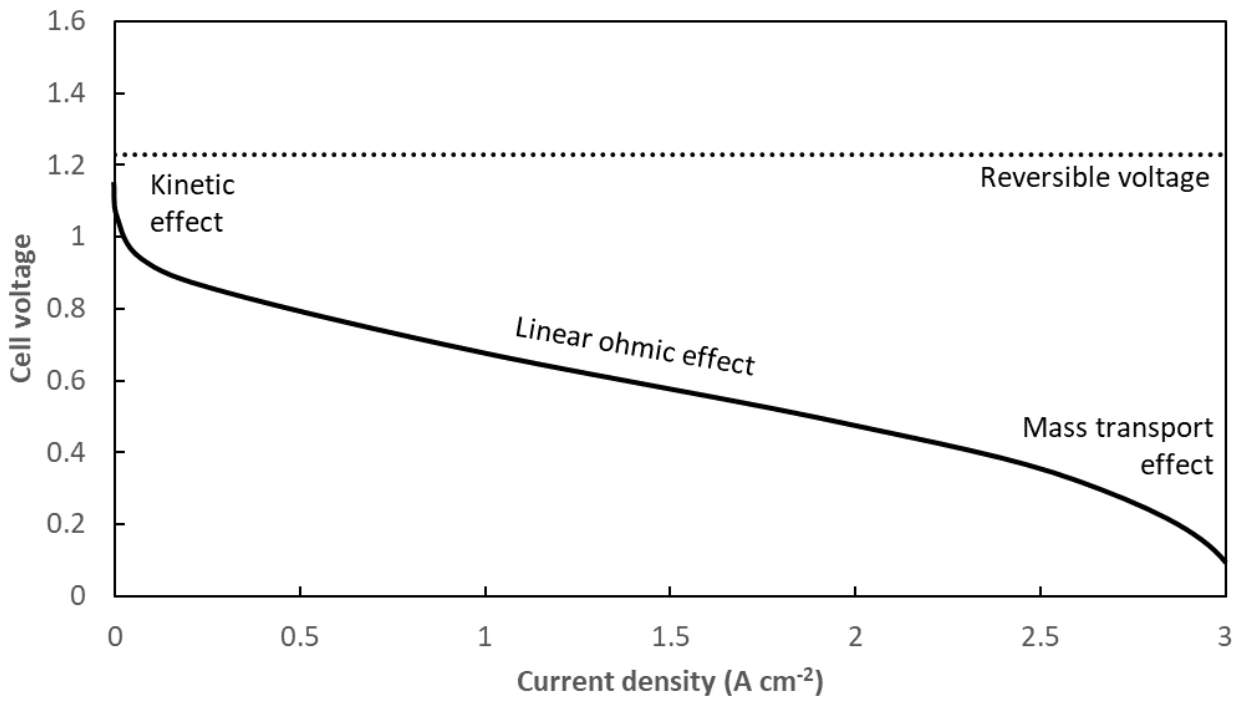
The polarization curve

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightleftarrows H_2O(l)$$

Electrolyzer



Fuel cell



Efficiencies

Efficiency	Description	Applies to
Thermodynamic efficiency	The thermodynamic limit for the electrical efficiency	Fuel cells Electrolyzer cells
Faradaic efficiency (current efficiency)	The degree of fulfilment of Faraday's law.	Fuel cells: The efficiency of conversion of the fuel molecules to electrons. Electrolyzers: The efficiency of conversion of electrons to product molecules
Voltage efficiency	The ration between the actual cell voltage and the idealized voltage determined by thermodynamics.	Fuel cells and electrolyzer cells.
Fuel efficiency	The fraction of the fuel that is consumed in a fuel cell.	Fuel cells only. In some cases, a small fraction of the fuel is passing unconverted.
System efficiency	The total electrical efficiency of the entire system including the balance of plant	Fuel cell systems and electrolyzer systems.

The thermo-neutral voltage – the voltage of 100% efficiency

$$H_2O(l) \rightleftarrows H_2(g) + ½O_2(g)$$

$$E_{TN} = \frac{\Delta H_r^0}{nF} = 1.48\ \text{V (HHV, liquid water)}$$

$$H_2O(g) \rightleftarrows H_2(g) + ½O_2(g)$$

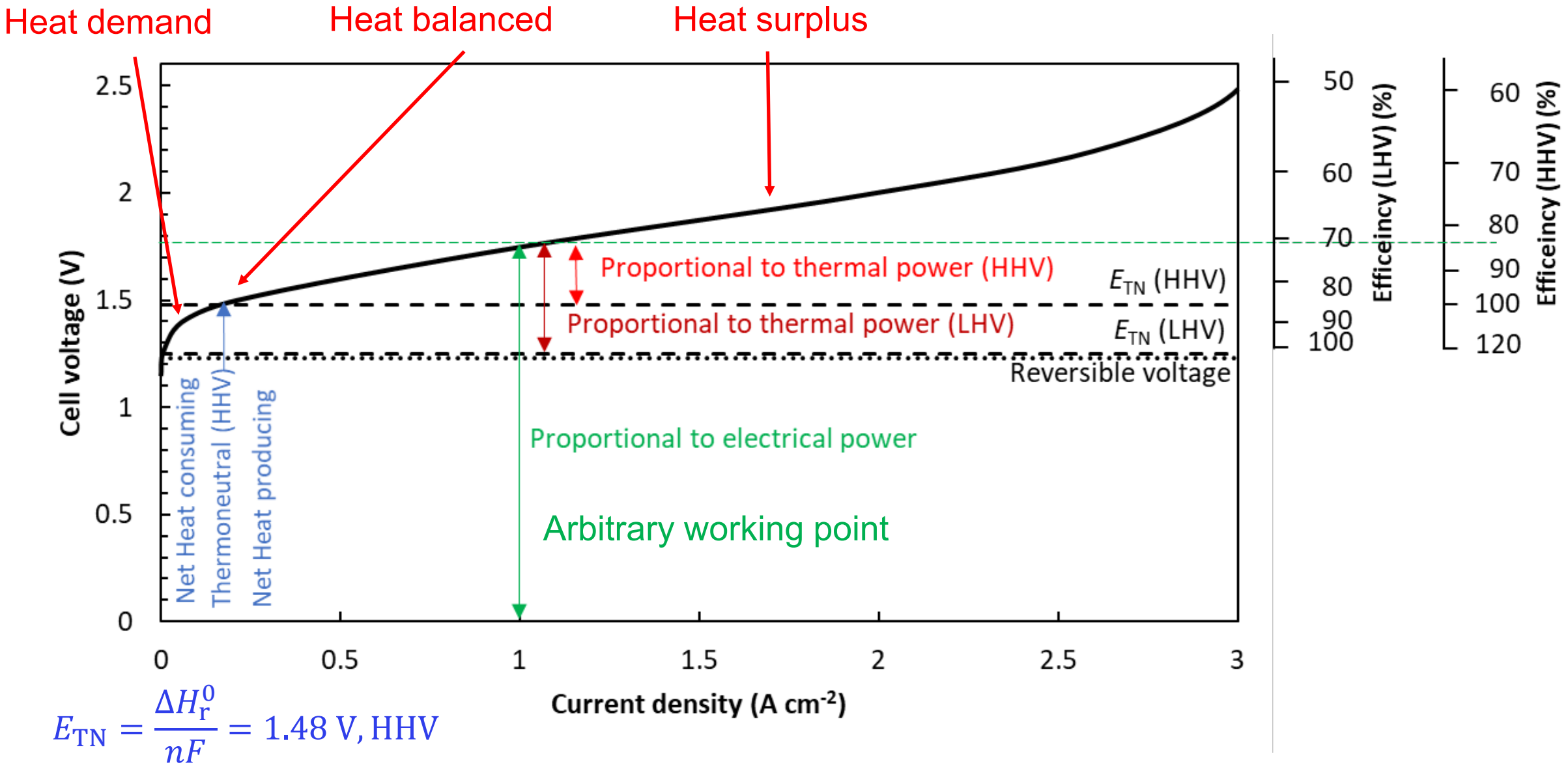
$$E_{TN} = \frac{\Delta H_r^0}{nF} = 1.25\ \text{V (LHV, water vapour)}$$

The voltage efficiency (*fuel cells and electrolyzers*)

Real fuel cell (voltage) efficiency: $$\eta_{\mathrm{FC}} = \frac{E_{\mathrm{cell}}}{E_{\mathrm{TN}}}$$

Real electrolyzer (voltage) efficiency: $$\eta_{\mathrm{EC}} = \frac{E_{\mathrm{TN}}}{E_{\mathrm{cell}}}$$

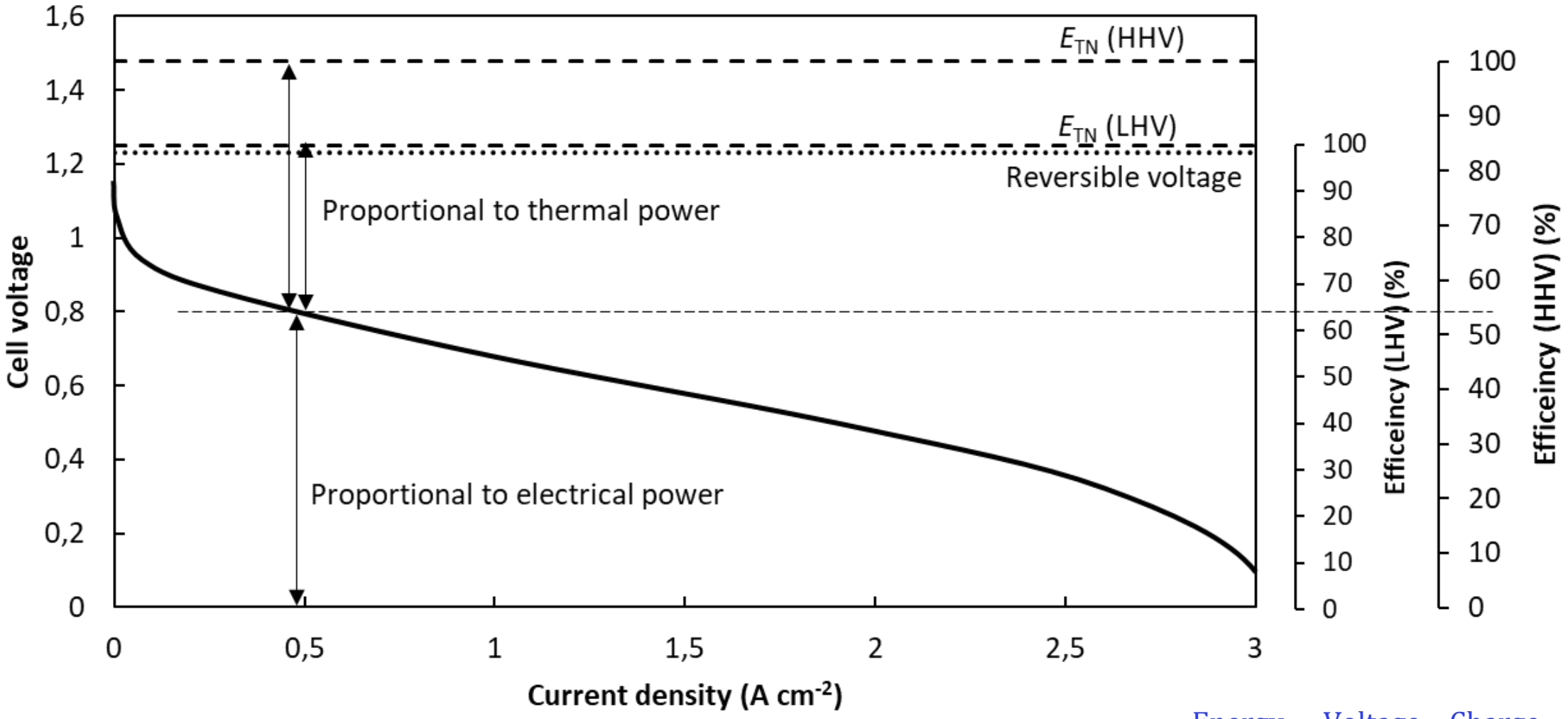
The polarization curve



$$E_{\mathrm{TN}} = \frac{\Delta H_{\mathrm{r}}^{0}}{nF} = 1.48\ \mathrm{V, HHV}$$

Efficiency and heat evolved

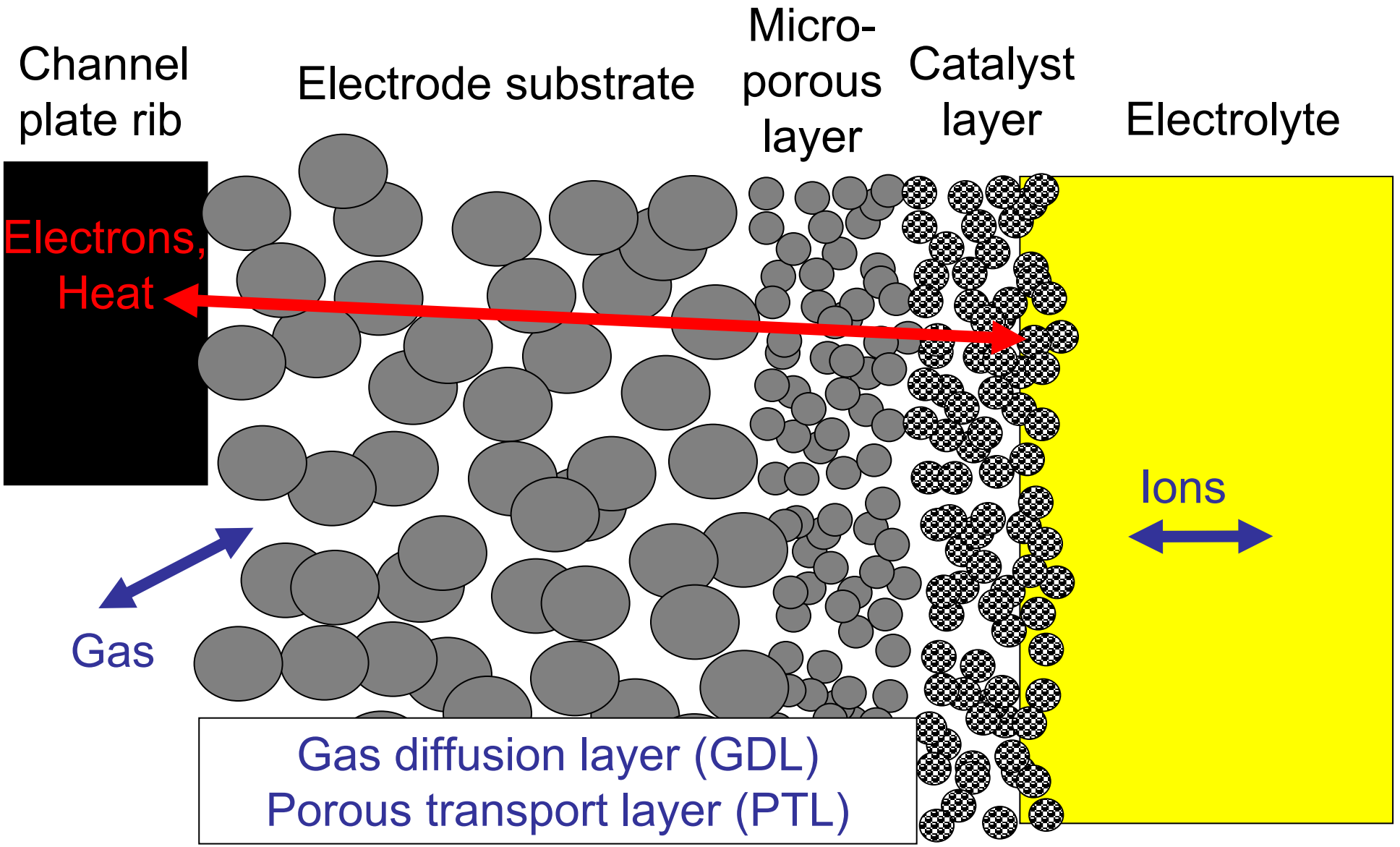
$$E_{\text{whatever}} = \frac{\text{Energy}}{\text{Charge}} = \frac{HHV \text{ or } LHV \text{ or } \Delta G \text{ or } \Delta H}{nF}$$

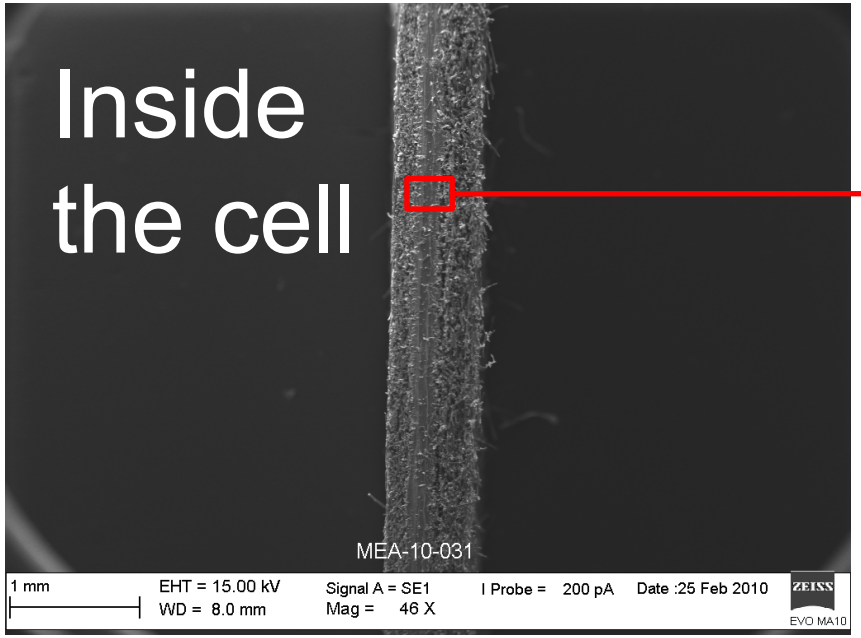


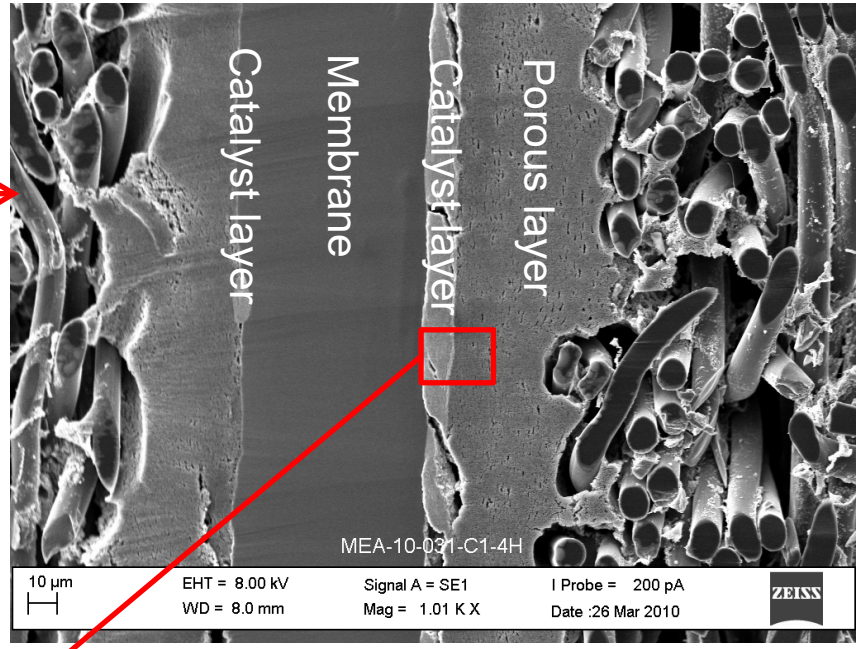
$$\text{Energy} = \text{Voltage} \cdot \text{Charge}$$

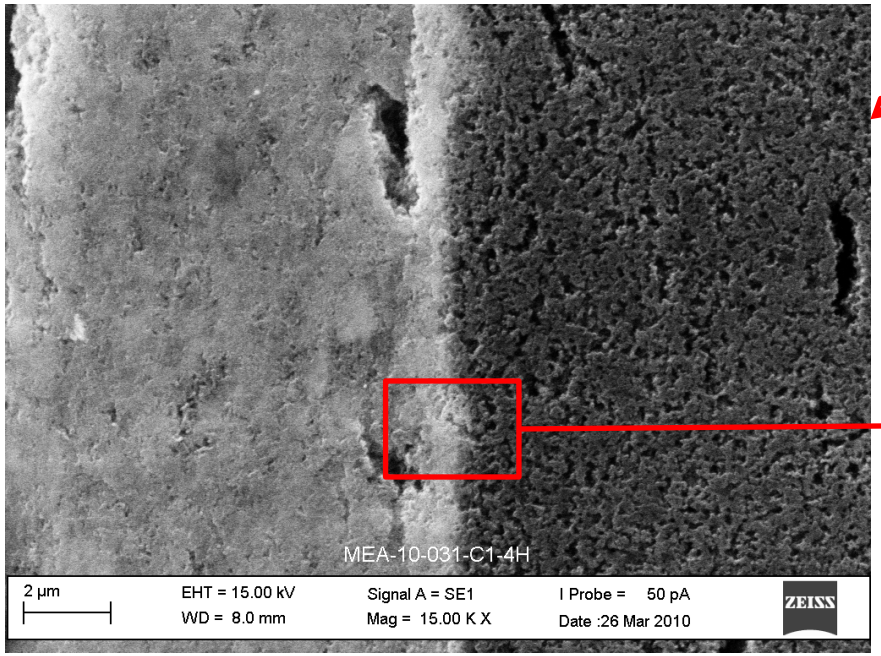
The function of fuel cell

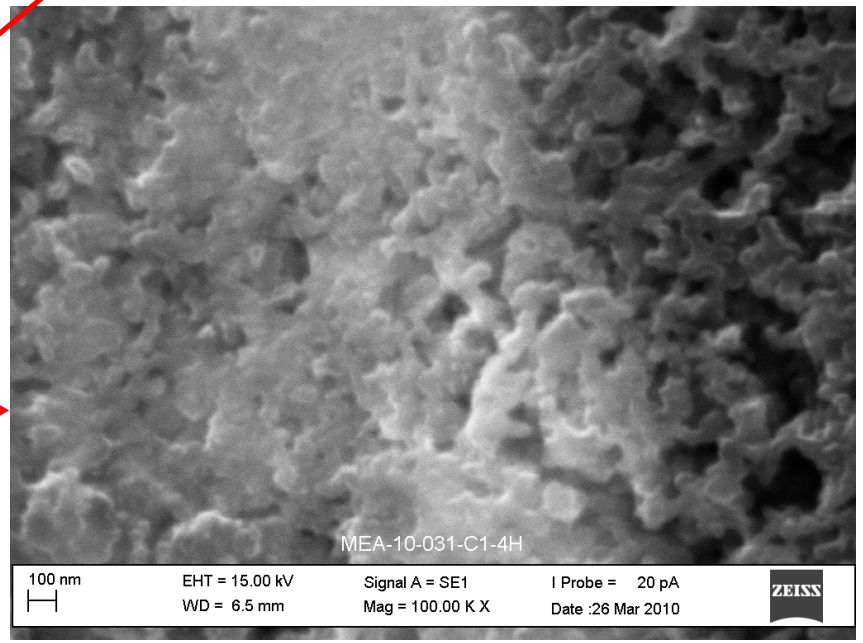
Electrode structure and transport processes



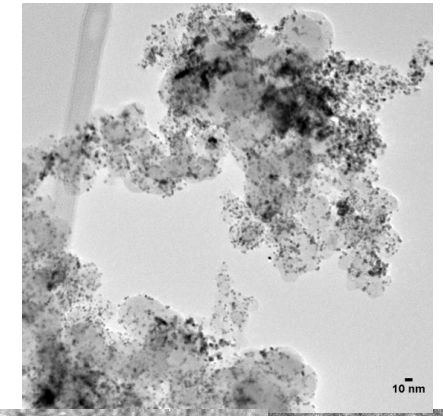

Inside
the cell
MEA-10-031
1 mm
EHT = 15.00 kV
WD = 8.0 mm
Signal A = SE1
Mag = 46 X
I Probe = 200 pA
Date :25 Feb 2010
ZEISS
EVO MA10


Catalyst layer
Membrane
Catalyst layer
Porous layer
MEA-10-031-C1-4H
10 µm
EHT = 8.00 kV
WD = 8.0 mm
Signal A = SE1
Mag = 1.01 K X
I Probe = 200 pA
Date :26 Mar 2010
ZEISS

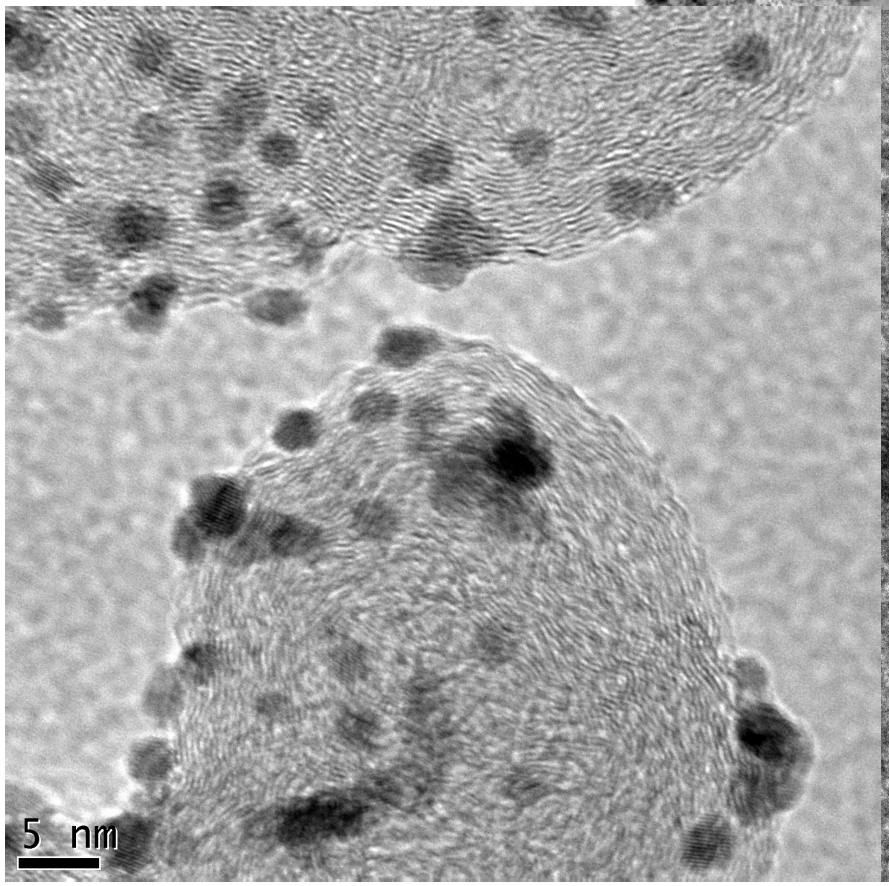

MEA-10-031-C1-4H
2 µm
EHT = 15.00 kV
WD = 8.0 mm
Signal A = SE1
Mag = 15.00 K X
I Probe = 50 pA
Date :26 Mar 2010
ZEISS

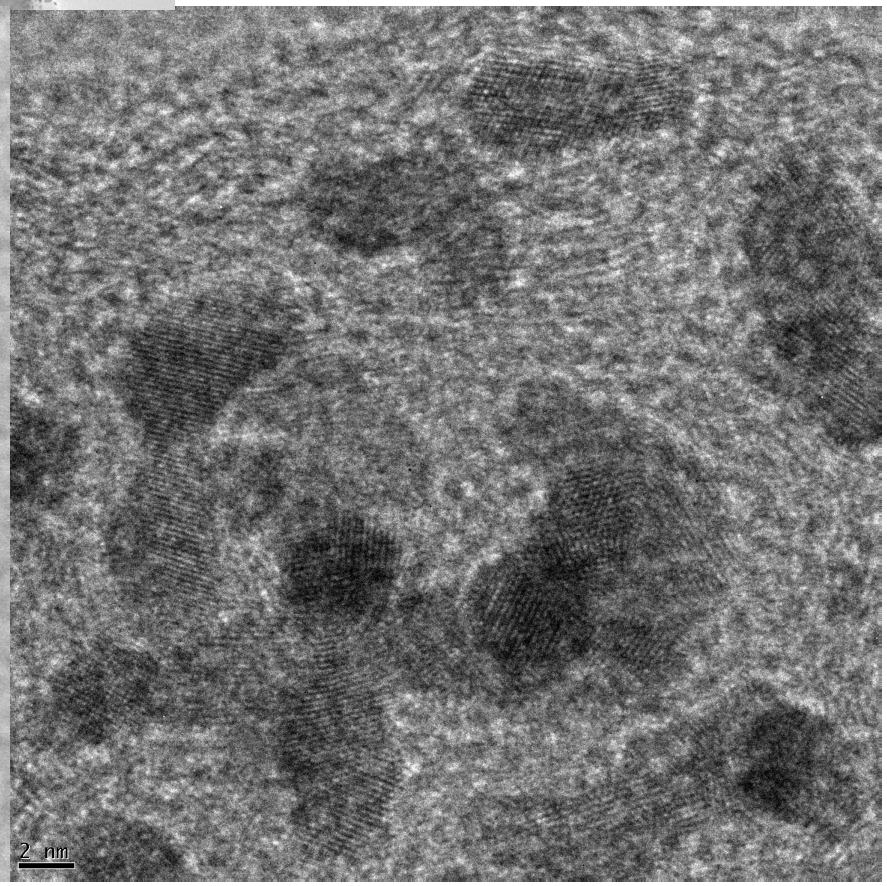

MEA-10-031-C1-4H
100 nm
EHT = 15.00 kV
WD = 6.5 mm
Signal A = SE1
Mag = 100.00 K X
I Probe = 20 pA
Date :26 Mar 2010
ZEISS

Catalyst

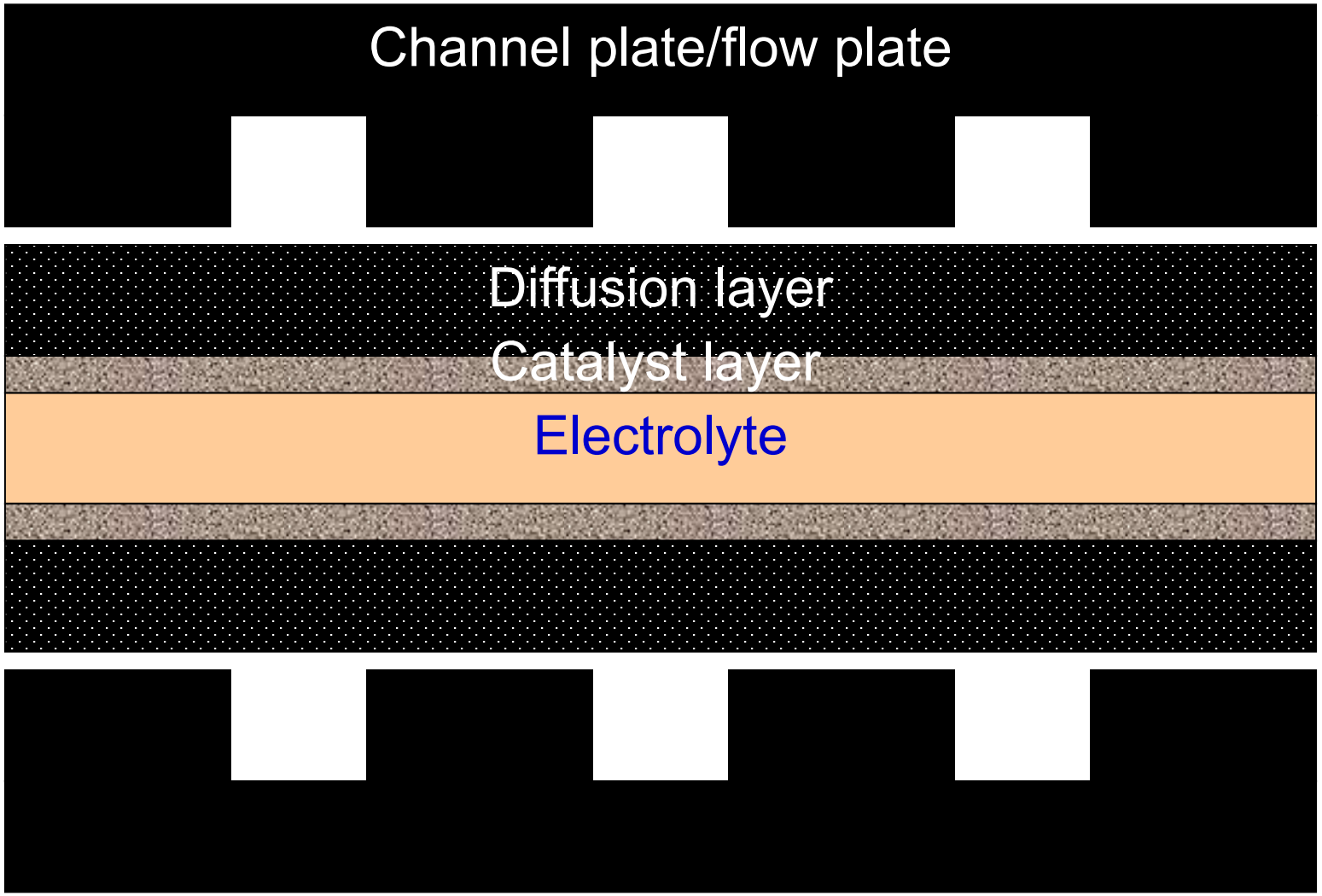


Platinum nanoparticles on carbon

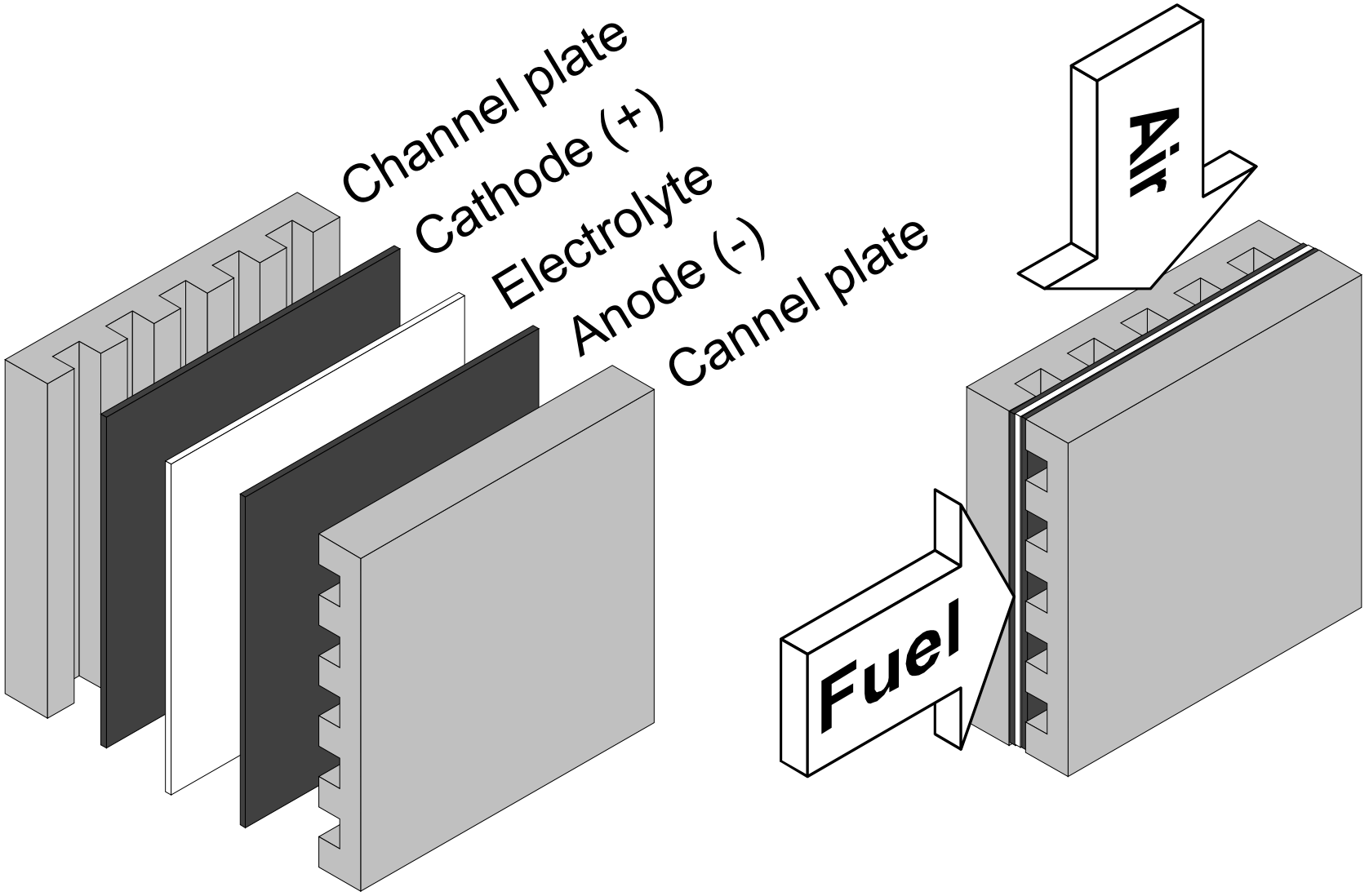




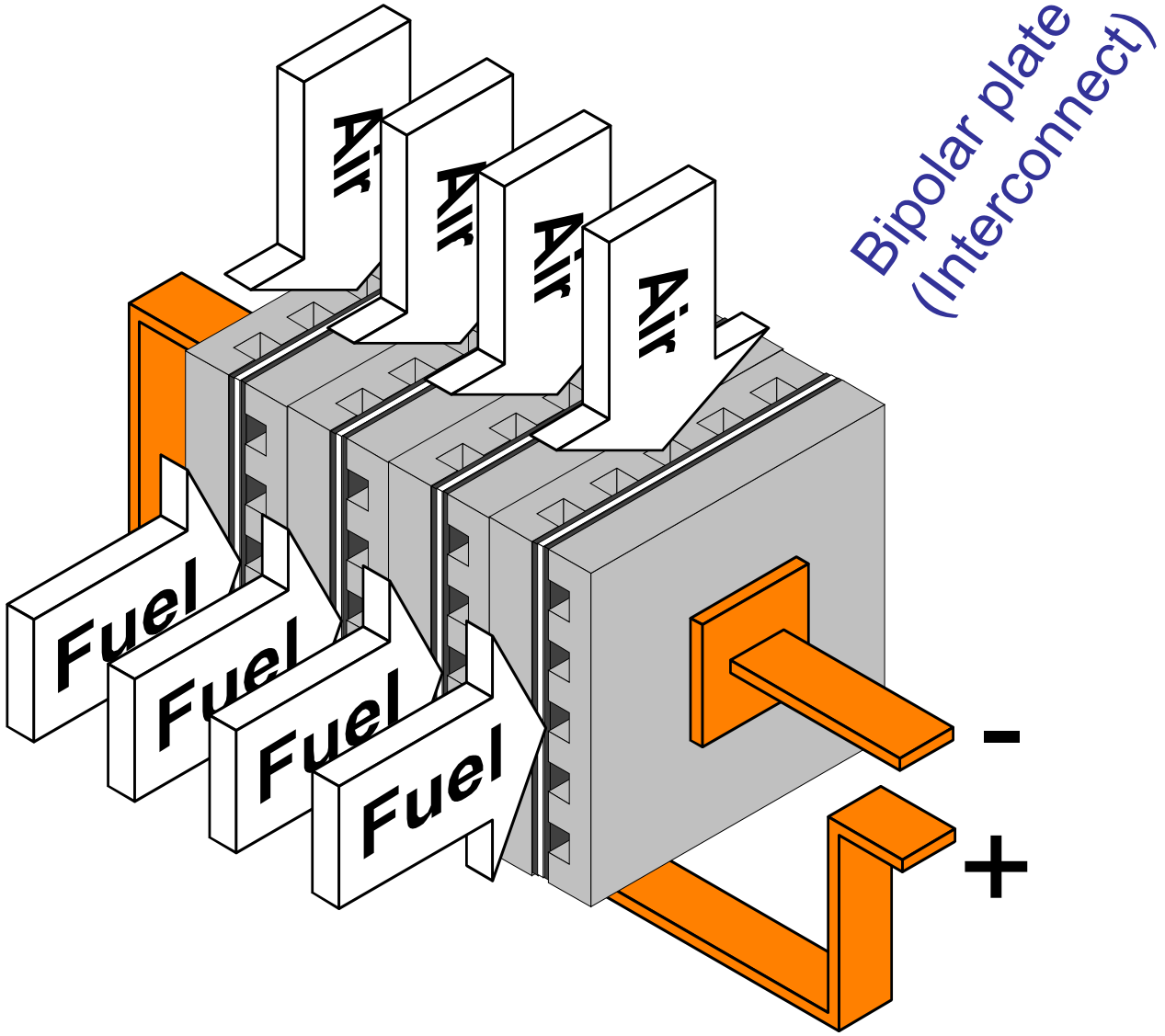
The single cell

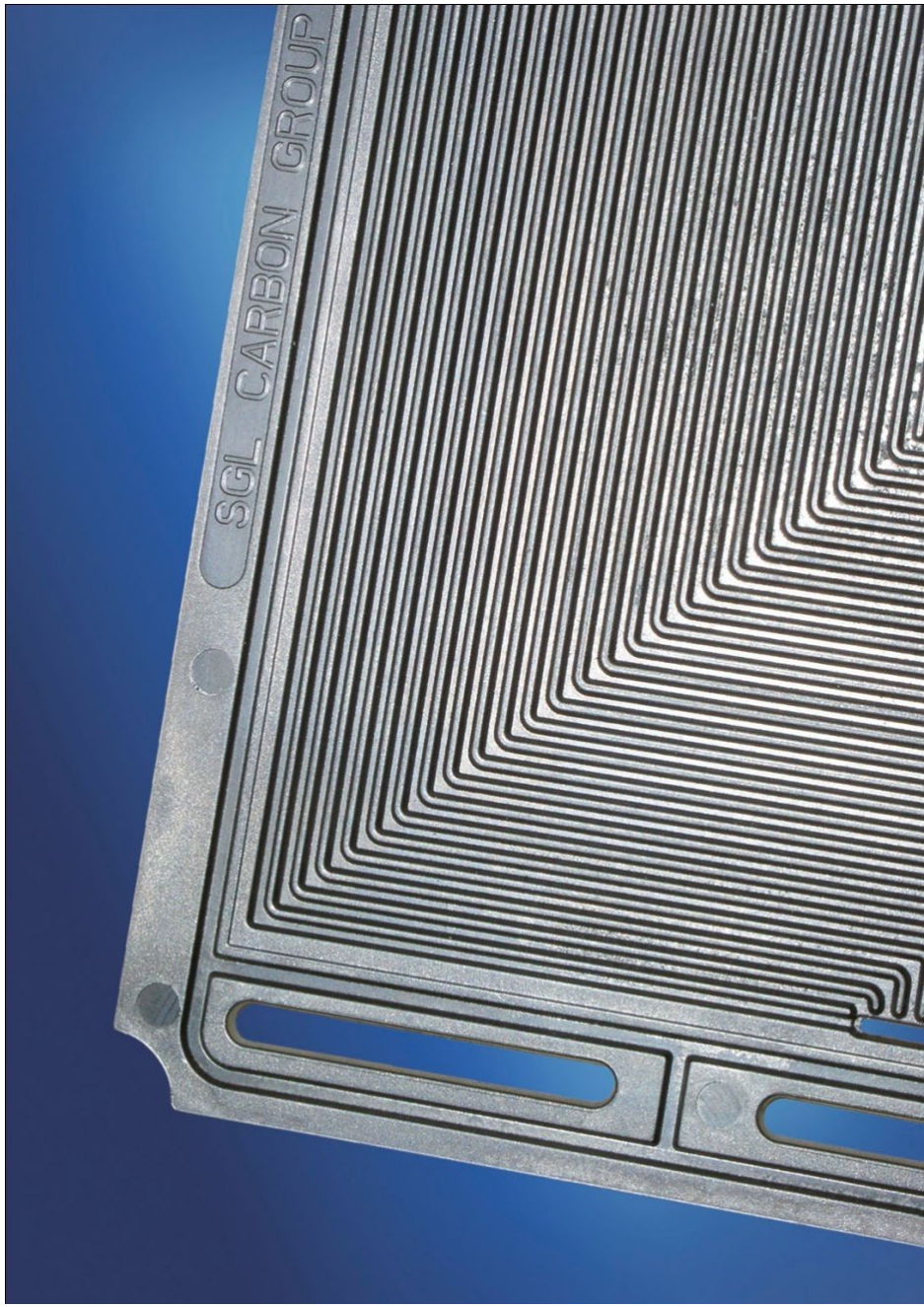


The single cell (fuel cell)

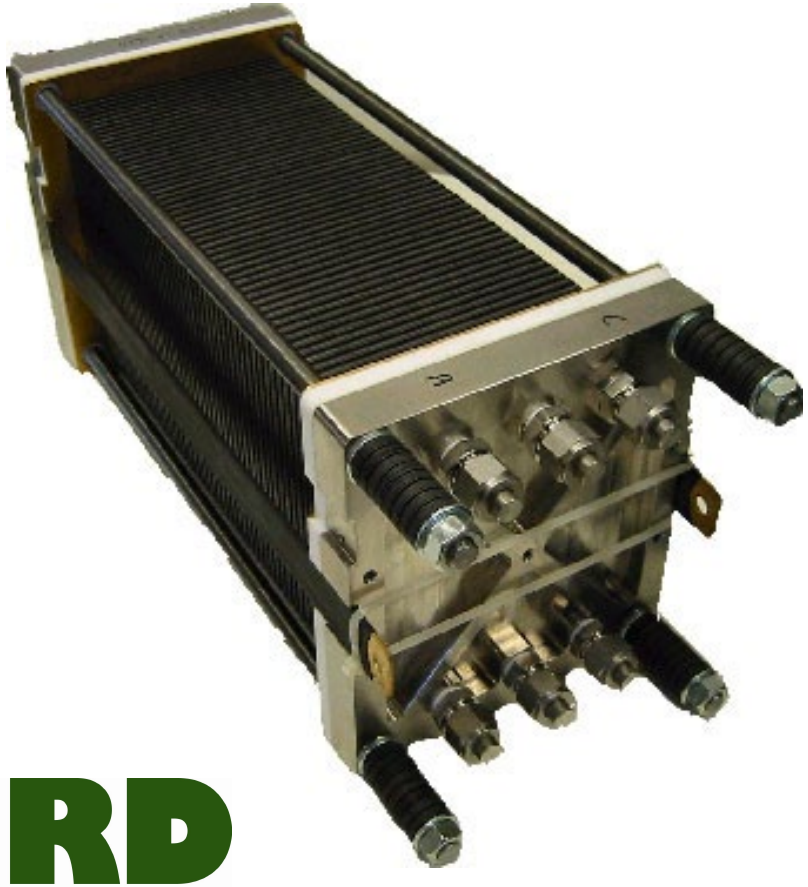


Stacking


Air
Air
Air
Air
Fuel
Fuel
Fuel
Fuel
Bipolar plate
(Interconnect)
-
+

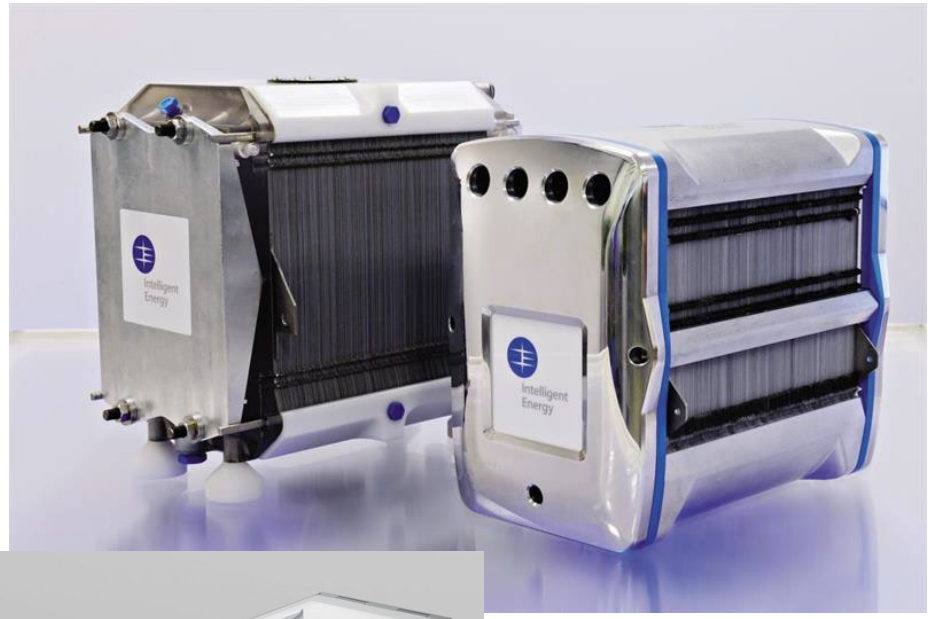

SGL CARBON GROUP

Fuel cell stacks

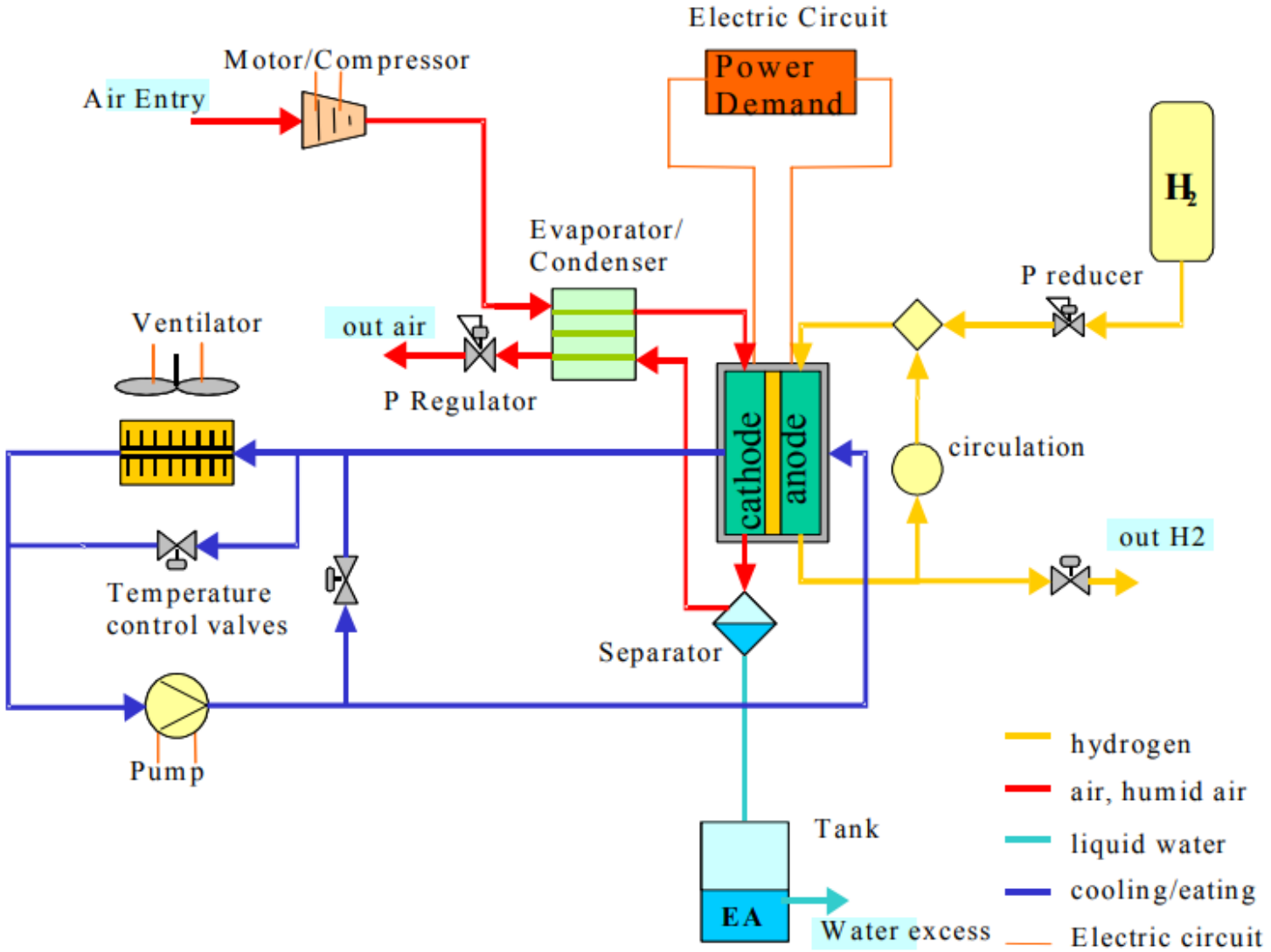


IRD
Fuel Cell Technology

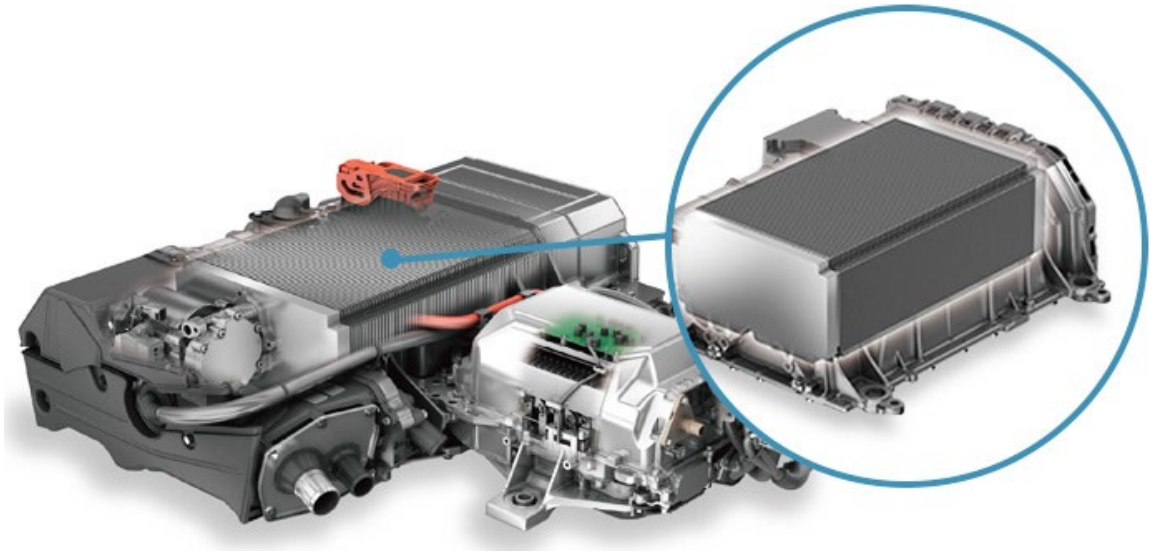
Modern automotive stacks

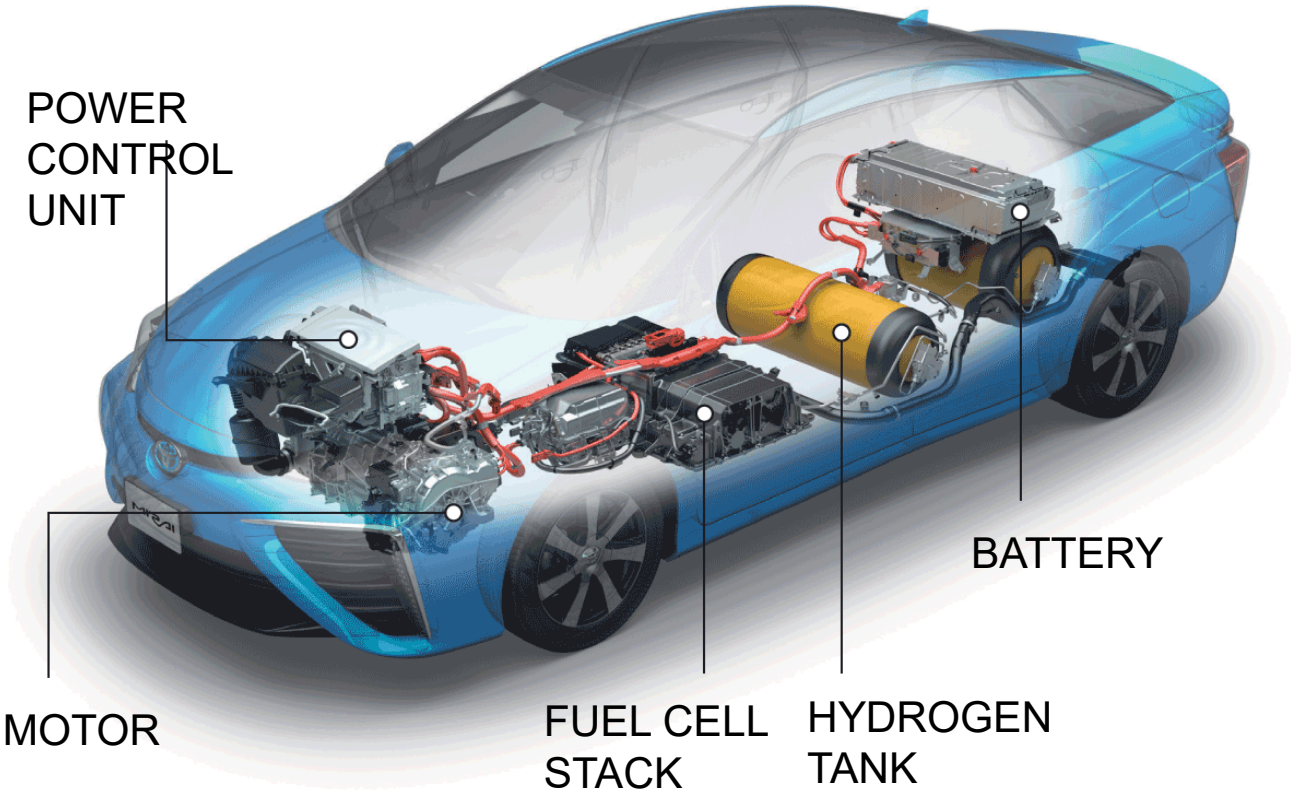


The complete system – balance of plant



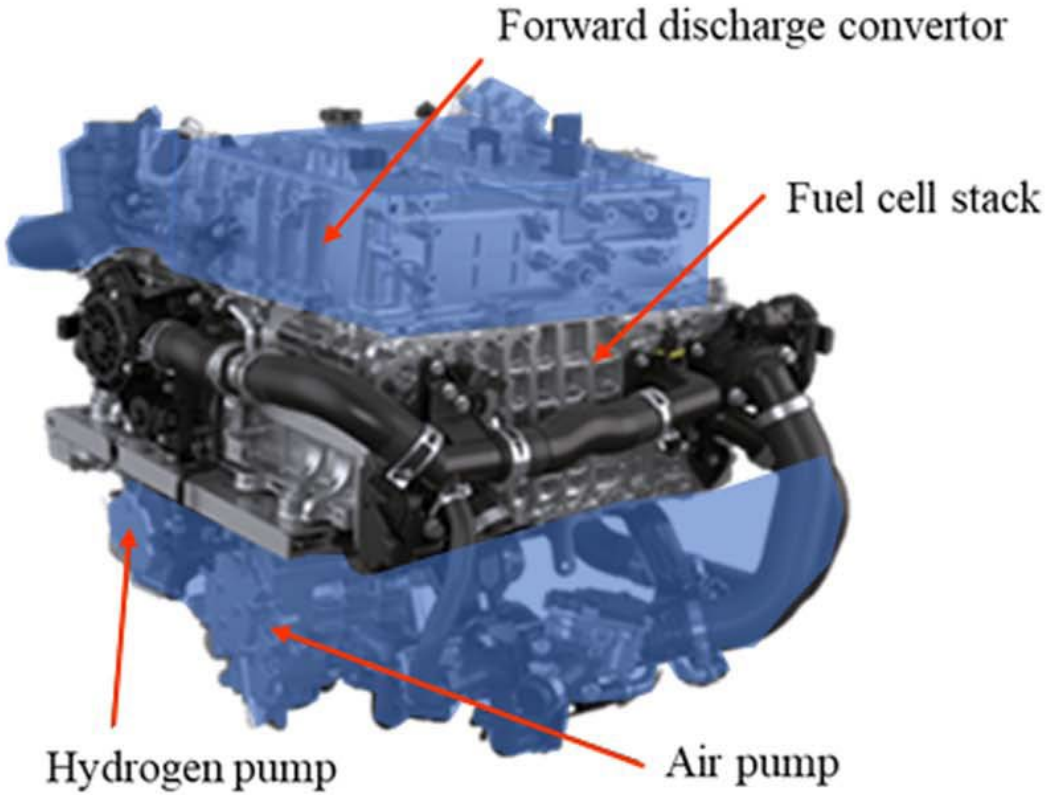
Toyota Mirai from 2015





Maximum output	114 kW (155 hp)
Volume power density	3.1 kW/L (global top level)
Number of cells in one stack	370 cells (single-line stacking)

Toyota Mirai 2020 model

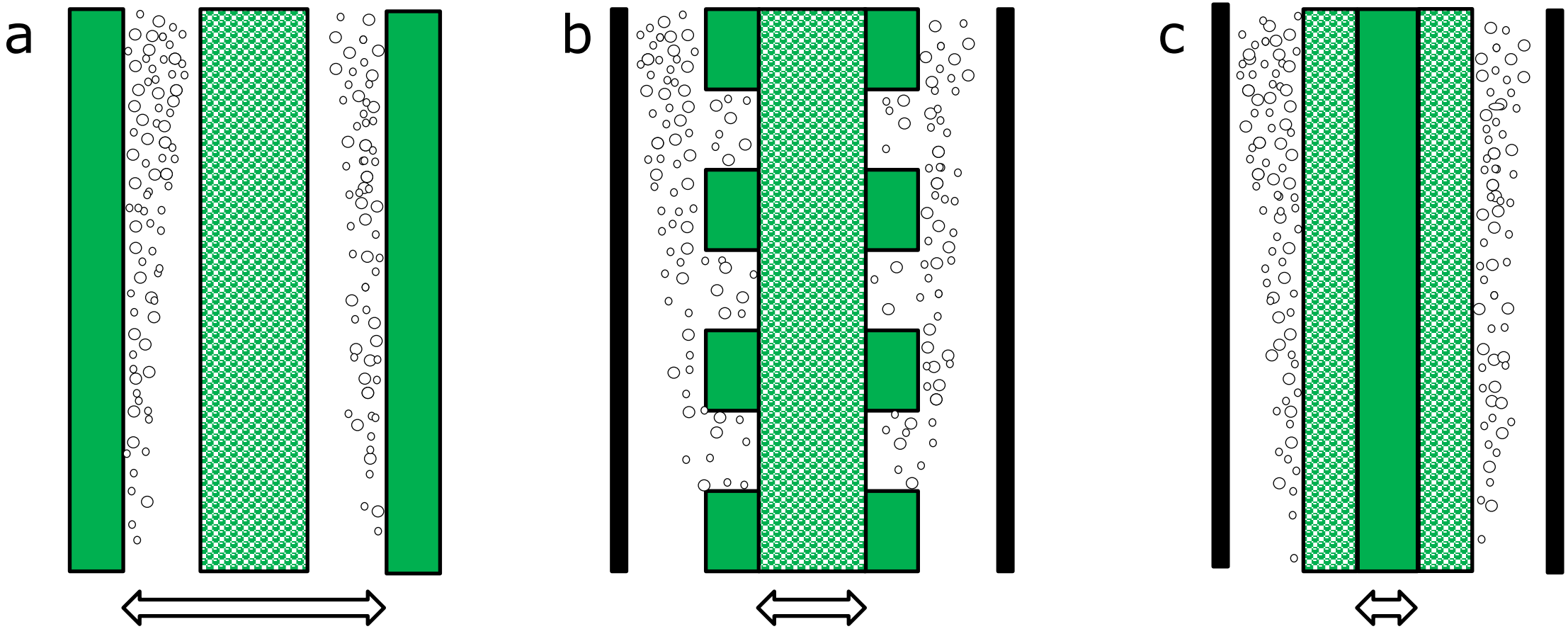


Volume power density: 5.4 kW/L (excluding end plates)

Max output: 128 kW (172 hp)

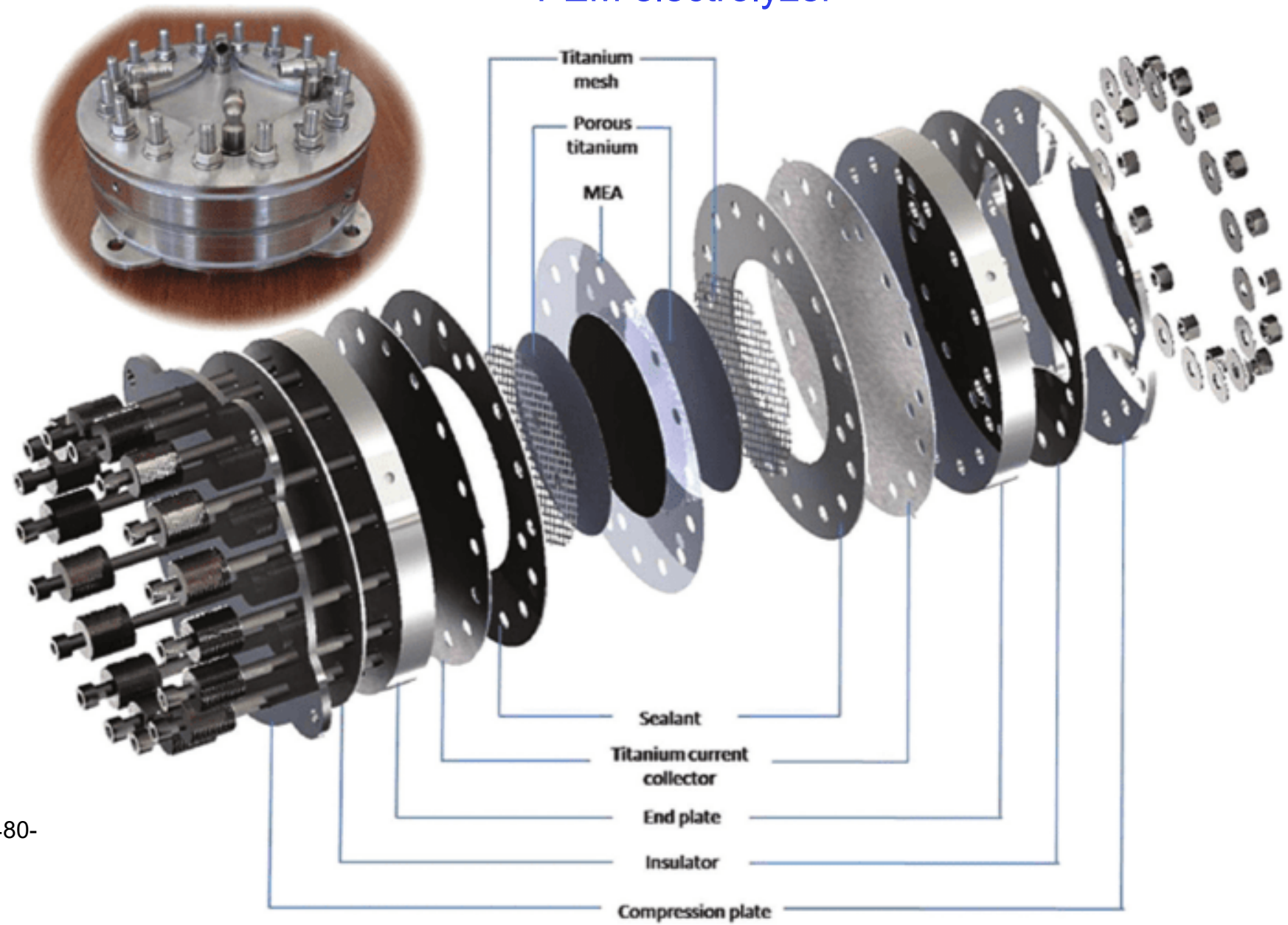
The function of electrolyzers

Electrolyzer concepts



Singel cell

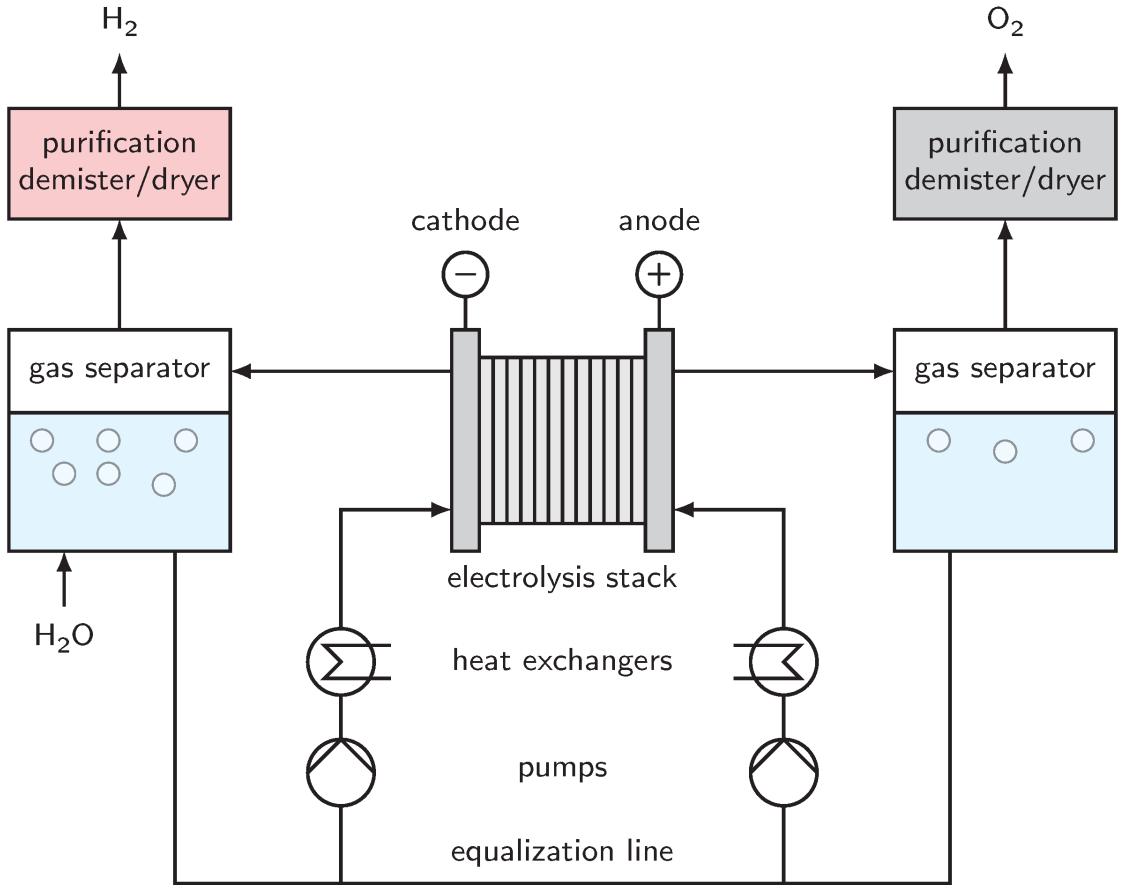
PEM electrolyzer

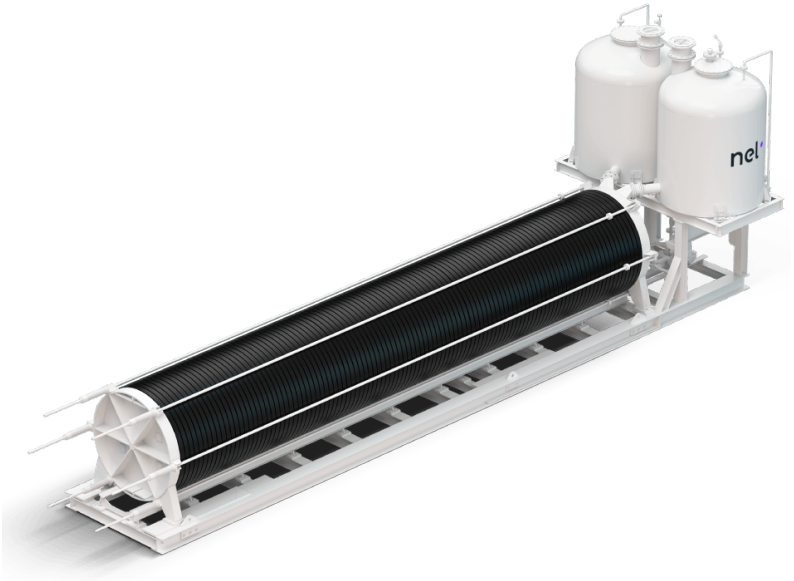


Selamet, O.F. et al., *Int. J. Hydr. Energ.,* 36, 11480-11487. 2011
https://doi.org/10.1016/j.ijhydene.2011.01.129

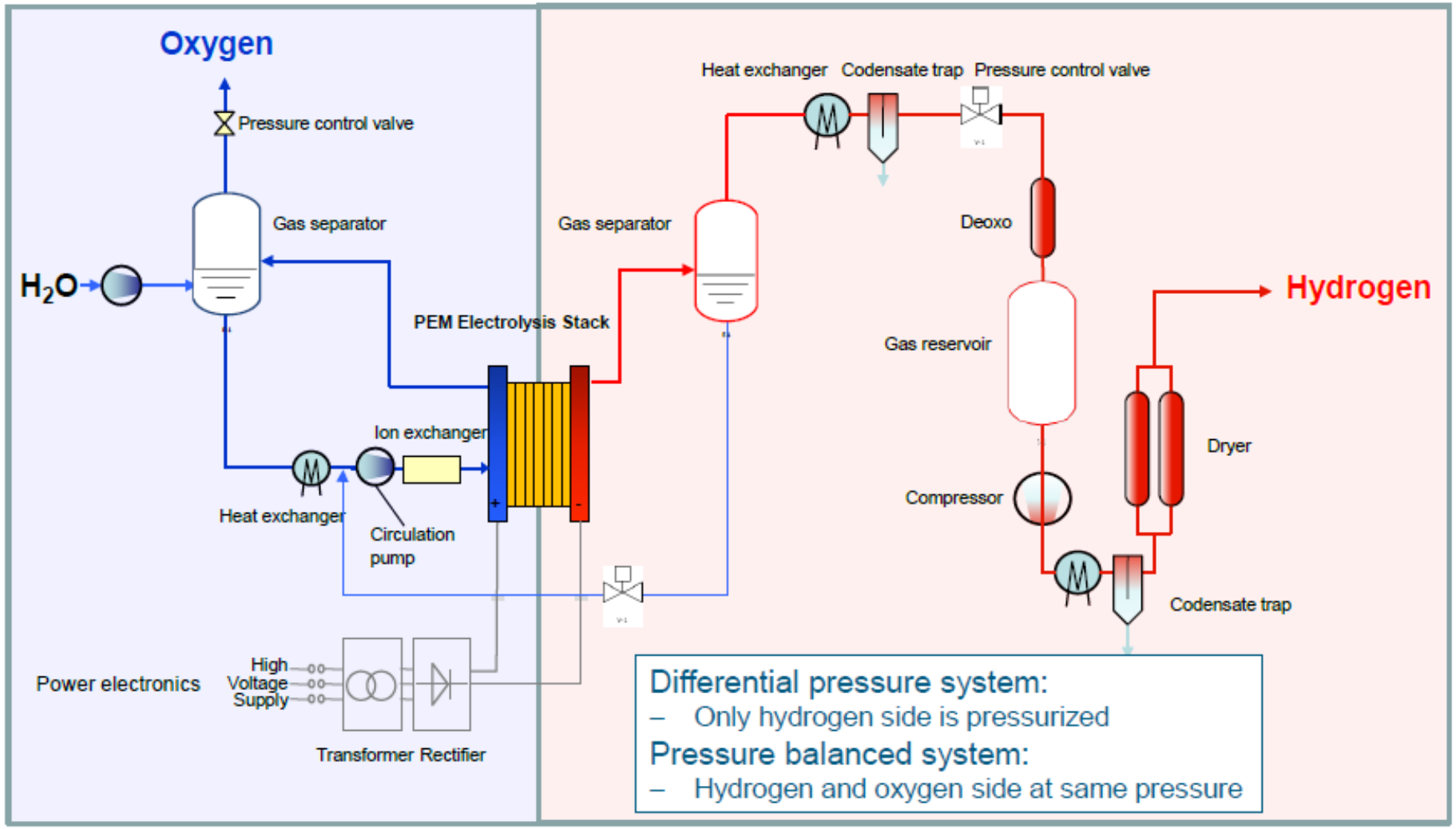
Stacking electrolyzer cells

The electrolyzer system

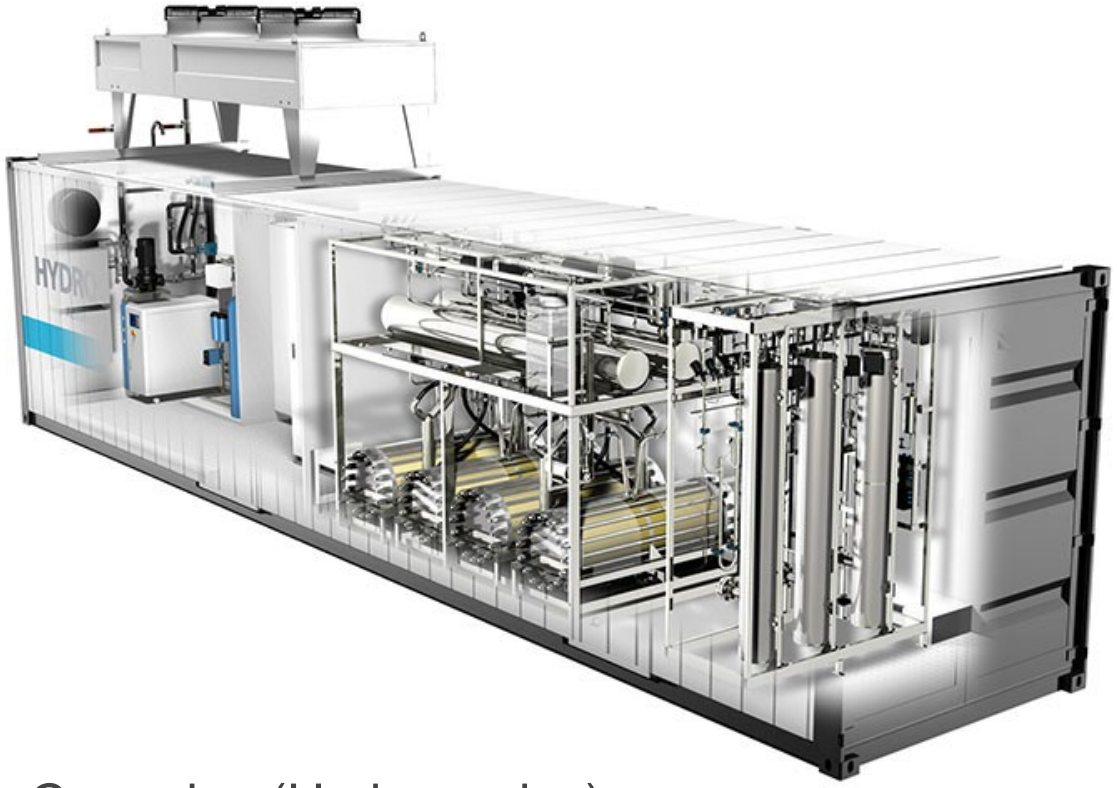




The electrolyzer system

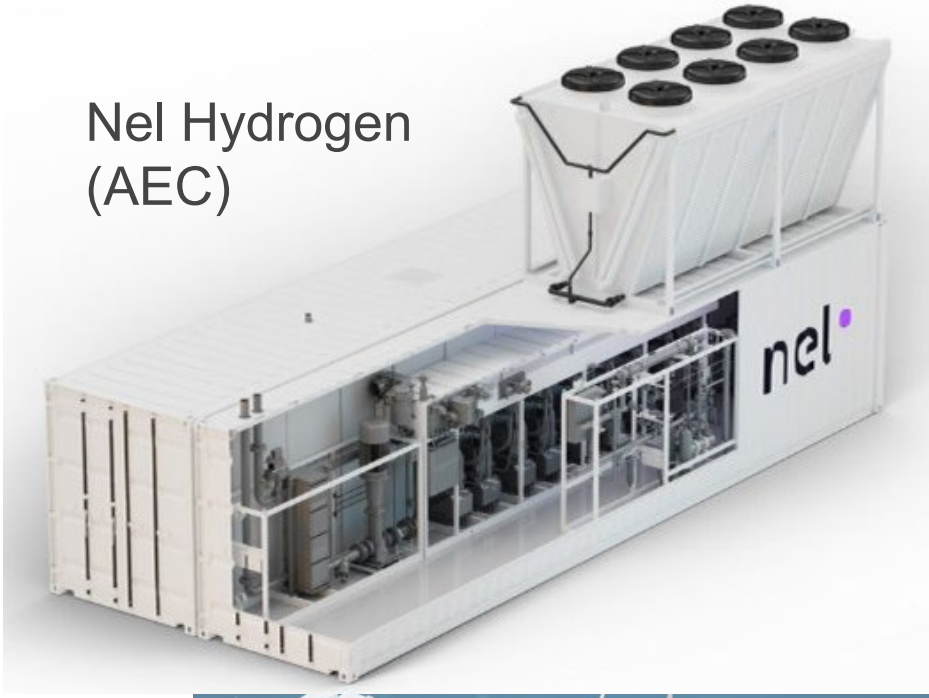


Container solutions

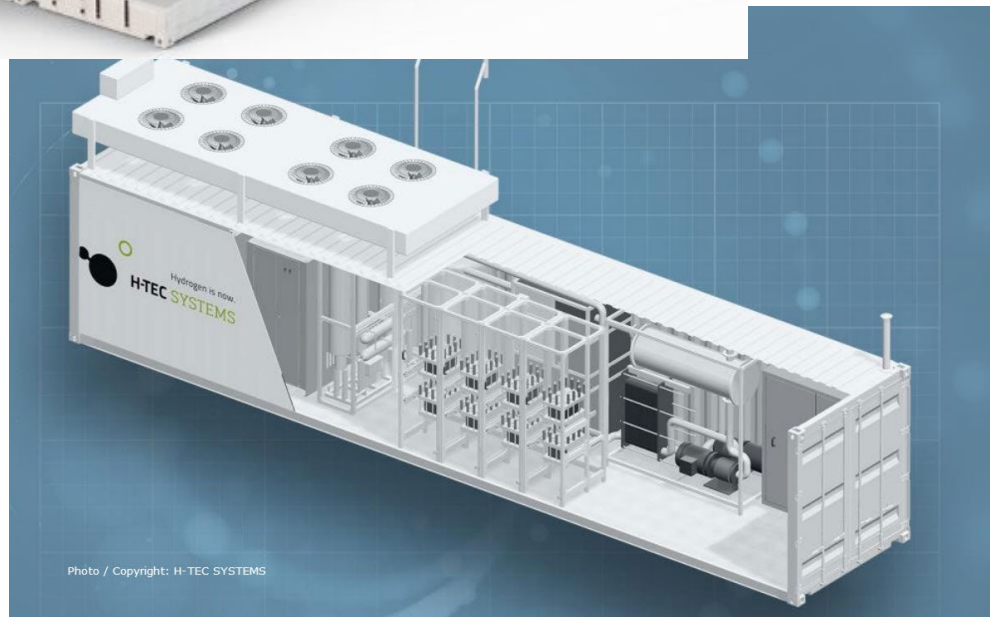


Cummins (Hydrogenics)
(PEMEC)

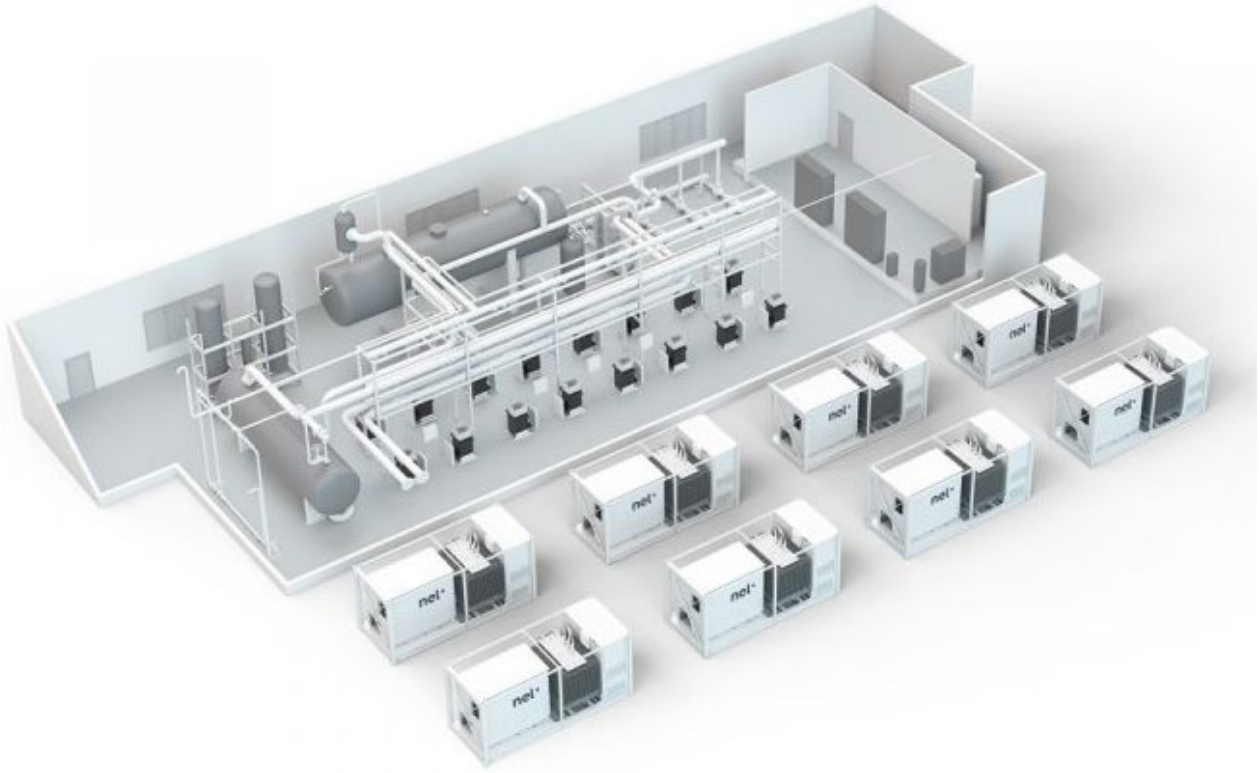
Nel Hydrogen
(AEC)



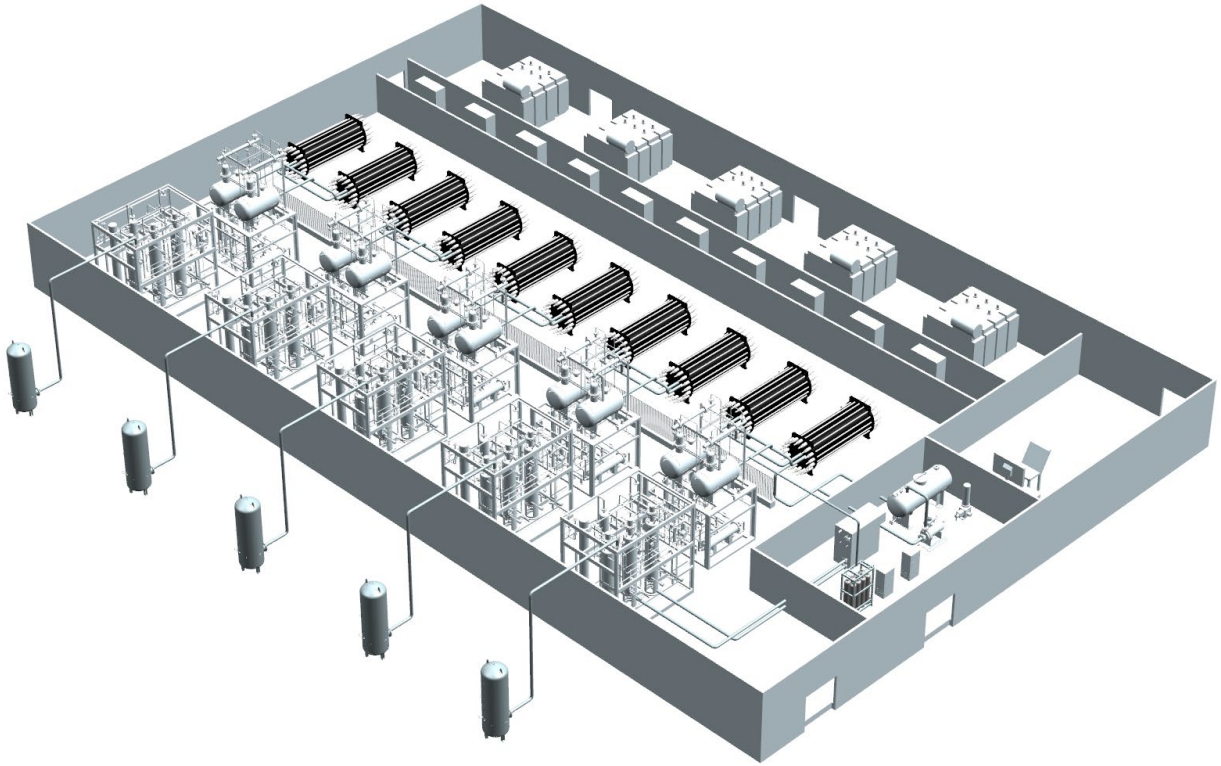
H-Tech
(PEMEC)



Hydrogen plants



Nel Hydrogen



SinoHy: 100 MW

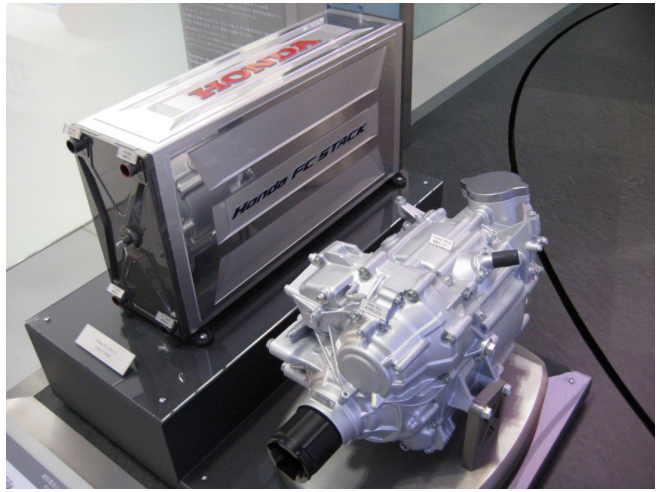
Types of fuel cells

Types of fuel cells (FC)

Type	Abbrev.	Temperature	Electrolyte (charge carrier)
Low temperature systems			
Alkaline fuel cell	AFC	60-80°C	aq. KOH (OH^-)
Proton exchange membrane fuel cell	**PEMFC** (PEFC)	**60- 80°C**	**Polymer (H^+)**
Phosphoric acid fuel cell	PAFC	200°C	H_3PO_4 (H^+)
High temperature systems			
Molten carbonate fuel cell	MCFC	650°C	Molten salt (CO_3^{-2})
Solid oxide fuel cell	**SOFC**	**700-900°C**	**Ceramic (O^{2-})**

Polymer fuel cell (PEMFC*)

- Most common fuel cell today
- Electrolyte: Proton conducting polymer membrane
- Electrodes: Carbon-based with Pt catalysts
- Uses: mobile/stationary/portable

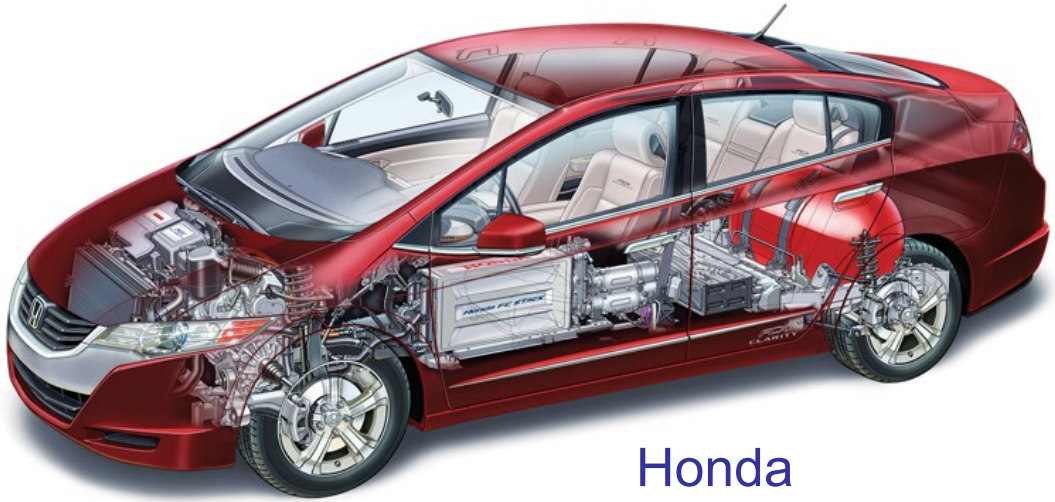


Proton conducting electrolyte

Anode: $2H_2 \rightarrow 4H^+ + 4e^-$

Cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

Total: $2H_2 + O_2 \rightarrow 2H_2O$

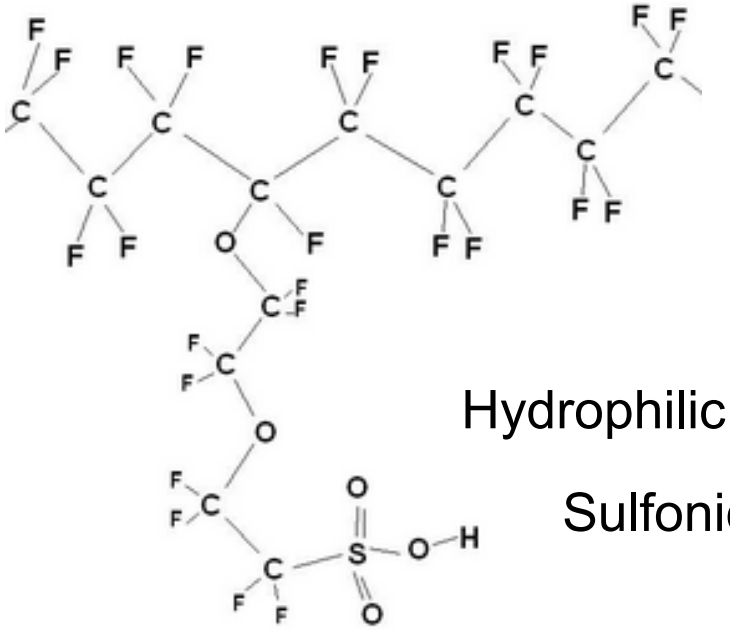


Honda

Proton Exchange Membrane Fuel Cell

Polymer Electrolyte Membrane Fuel Cell

Proton conduction in polymer

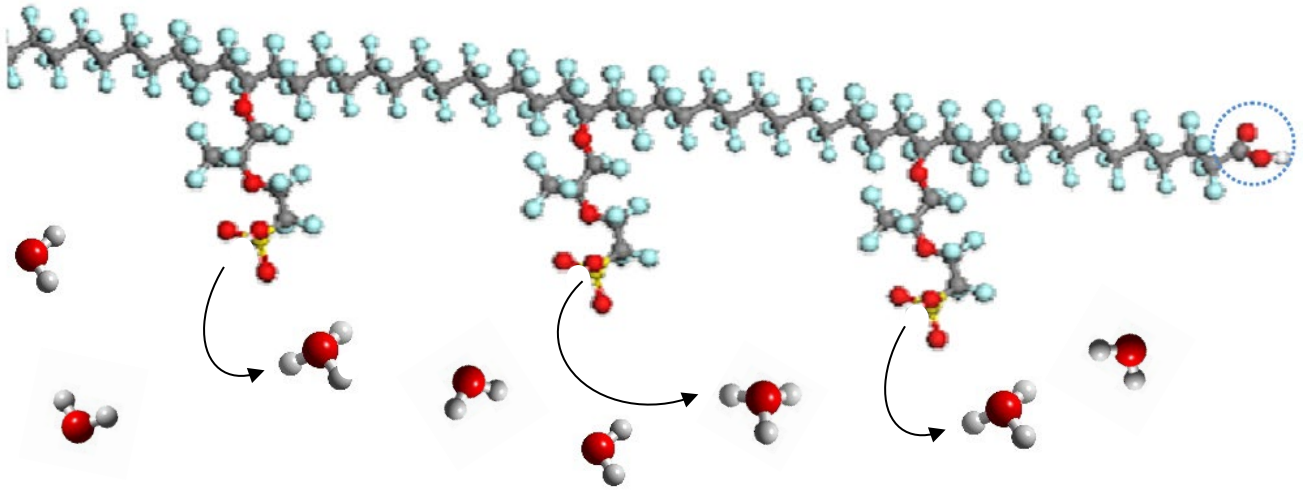


Hydrophobic backbone

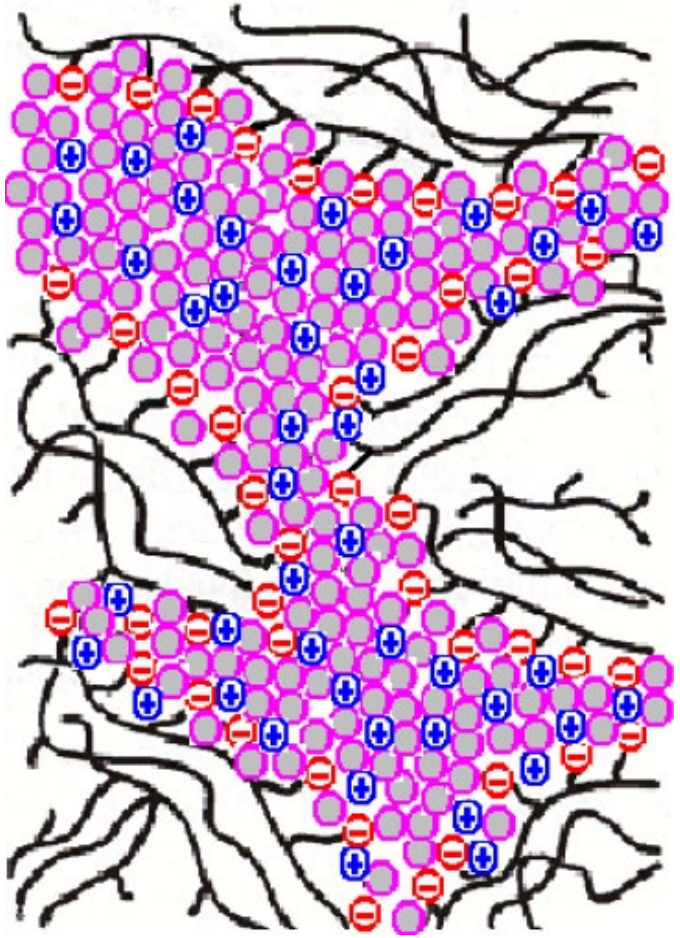
Hydrophilic side chains

Sulfonic acid

Humidification of membrane is key



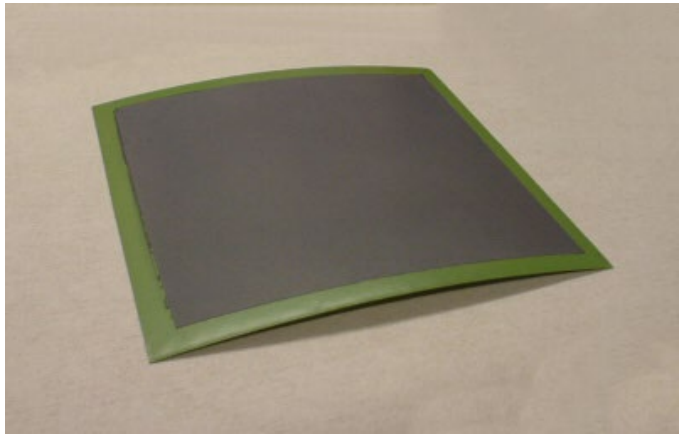
Hydrophobic backbone



Perfluorosulfonic acid, PFSA

Solid oxide fuel cell (SOFC)

- Entering the market
- Electrolyte: Oxide ion conducting ceramic (700-800 °C)
- Electrodes: Ceramic
- Can be powered by natural gas and ammonia
- So far mostly stationary applications
- Can operate as electrolyzer as well
- Most important high-temperature fuel cell

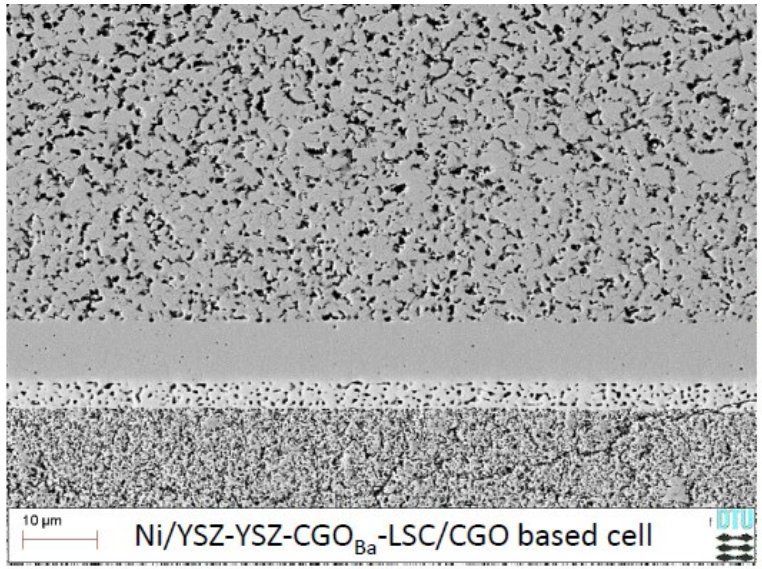


Oxide ion conducting electrolyte

Anode: $2H_2 + 2O^{2-} \rightarrow 2H_2O + 4e^-$

Cathode: $O_2 + 4e^- \rightarrow 2O^{2-}$

Total: $2H_2 + O_2 \rightarrow 2H_2O$



Ni/YSZ support layer

Ni/YSZ electrode

YSZ electrolyte

CGO barrier layer

LSC/CGO electrode

Ref. Brodersen, Hauch, Chen and Hjelm, Patent no.: WO201702935 (2017)

Alkaline fuel cell (AFC)

- First fuel cell for practical use
- Electrolyte: Aqueous KOH (ca. 30 w%)
- Electrodes: Ni based with PGM* or Ni based catalysts
- **Disadvantage**: Carbonization by CO_2 from air: $CO_2 + 2K^+ + 2OH^- \rightarrow K_2CO_3$ (s)
- Used for speciality purposes.
- Not common today

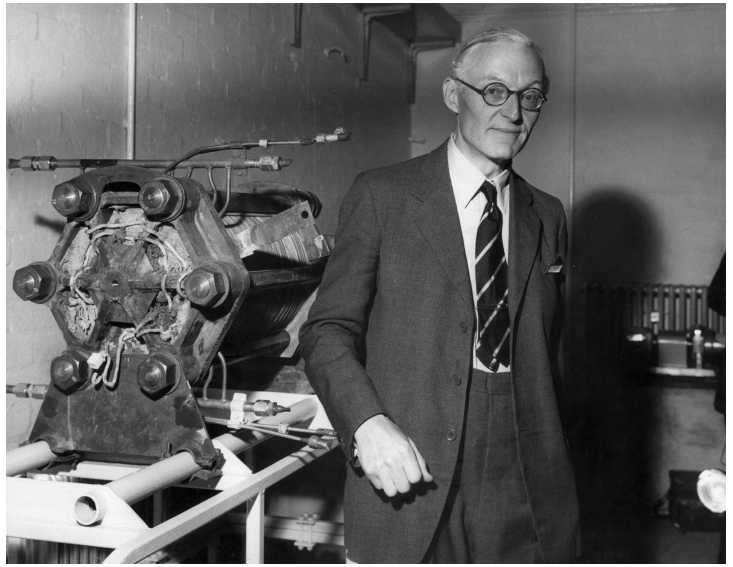
Hydroxide ion conducting electrolyte

Anode: $2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$

Cathode: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

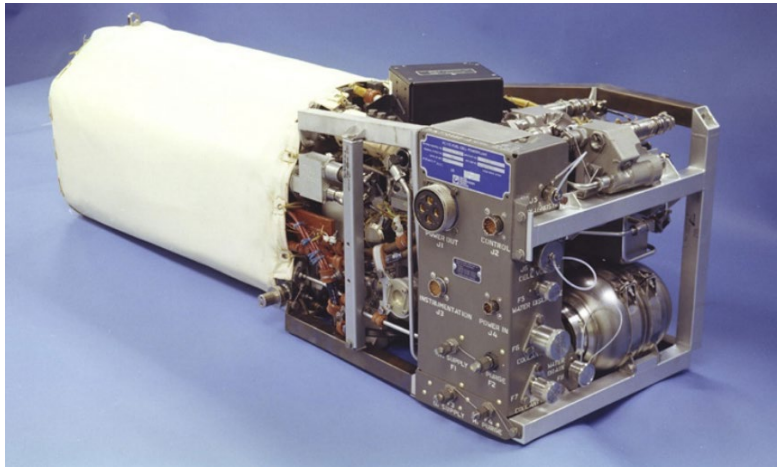
Total: $2H_2 + O_2 \rightarrow 2H_2O$

* PGM = Platinum group metals (mostly Pt and Ir)



Thomas Francis Bacon and his AFC

Apollo AFC 1964



AFC from UTC for the space shuttle

Phosphoric acid fuel cell (PAFC)

- First large-scale commercial fuel cell
- Electrolyte: Phosphoric acid
- Electrodes: Carbon-based with Pt catalysts
- Heat and power from natural gas

"PC25" 200 kW from ONSI/UTC
Later 400 kW

Proton conducting electrolyte

Anode: $2H_2 \rightarrow 4H^+ + 4e^-$

Cathode: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

Total: $2H_2 + O_2 \rightarrow 2H_2O$

Molten carbonate fuel cell (MCFC)

- Not common fuel cell today
- Electrolyte: Molten carbonate salt (650 °C)
- Electrodes: ceramic/metallic
- Needs CO_2 in the conduction mechanism
- Used in power plants
- Can be powered directly with natural gas

Carbonate ion conducting electrolyte

Anode: $2CO_3^{2-} + 2H_2 \rightarrow 2H_2O + 2CO_2 + 4e^-$

Cathode: $O_2 + 2CO_2 + 4e^- \rightarrow 2CO_3^{2-}$

Total: $2H_2 + O_2 \rightarrow 2H_2O$

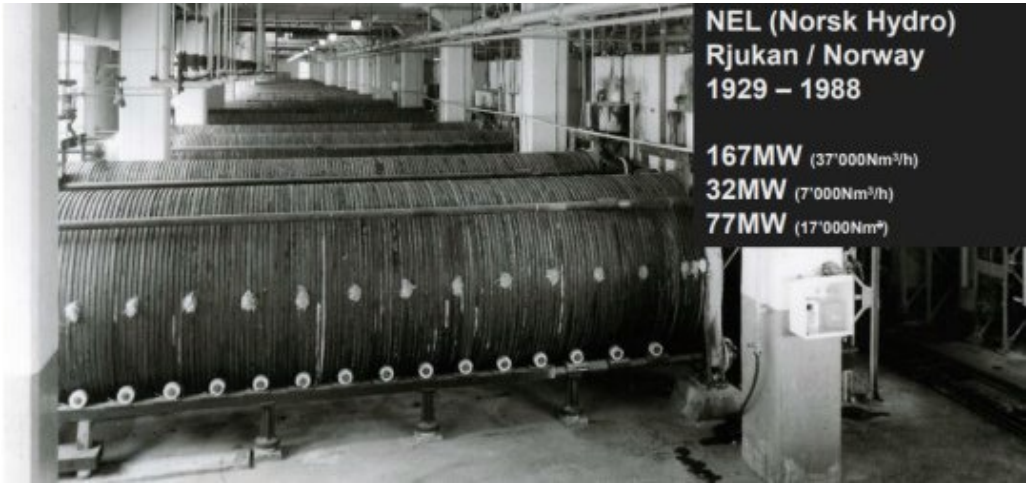
Types of electrolyzers

Types of electrolyzer cells (EC)

Type	Abrev.	Temperature	Electrolyte (charge carrier)
Low temperature systems			
Liquid alkaline electrolyzer cell	**AEC** (AWE, AEL)	60 - 80°C	aq. KOH (OH^-)
Anion-exchange membrane electrolyzer cell	**PEMEC** (PEMWE, PEMEL)	60 - 80°C	Polymer (H^+)
Anion-exchange membrane electrolyzer cell	**AEC** (AWE, AEL)	~ 60 °C	Polymer (OH^-)
High temperature systems			
Solid oxide electrolyzer cell	**SOEC** (SOWE, SOEL)	700-900°C	Ceramic (O^{2-})

Alkaline electrolyzer (AEC, AEL, AWE)

- First electrolyzer cell for practical use
- Electrolyte: Aqueous KOH (ca. 30 w%)
- Electrodes: Ni based with Ni based catalysts
- No need for PGM*
- Largest market share
- Largest stacks
- Low to moderate power density
- Cheapest technology



Hydroxide ion conducting electrolyte

Cathode : $4H_2O + 4e^- \rightarrow 2H_2 + 4OH^-$

Anode: $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

Total: $2H_2O \rightarrow 2H_2 + O_2$

* PGM = Platinum group metals (mostly Pt and Ir)

Polymer electrolyzer (PEMEC)

- Second most common electrolyzer
- Electrolyte: Proton conducting polymer membrane
- Electrodes: Titanium/Carbon-based with Pt and IrO_2 catalysts
- Compact, high power density
- Pressurizeable
- Expensive

Giner Inc.
1200 psig
(82.7 bar)
PEMEC

Proton conducting electrolyte

Cathode: $4H^+ + 4e^- \rightarrow 2H_2$

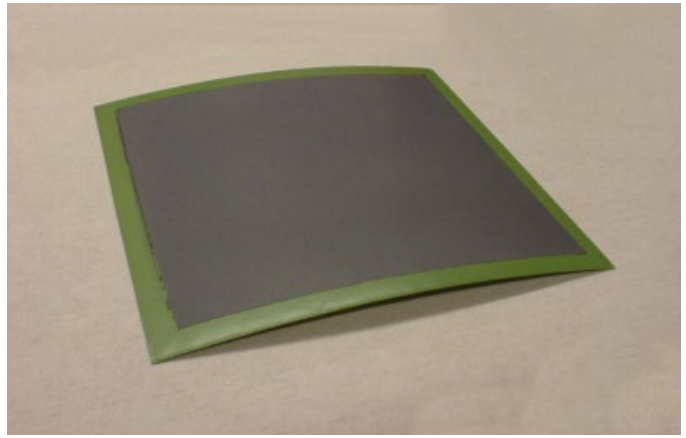
Anode: $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

Total: $2H_2O \rightarrow 2H_2 + O_2$

ITM
2 MW
PEMEC
stack

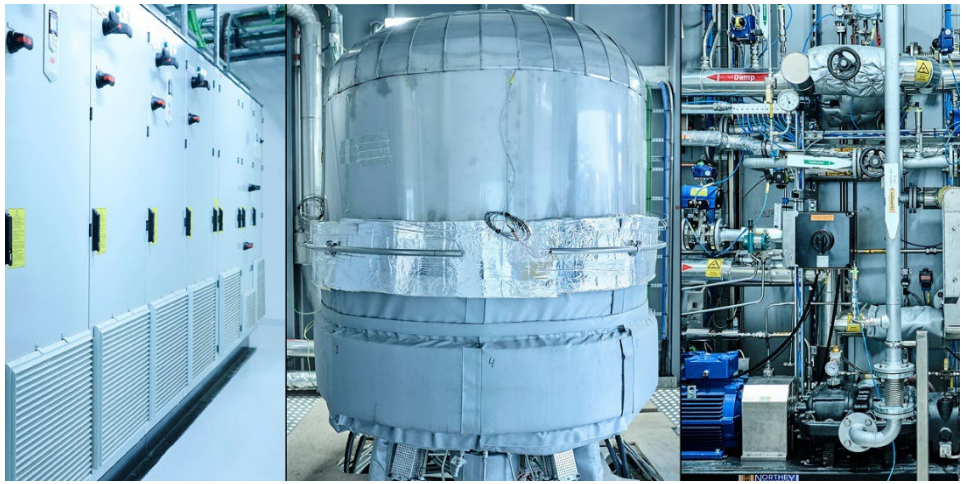
Solid oxide electrolyzer (SOEC)

- Entering the market
- Electrolyte: Oxide ion conducting ceramic (700-800 °C)
- Electrodes: Ceramic
- Can operate as fuel cell
- Most important high-temperature electrolyzer
- Can utilize heat as part of the energy
- Very high efficiency



DTU/Topsoe

Topsoe SOEC system



Oxide ion conducting electrolyte

Cathode: $2H_2O + 4e^- \rightarrow 2H_2 + 2O^{2-}$

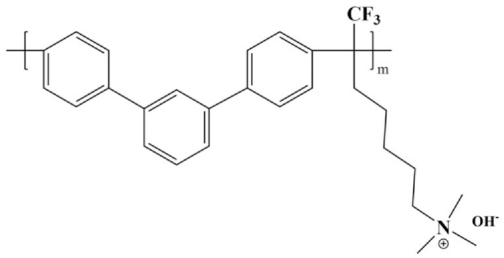
Anode: $2O^{2-} \rightarrow O_2 + 4e^-$

Total: $2H_2O \rightarrow 2H_2 + O_2$

Anion-exchange membrane electrolyzer (AEMEC, AEMEL, AEMWE)

- PEM-like alkaline technology
- Electrolyte: OH^- conducting membrane
- Electrodes: Ni based with PGM* or Ni based catalysts
- Active field of development
- Not really commercial yet

ORION



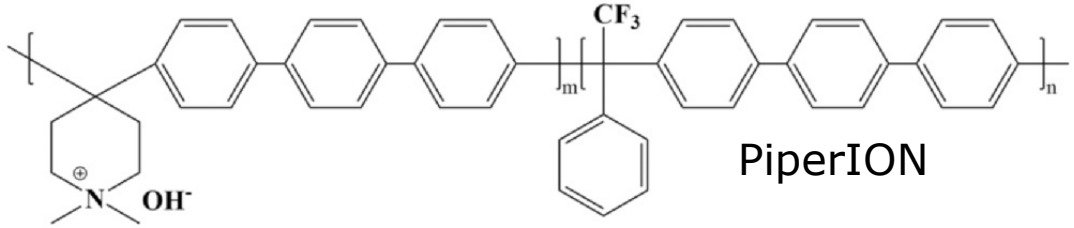
Hydroxide ion conducting electrolyte

Cathode : $4H_2O + 4e^- \rightarrow 2H_2 + 4OH^-$

Anode: $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

Total: $2H_2O \rightarrow 2H_2 + O_2$

Versogen
The Hydrogen Generation.



PiperION

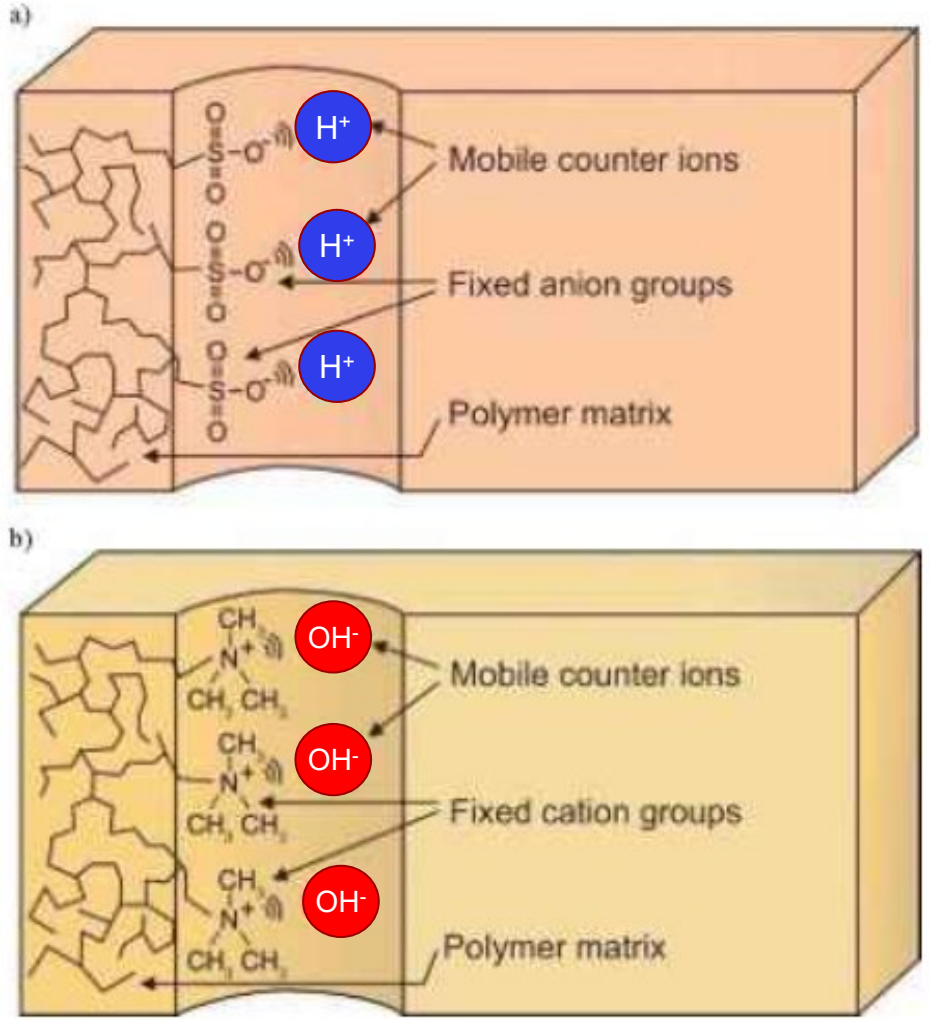
* PGM = Platinum group metals (mostly Pt and Ir)

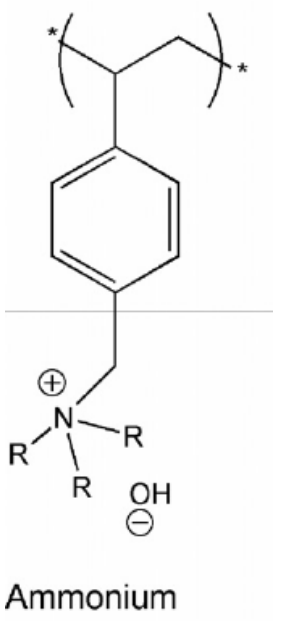
Anion-exchange membranes (AEM)

Humidification of membrane is key

PEM

AEM



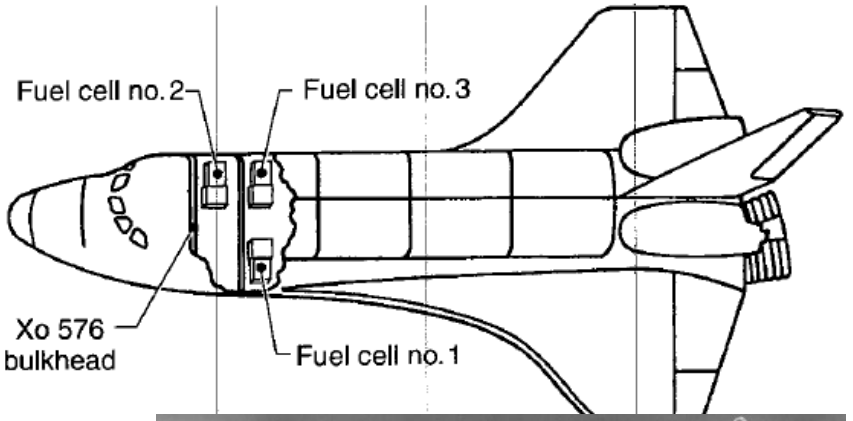


Under development

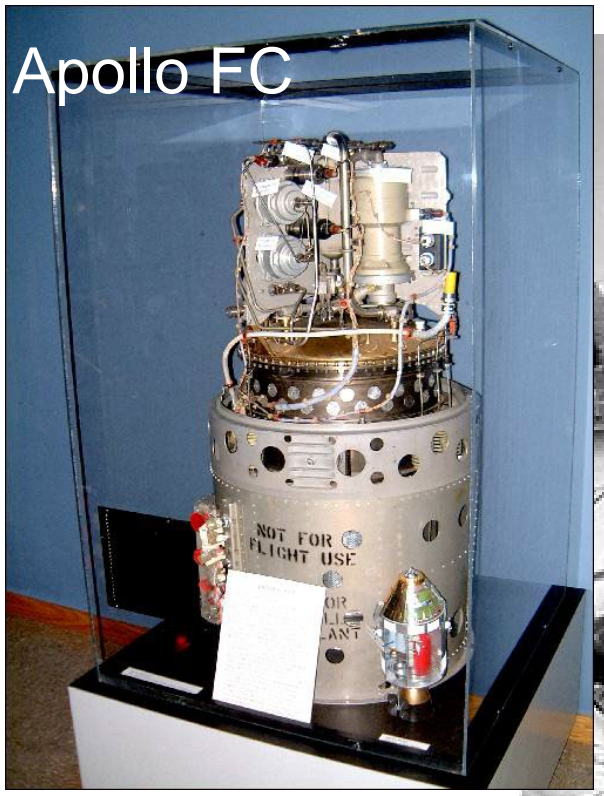
Implementation of fuel cells

First applications

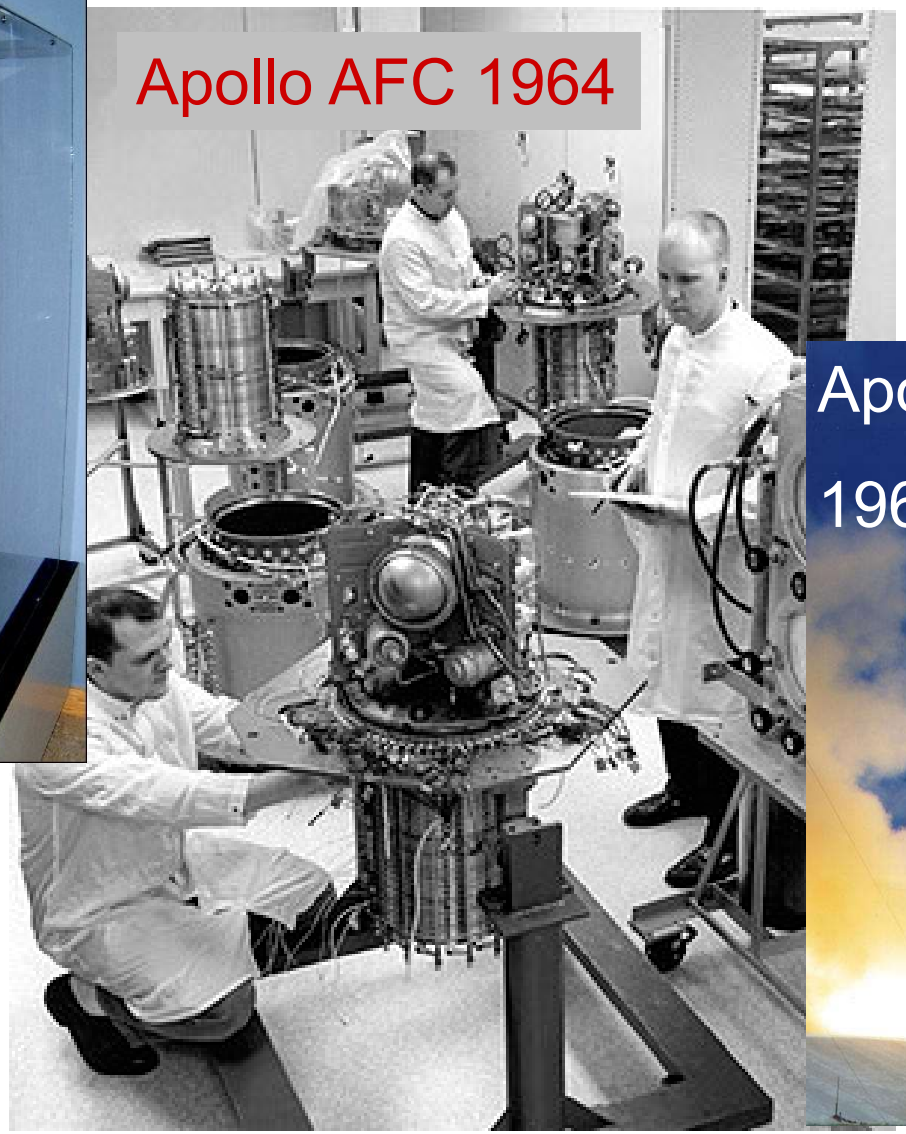
Space shuttle



Apollo FC

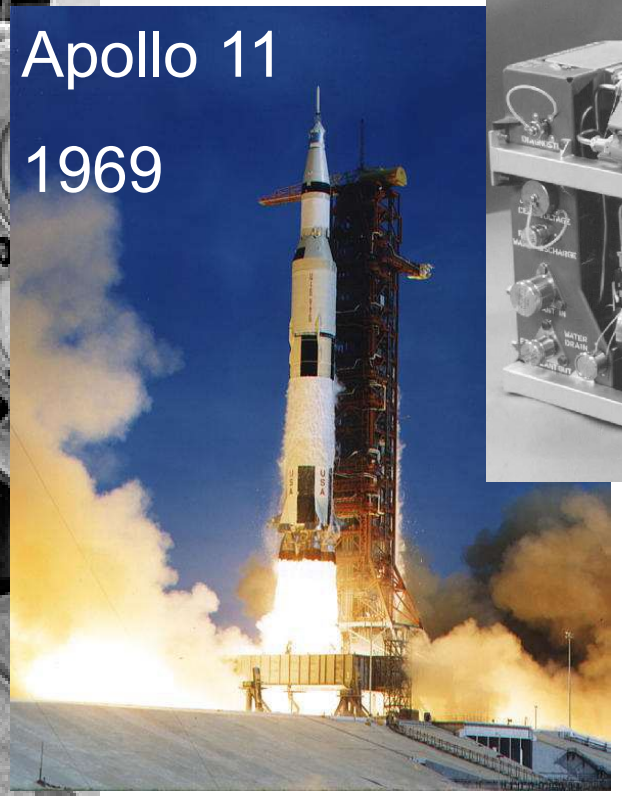


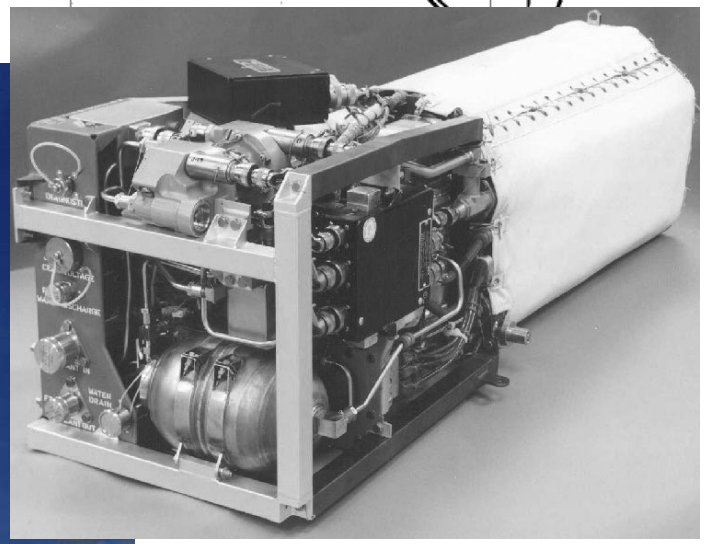
Apollo AFC 1964



Apollo 11

1969

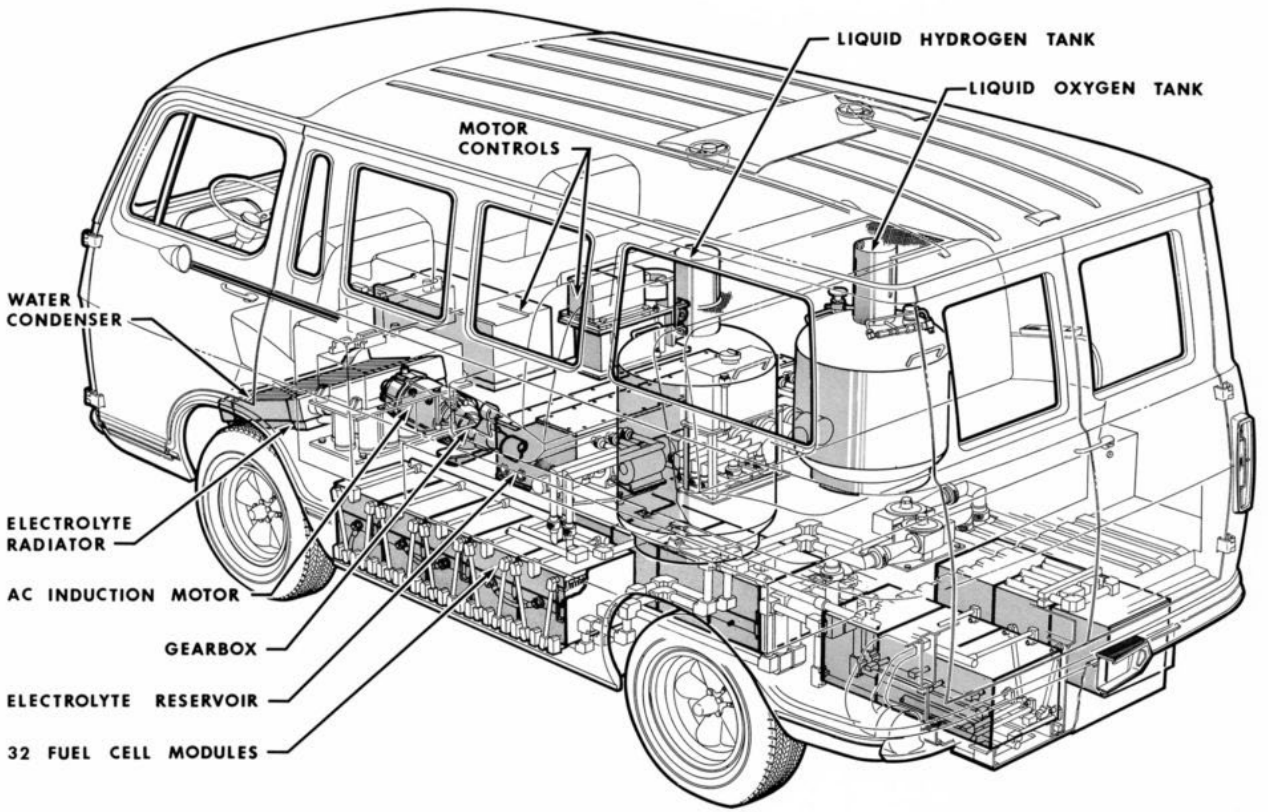




Early vehicular applications

1966 GM Electrovan

First hydrogen fuel cell car



CUTAWAY VIEW OF GENERAL MOTORS ELECTROVAN

Fuel cell electric vehicles

Hyundai Nexo 2018

Toyota Mirai 2020

Honda clarity 2018

Toyota Crown 2023

BMW NEXT FCEV
Concept car

Mercedes Benz
Plug-in hybrid

Grove fuel cell car
(China)

The battle with the batteries

Driving range	√	√
Efficiency	÷	√
Fuelling time	√	÷
Emissions	√	√
Weight	(√)	(√)
Infrastructure	÷	÷
Home charging	÷	√
Apartment charging	√	÷

Changing focus for hydrogen vehicles

FC trucks

Hyzon (US)

Mercedes-Benz (Germany)

Nikola (US)

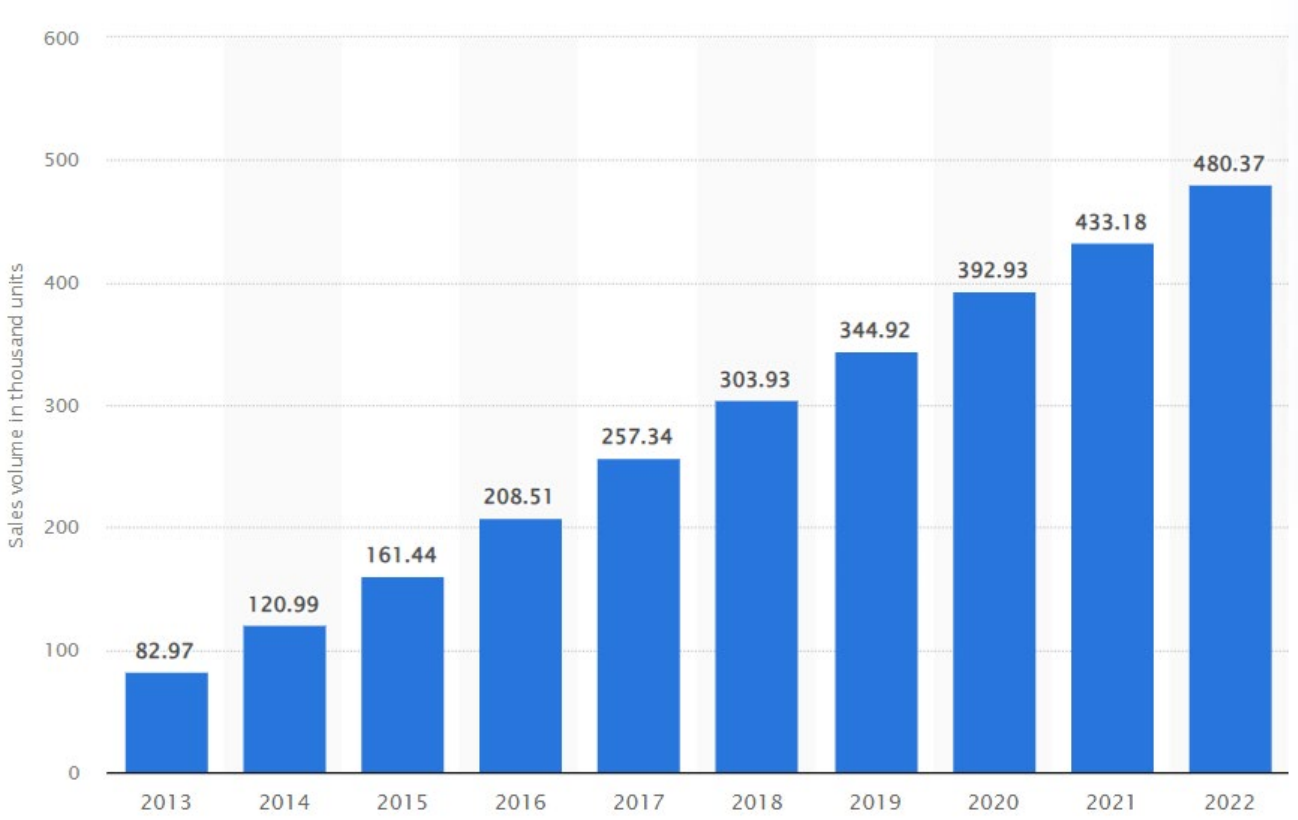
Hyundai (Korea)

Hongyan (China)

Foton (China)

Status for micro-CHP in Japan (ENE-FARM)

Panasonic μCHP
700W_e, 1000 W_h



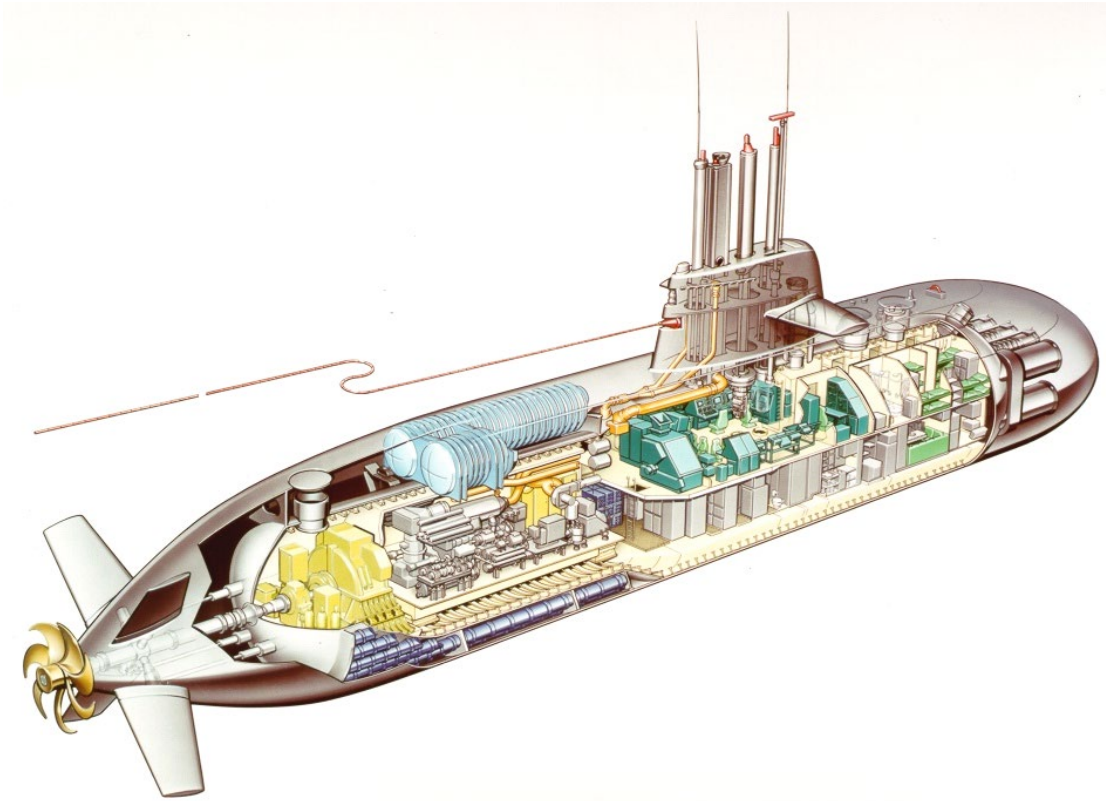
Japan LP Gas Association

Japan:
ENE-FARM programme from 2009.
2024: +540.000 units installed. Most PEMFC and some SOFC

Forklifts and submarines

Hyster N-ZR forklift

Work continuously, no battery charging
Ca. 50,000 units, mostly North America.
(Amazon and Walmart)



German submarine U - 212
Siemens PEM fuel cell
GfE metal hydride storage tank

45 vessels of the German design are said to be in service, or on order, as of mid-2017.

A. Al-Douri et al. Int. J. Hydrogen Energy, **48** (50), 19340-19355 (2023)

Hydrogen powered aircrafts

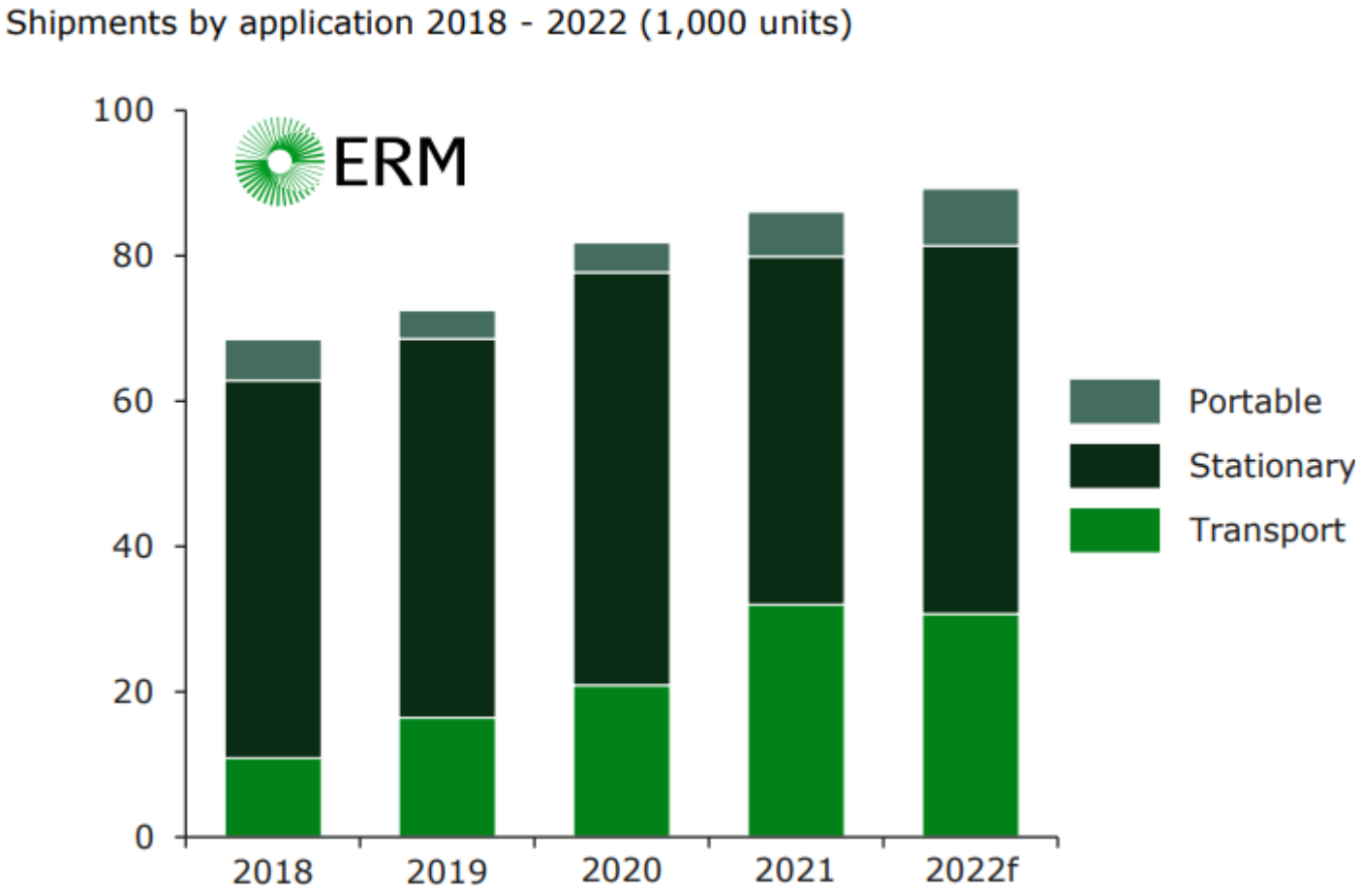
Vertiia zero emission aircraft

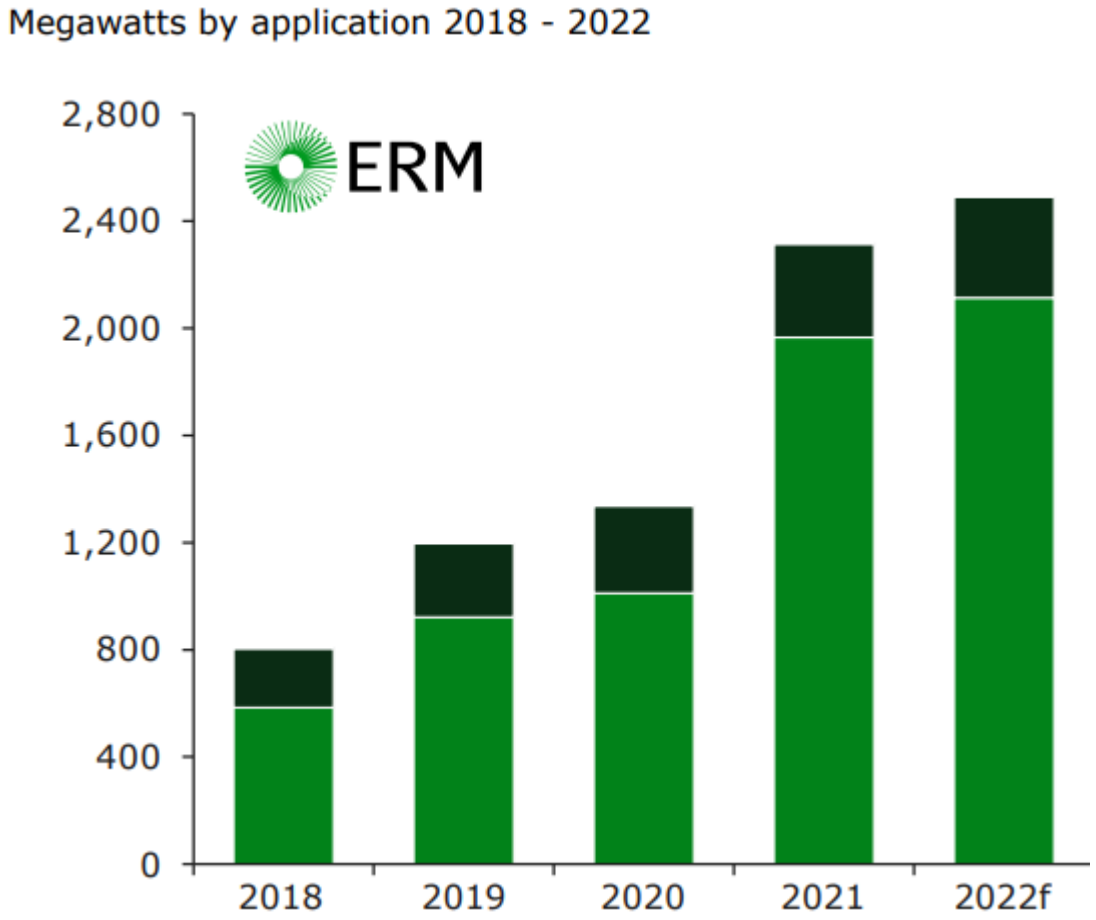
VTOL Aircraft (Vertical Takeoff & Landing)

Airbus

Concept 2020, Flights 2035 (or later)

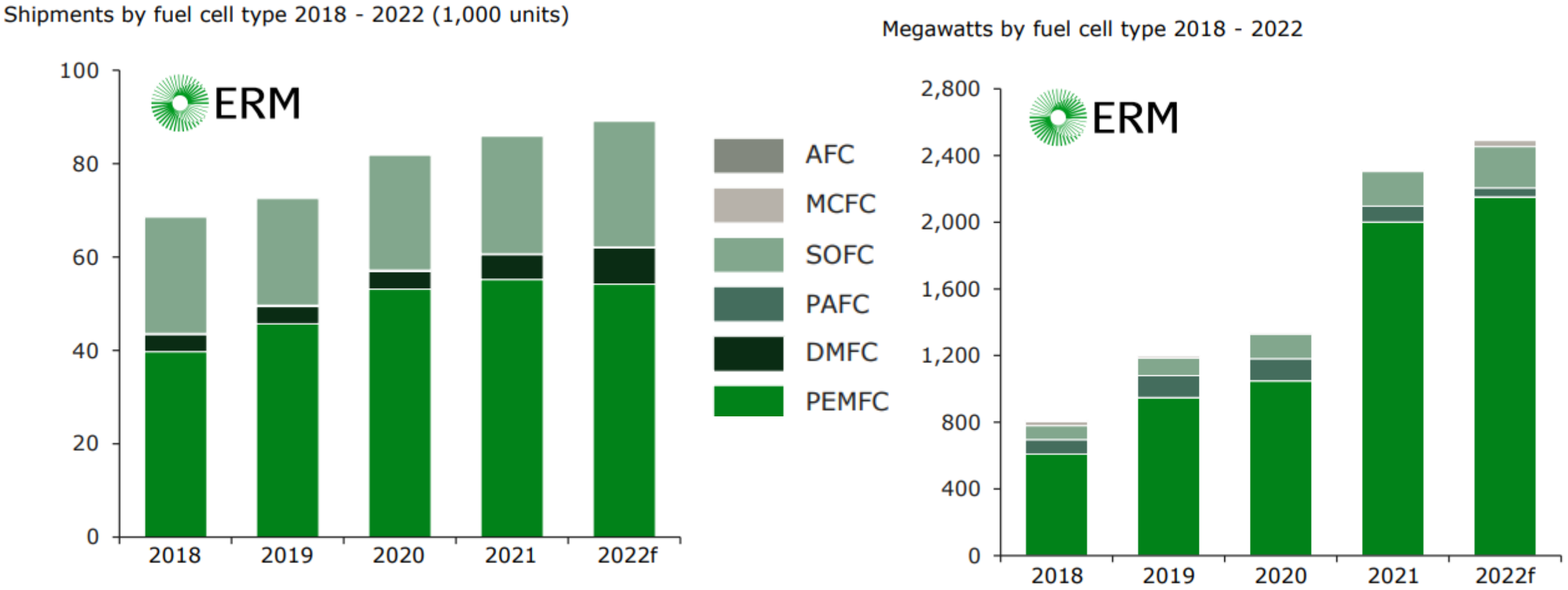
Shipment of fuel cells (by application)





https://www.erm.com/solutions/climate-net-zero/the-fuel-cell-industry-review/

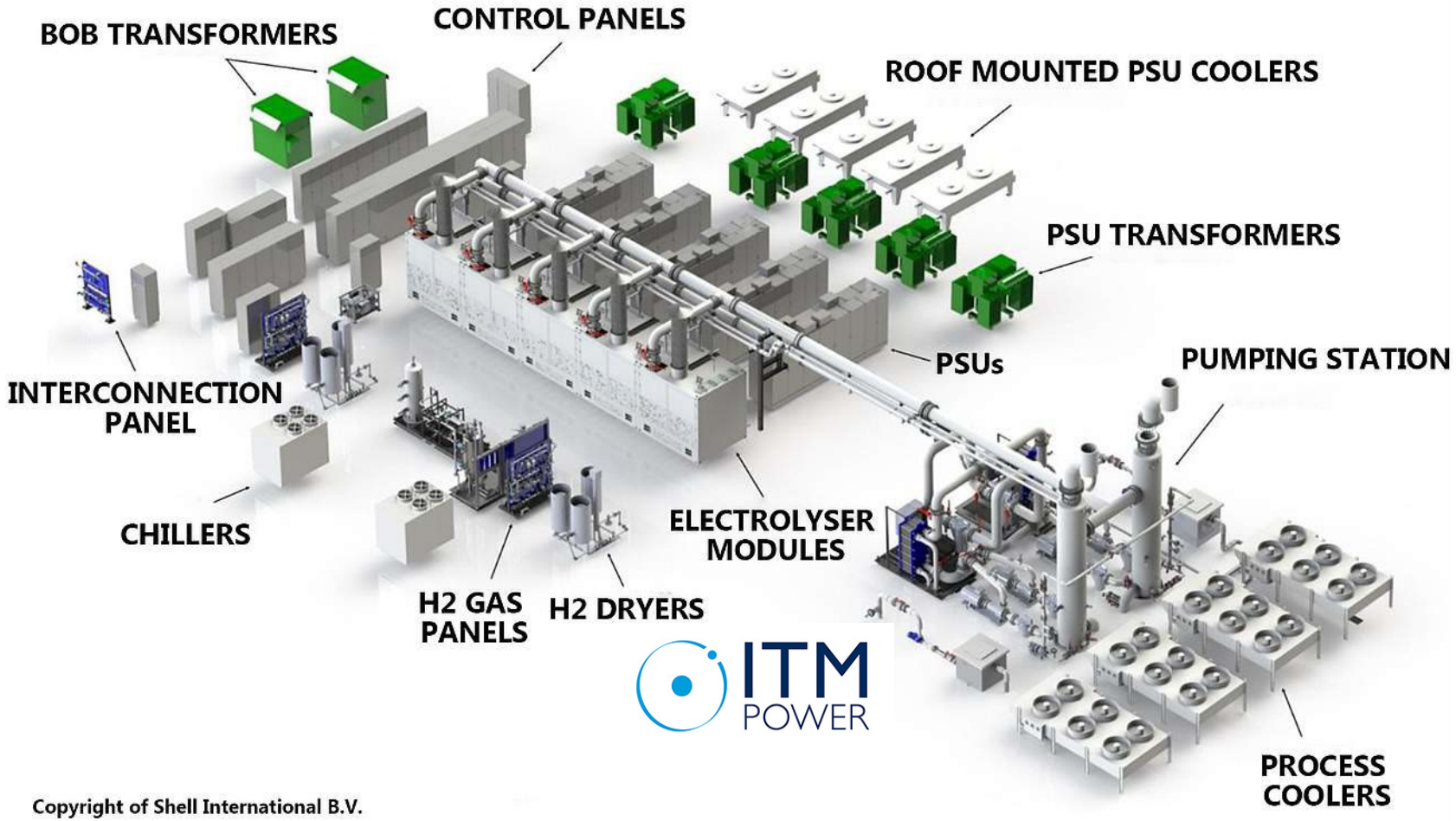
Shipment of fuel cells (by type)



https://www.erm.com/solutions/climate-net-zero/the-fuel-cell-industry-review/

Implementation of electrolyzers

REFHYNE: hydrogen for refinery



Copyright of Shell International B.V.

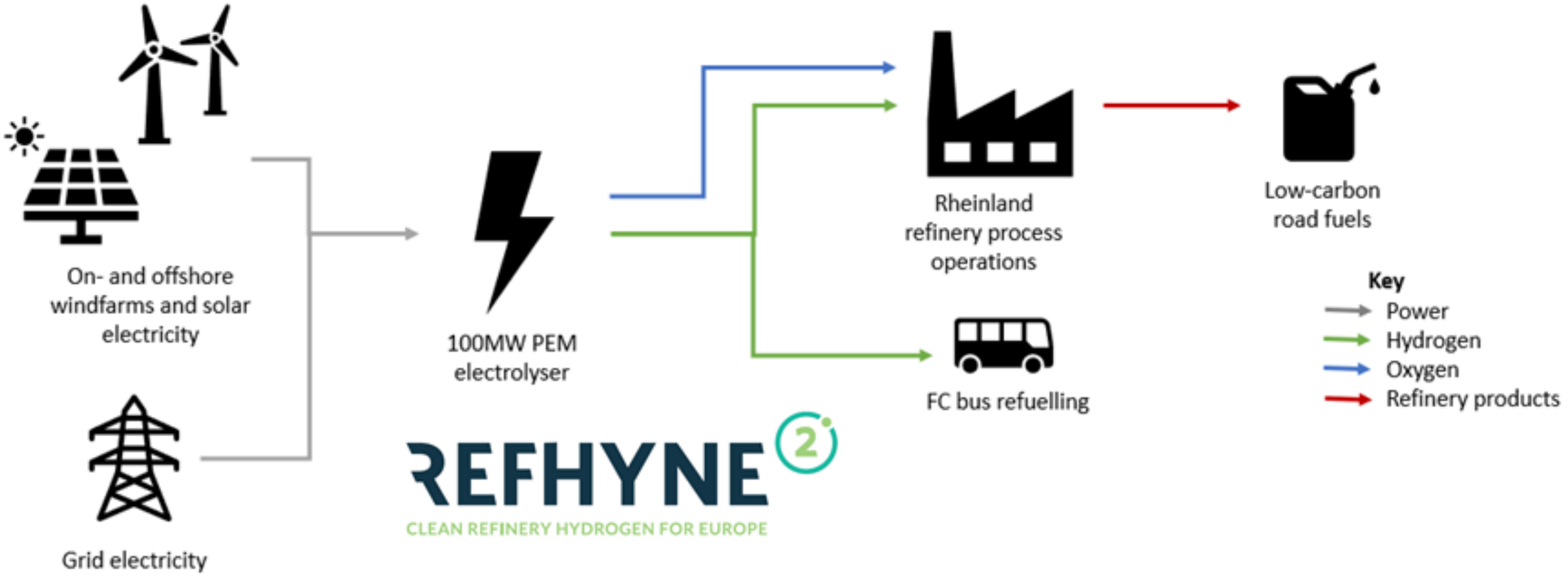
- 10 MW electrolyzer
- 1,300 tons H_2 per year
- The world's largest PEM electrolyser,
- At Shell Rheinland Refinery.

Germany's largest refinery, requires approximately 180,000 tons of hydrogen annually

Funded by the European Union

REFHYNE 2 (2021-2026)

Funded by the European Union



100 MW PEMEC, 15,000 tonnes of green hydrogen per year.

Funded by the European Union

The largest electrolyzer stack (Longi, China)

Capacity:

5,000 $NM^3\ hr^{-1}$

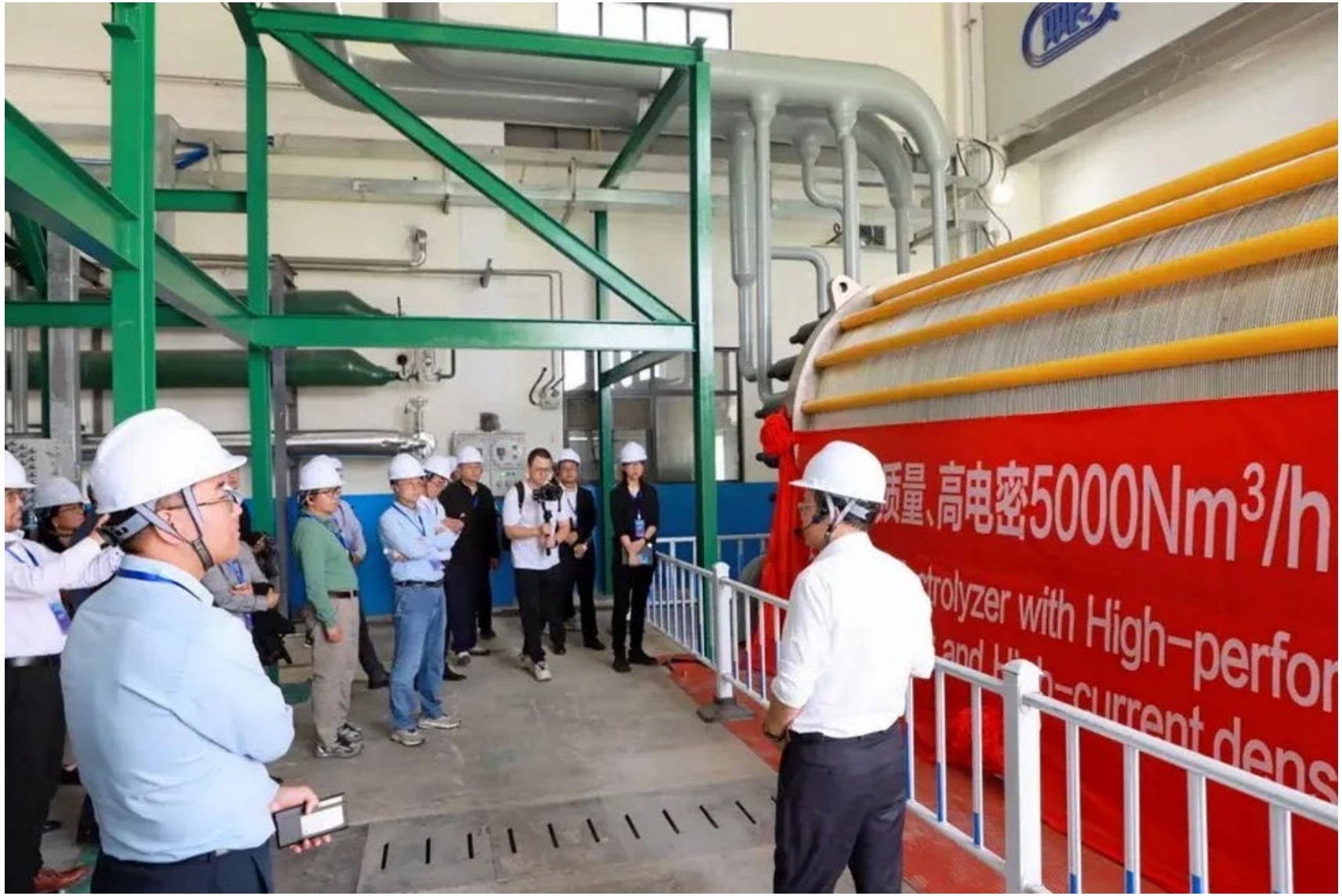
450 $kg\ hr^{-1}$

Claimed efficiency:

43.09 kWh/kg

Power rating:

Ca. 20 MW_e (calculated)



Chifeng Net Zero Hydrogen Industrial Park

- World’s largest electrolysis project
- 500 MW electrolysis from off-grid renewable electricity
- Conversion to green ammonia
- 320,000 tons/year of ammonia (2025)
- Real-time optimization via AI
- Chifeng in North China's Inner Mongolia
- By Envision Energy
- Offtake agreement with Japanese Marubeni Corporation

People's Daily, Chinas

Chifeng Net Zero Hydrogen Industrial Park

- Phase one: 300.000 tons/year ammonia
- Phase two: 600.000 tons/year ammonia
- Phase three: 1.500.000 tons/year ammonia (2028)
- Electrolysis increase from 500 MW to 2.5 GW
- Ammonia pipeline

NEOM Green Hydrogen Project

2026

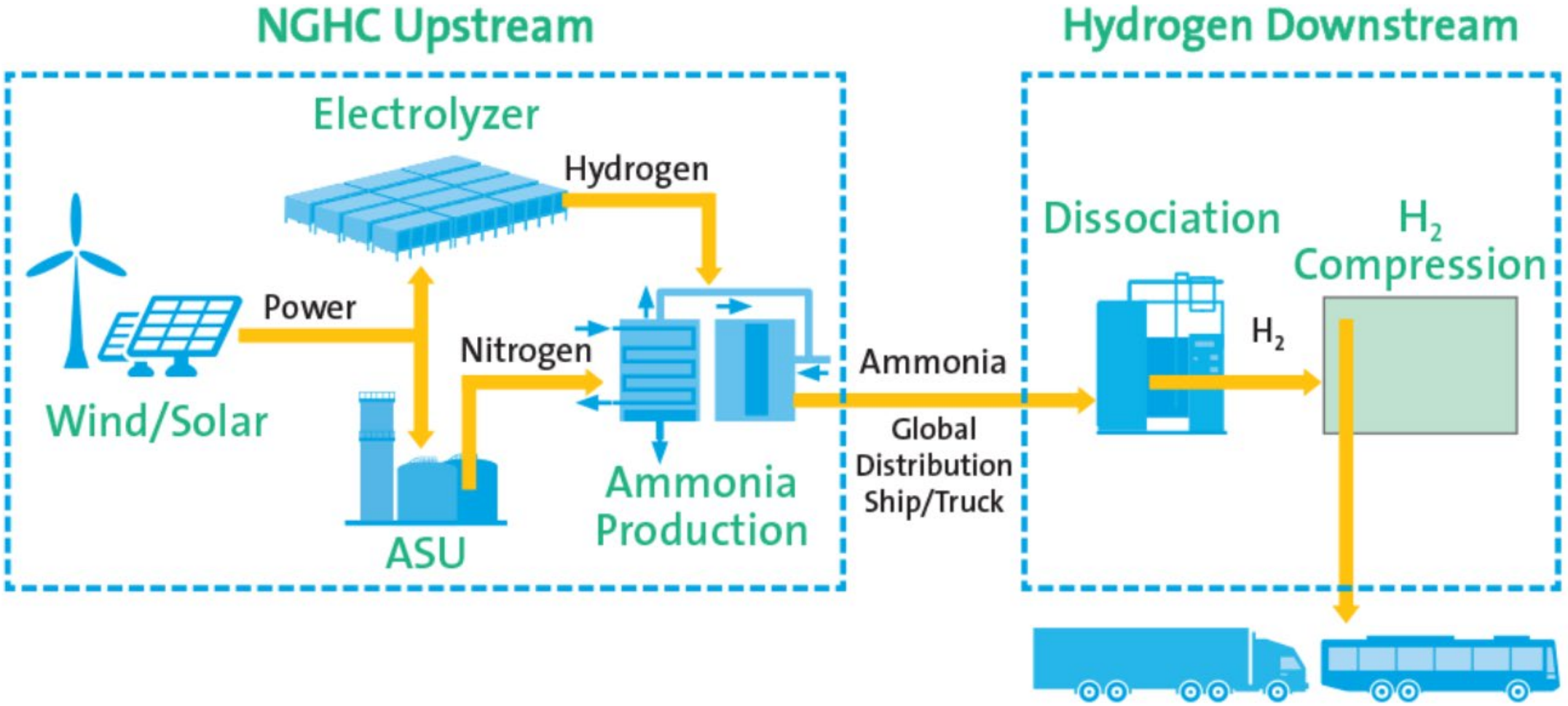
- 4 GW of solar and wind power

2027

- 600 tons hydrogen a day (220,000 tons a year) by electrolysis
- Up to 1.2 million tons a year of green ammonia.
- NEOM Green Hydrogen Company
- Thyssenkrupp electrolyzers, 2,2 GW.
- Air separation by Air Products technology

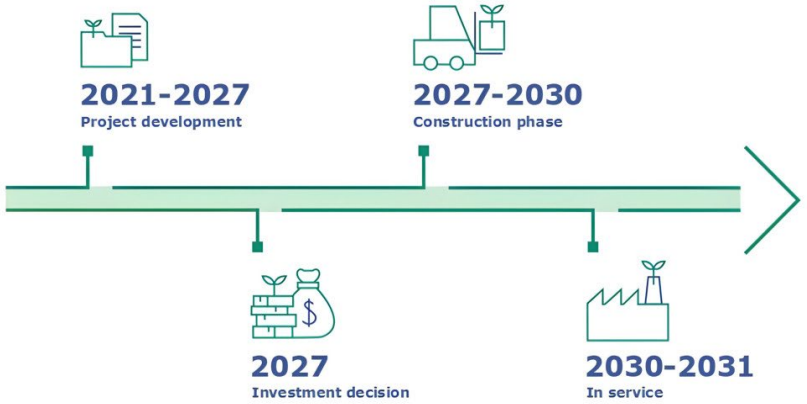
NEOM Green Hydrogen Company is an equal joint venture between ACWA Power, Air Products and NEOM

NEOM Green Hydrogen Project



HØST PtX Esbjerg (Denmark)

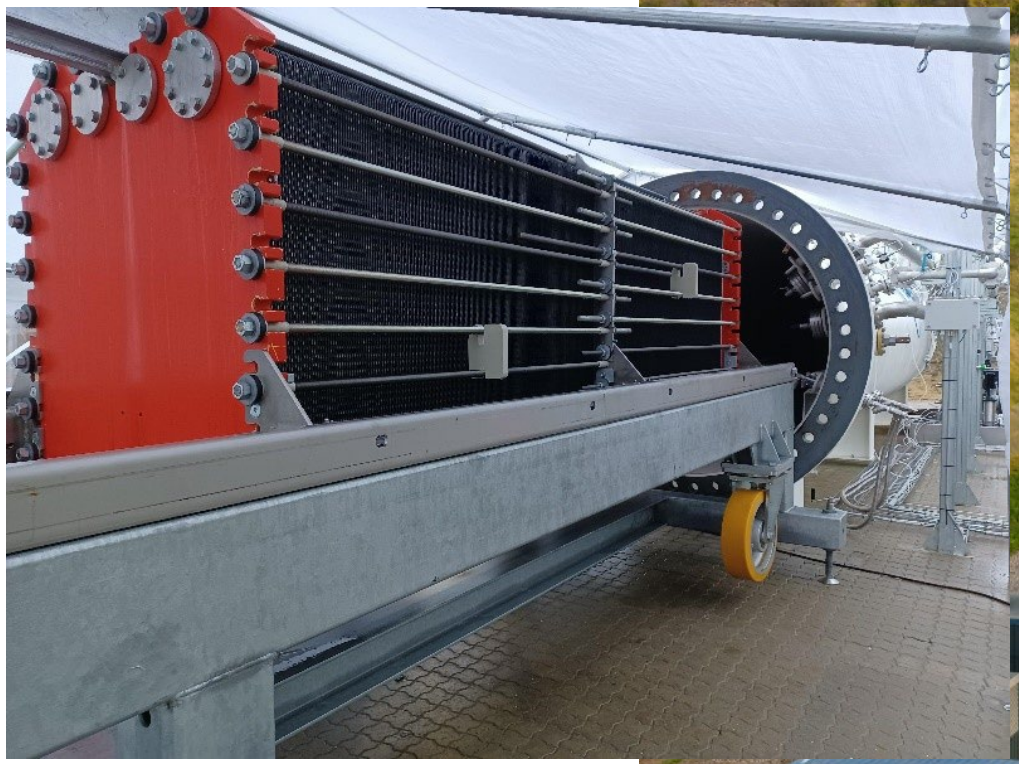
- 1GW electrolyzers
- Expected annual electricity consumption of 6000 GWh
- Expected annual production of 100,000 t green hydrogen *or*
- Expected annual production of 600,000 t green ammonia
- Emission-free district heating of app. 15,000 households



Lead by Copenhagen Infrastructure Partners (CIP)

https://hoestptxesbjerg.dk/about-ptx/

Stiesdal Hydrogen as an example



Glansager biomethanisation plant (DK) – 3 Stiesdal electrolyzer Units 3 X 3.1 MW

Installed electrolyzer capacity (GHR 2026)

Global Hydrogen Review 2026

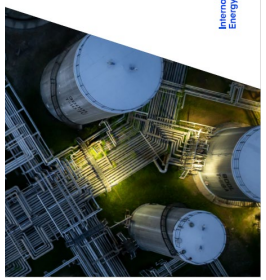
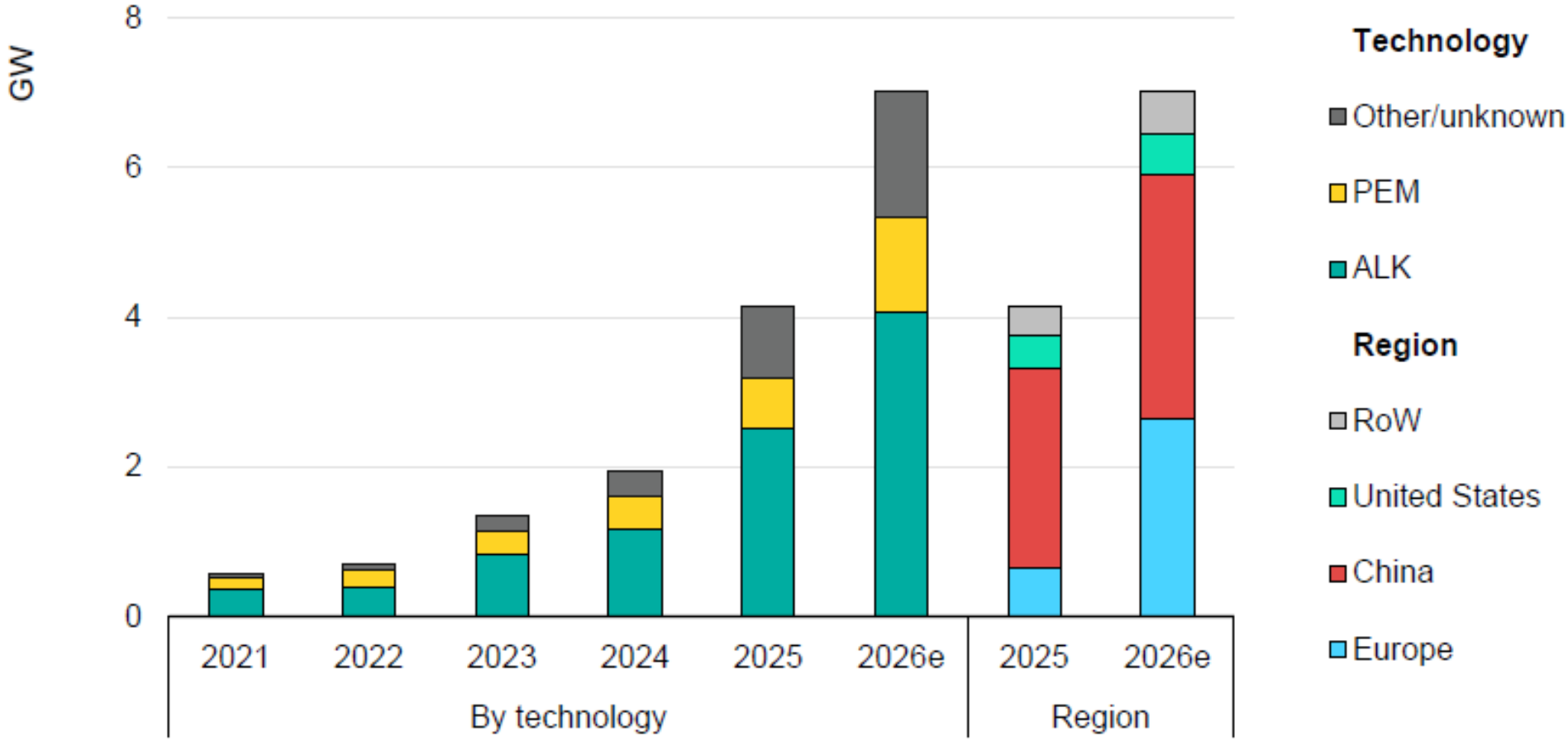


Figure 3.8 Installed electrolyser capacity by technology and region, 2021-2026e



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A tremendous need for electrolysis

EU Hydrogen strategy 2020	
2020-2024	6 GW of electrolysis, 1 million tonnes of hydrogen
2025-2030	40 GW of electrolysis, 10 million tonnes of hydrogen
2030-2050	13-14 % of total energy mix - hydrogen

Repower EU 2022	
2030	10 million tonnes of hydrogen internally produced + 10 Mt imported hydrogen. 90-100 GW_{LHV} Electrolysis, 140 GW electricity rated (utilization 43 %, eff. 70%)

BloombergNEF's New Energy Outlook (NEO)	
2030	Green Scenario: 1.9 TW of electrolyzers to kick-start the hydrogen sector.
2050	Red (nuclear) Scenario 3.572 TW of nuclear capacity to power electrolyzers

*Joint Declaration. European Electrolyser Summit, Brussels 5 May 2022, Hydrogen Europe.

Synthetic fuells

Synthetic fuels - hydrocarbons

The Sabatier process

Synthesis of methane from hydrogen and carbon dioxide

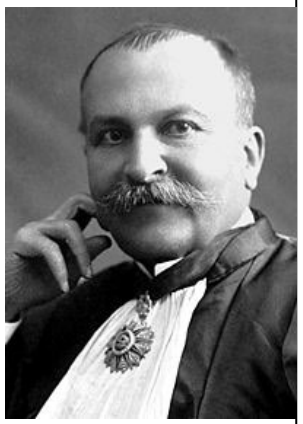
$$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O \qquad \Delta H = -165.0 \text{ kJ/mol}$$

300-400 ºC with a nickel catalyst

Uses:

- Production of "*synthetic natural gas*"
- Upgrading of biogas by conversion of CO_2 to more methane.

The Sabatier process was discovered by the French chemist *Paul Sabatier* in the 1910s



The Fischer–Tropsch process

Synthesis of liquid hydrocarbons from hydrogen and carbon monoxide

Carbon monoxide is formed from carbon dioxide by the *reverse water gas shift* reaction:

$$CO_2 + H_2O \rightarrow CO + H_2 \qquad \Delta H = +41 \text{ kJ mol}^{-2}$$

Hydrocarbons by the Fischer–Tropsch process:

$$(2n + 1)\, H_2 + n\, CO \rightarrow C_nH_{2n+2} + n\, H_2O \qquad \Delta H << 0$$

Developed by *Franz Fischer* and *Hans Tropsch* in 1925
Used in Germany during World War II and in South Africa during the embargo in the apartheid days

Methanol

Methanol synthesis from syngas ($CO + H_2$ and some CO_2) or just $CO_2 + H_2$ at 200-300 ºC and 50–100 bar

1) $CO + 2H_2 \rightleftarrows CH_3OH$ $\quad \Delta H_r = -89$ kJ mol^{-1}

2) $CO_2 + 3H_2 \rightleftarrows CH_3OH + H_2O$ $\quad \Delta H_r = -48$ kJ mol^{-1}

3) $CO + H_2O \rightleftarrows CO_2 + H_2$ $\quad \Delta H_r = -41$ kJ mol^{-1}

Catalyst: $Cu/ZnO/Al_2O_3$ (same as in a methanol reformer)

Cu is the catalyst, but both ZnO and the carrier Al_2O_3 promotes the catalyst

Ammonia, the Haber–Bosch process

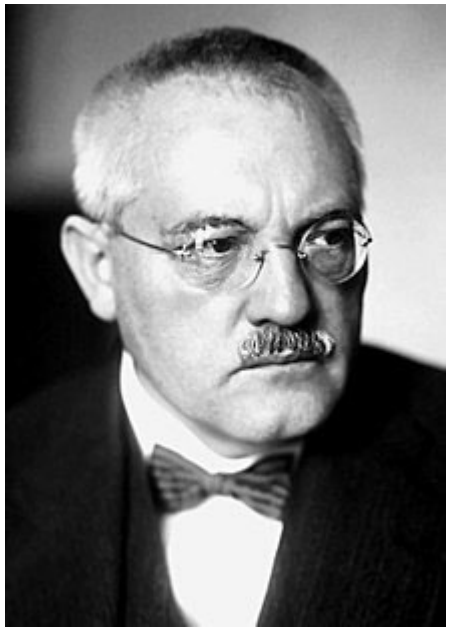
The Haber-Bosch process is still the state-of-art process for ammonia synthesis

Nitrogen isolated from air reacts with hydrogen under pressure after

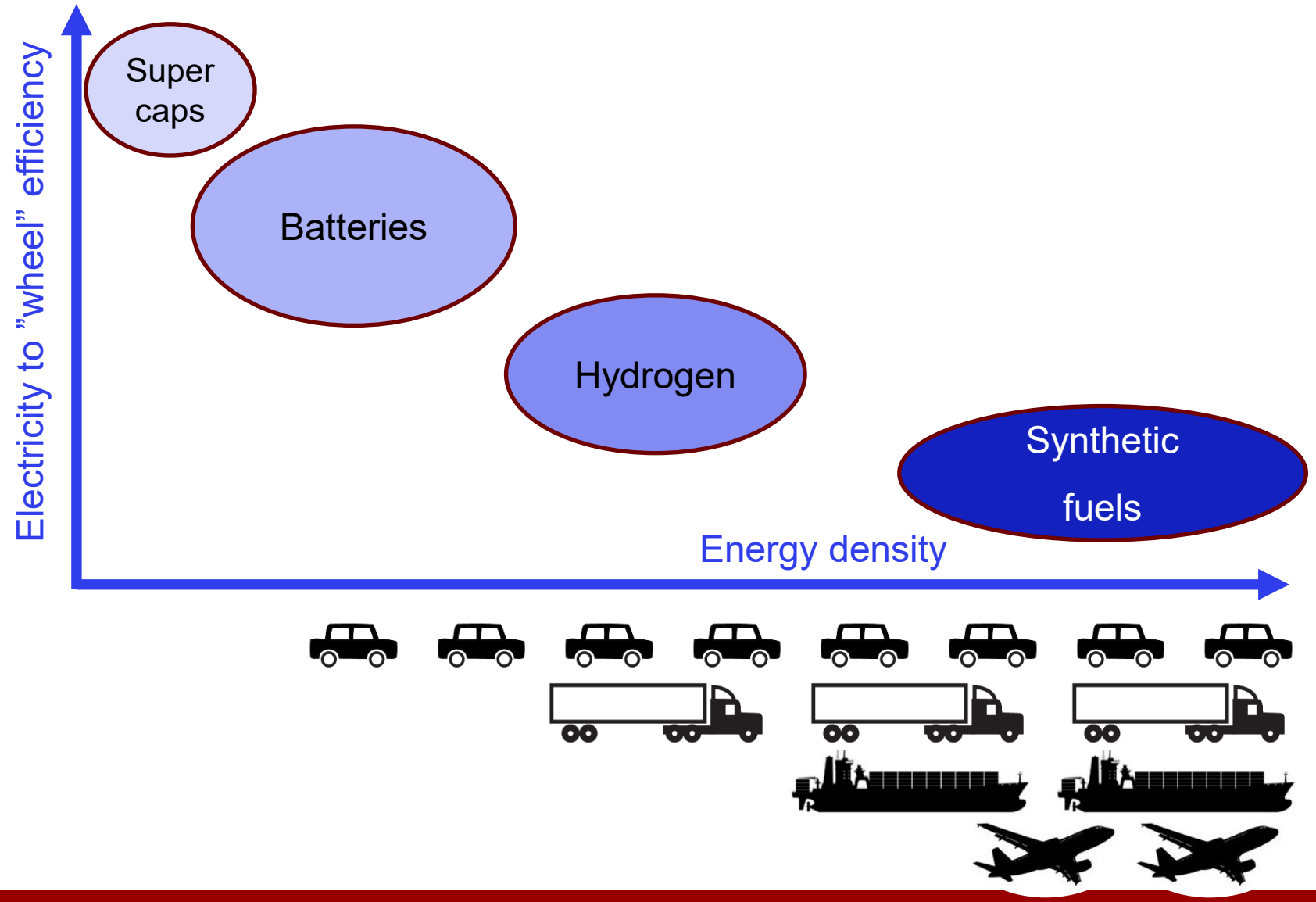
$$N_2 + 3H_2 \rightarrow NH_3 \qquad \Delta H = -91.8 \text{ kJ mol}^{-1}$$

100 – 250 bar and 400 - 500 ºC on an iron-based catalyst

Fritz Haber demonstrated it in 1909 on the lab scale and *Carl Bosch* (BASF) scaled it up 1910.



Efficiency vs. energy density (schematic)



Carbon capture CCS and CCU

CCS and CCU (and CCUS)

Carbon capture and storage (CCS)

Concept: collecting CO_2 and storing it somehow/somewhere forever

Motivation: Continued use of fossil fuels without increasing the atmospheric CO_2 concentration. We may even collect and store CO_2 from the atmosphere (negative emissions)

Carbon capture and utilization (CCU)

Concept: collecting CO_2 and using it for synthesizing fuels and chemicals

Motivation: A central part of power-to-X when hydrogen is not practical

Carbon capture, utilization and storage (CCUS)

The term covering both concepts

Ways to capture CO_2

Absorption (*bulk sorption*)

- Amines (pure liquid or solutions)
- Solutions of hydroxides

Adsorption on high surface area materials (*surface sorption*)

- Carbon materials
- Zeolites
- Metal-organic frameworks (MOF)

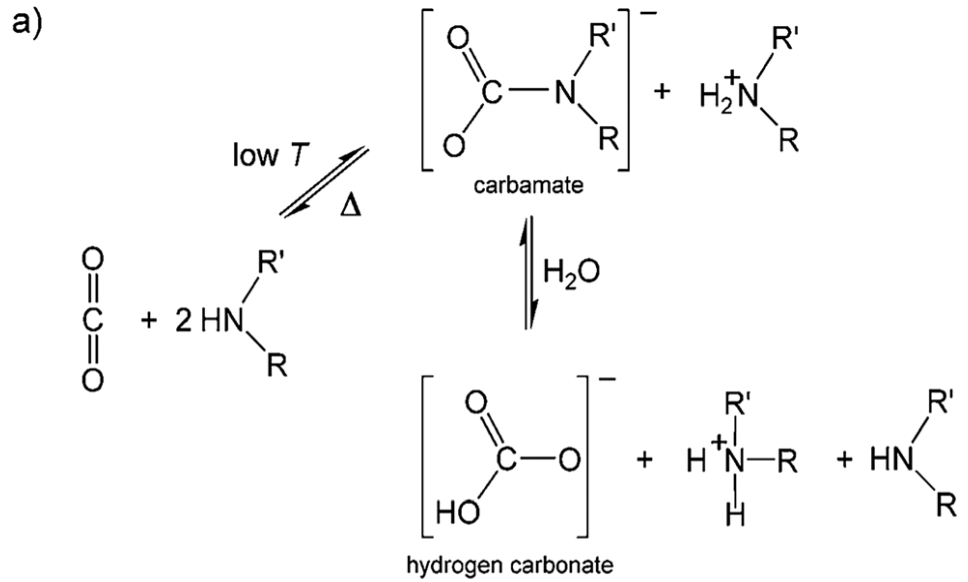
Filtration

- Selective membranes

Absorption/desorption

Amines (liquids or solutions, state-of-art)

- Monoethanolamine, $NH_2CH_2CH_2OH$ (MEA)
- Diethanolamine, $NH(CH_2CH_2OH)_2$ (DEA)
- Methyldiethanolamine, $CH_3N(C_2H_3OH)_2$ (MDEA)



Desorption at 85-120 ºC

Hydroxides (solutions)

- $Ca(OH)_2$
- NaOH

$$Ca(OH)_2 + CO_2 \rightleftarrows CaCO_3 + H_2O$$

Desorption: reverse of process by heat at around 800 ºC or more

$$CaCO_3 + heat \rightleftarrows CaO + CO_2$$

$$CaO + H_2O \rightleftarrows Ca(OH)_2$$

Direct air capture (DAC)

climeworks

First commercial CO_2 capturing project in Hinwil, Switzerland in 2017.

900 tons CO_2 each year (for greenhouses)

CarbFix project, Iceland

Deposition of CO_2 as carbonate in minerals

$$Mg_2SiO_4 + 2CO_2 \rightarrow 2MgCO_3 + SiO_2$$

DAC by Climeworks

With Audi 2020: 4,000 metric tons of CO2 will be stored as rock underground each year

The Role of PtX in the future

Application

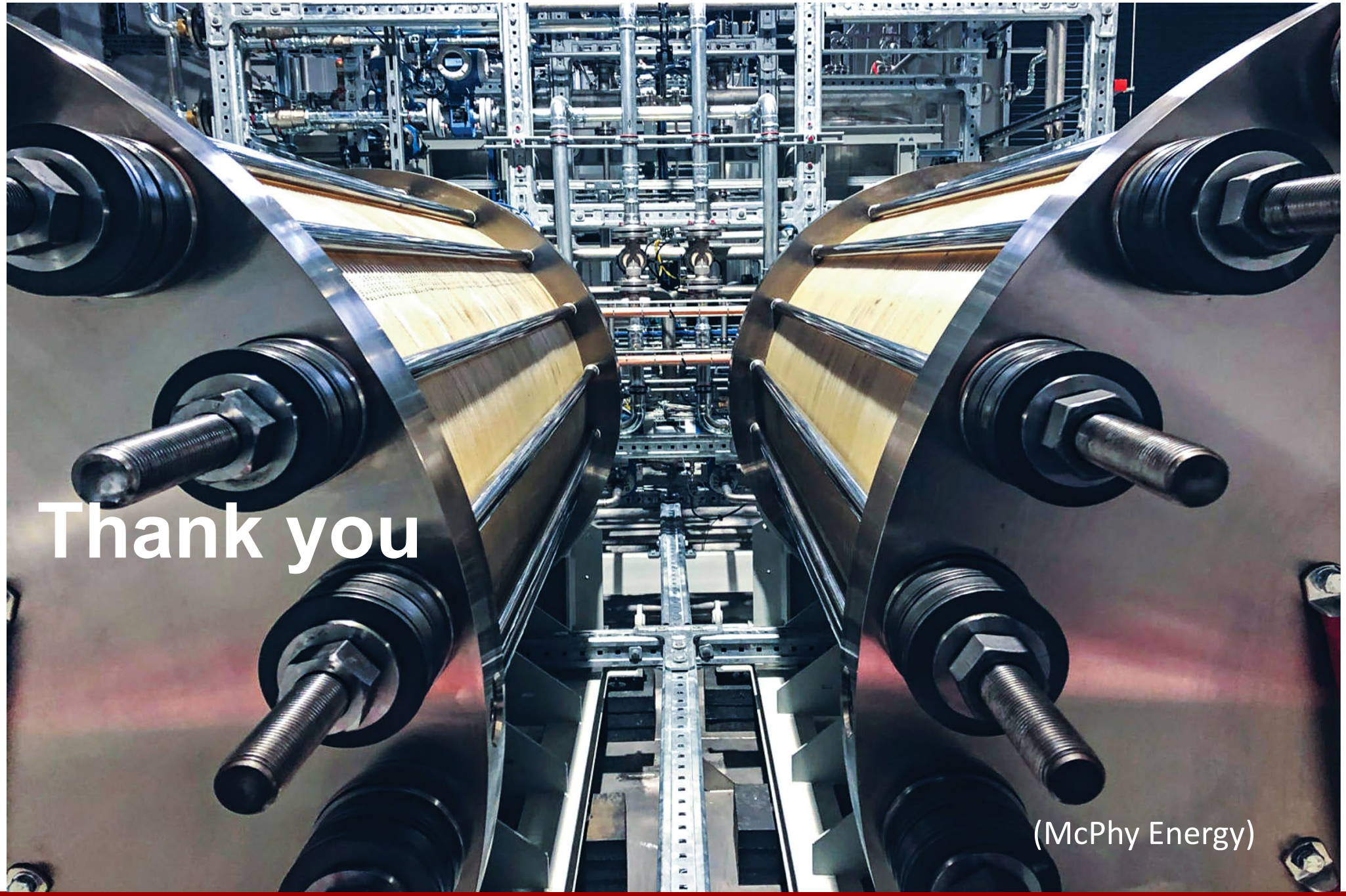
- Hydrogen for transport: uncertain
- Production of fuels for demanding transportation (aviation, shipping, long-distance trucking) (carbonaceous or ammonia)
- Production of carbon neutral chemicals
- Hydrogen as reducing agent in steel works and other industries
- Energy storage at large scale (e.g., seasonal storage)

Extent of PtX

- Likely at the TW scale globally

Carbon capture

- CCU: non-fossil carbon sources needed for chemical and advanced fuels
- CCS: possibly a short-term fix. Uncertain on the longer term


Thank you
(McPhy Energy)